Bioinformatics

KB8029



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Highlights:

- Databases, records and evidence
  - Choosing databases and connecting DNA, RNA and protein records
  - Using identifiers and cross-references and accessing evidence and versions
- Database access and retrieval
  - Command-line/programmatic access and database web pages
  - Genome browsers: DNA, RNA and protein in one interface
- Reading literature online
  - Database searches and AI-assisted discovery
  - Building queries and checking original sources

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Databases

- A database is an organised collection of data that supports searching and retrieval.
- Database management software stores, updates and retrieves the data.
- Hardware provides storage and computing power.
- A database website provides an interface for searching and viewing records.

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Growing data volumes

- *Decimal units and dated examples*

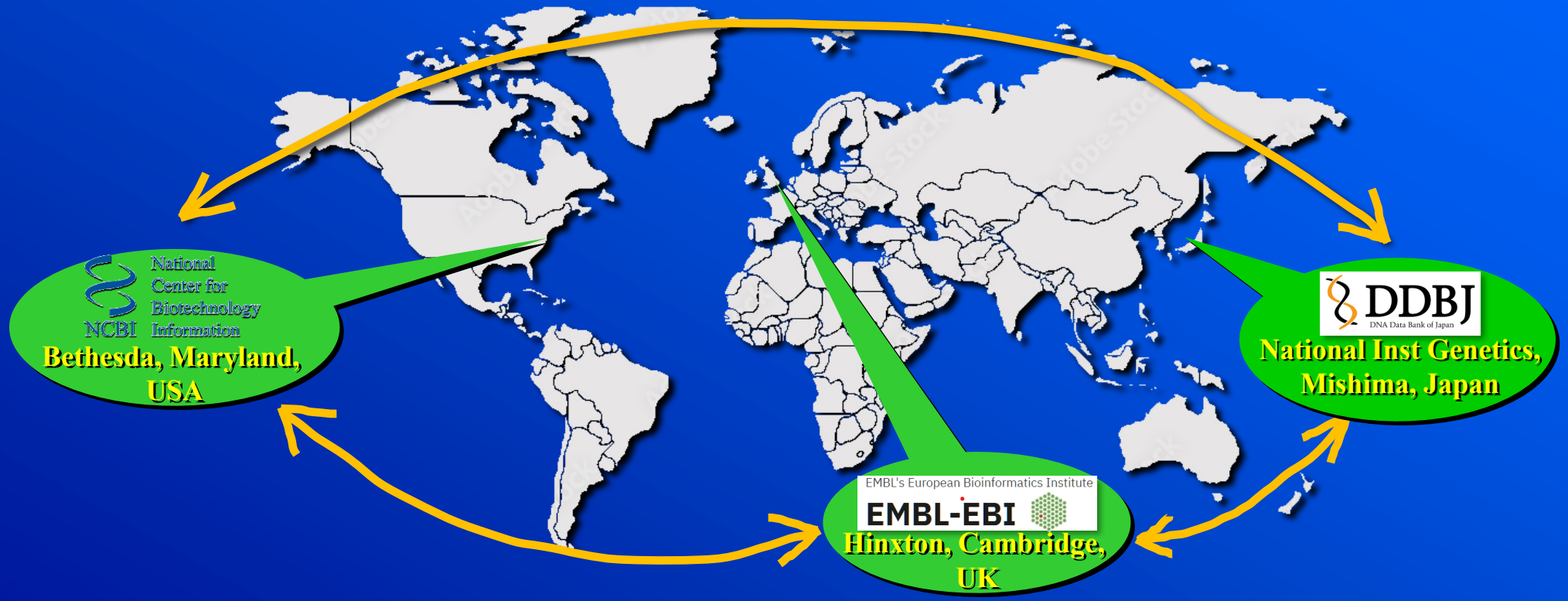
Unit	Bytes per unit	Symbol	Example	Size
Byte	10^{0}	B	One ASCII character	1 B
Kilobyte	10^{3}	kB	1,000 ASCII characters	1 kB
Megabyte	10^{6}	MB	Swiss-Prot, compressed text (2026_03)	~700 MB
Gigabyte	10^{9}	GB	TrEMBL, compressed text (2026_03)	~121 GB
Terabyte	10^{12}	TB	PDB Core snapshot (1 Jan 2026)	~1.5 TB
Petabyte	10^{15}	PB	SRA unique data (April 2026)	>50 PB

- *Full copies of the largest archives are impractical.*
- *Use shared databases efficiently.*

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Shared sequence archives

- International Nucleotide Sequence Database Collaboration (INSDC)



- Shared accessions and daily data exchange
- Common standards, different search interfaces

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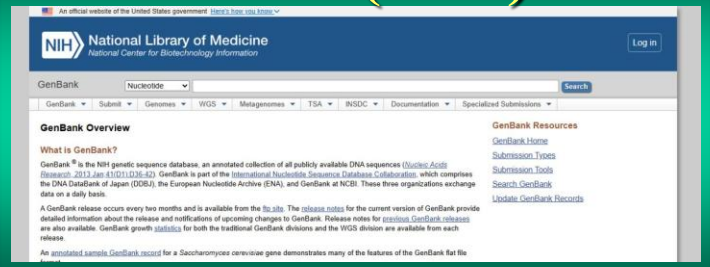
Shared sequence archives

- Access data through any partner.
- Submit data to only one.



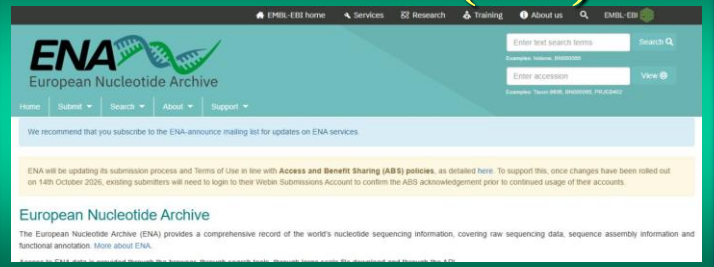


NCBI (USA)



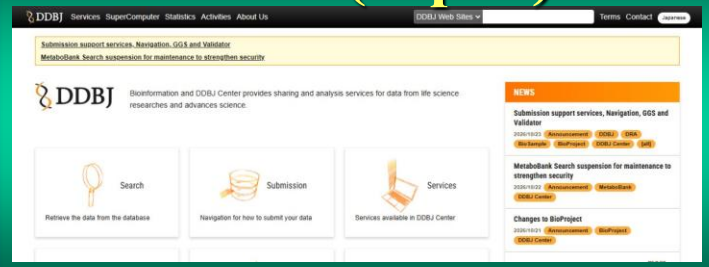
ncbi.nlm.nih.gov/genbank

EMBL-EBI (UK)



ebi.ac.uk/ena/browser

DDBJ (Japan)



ddbj.nig.ac.jp

Daily exchange

Daily exchange

Connected records

Read archives
SRA / ENA reads / DRA
Reads and quality data

Assembled sequences and annotations
GenBank / ENA / DDBJ

BioProject / Study
Project description

BioSample / Sample
Biological material

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Shared sequence archives

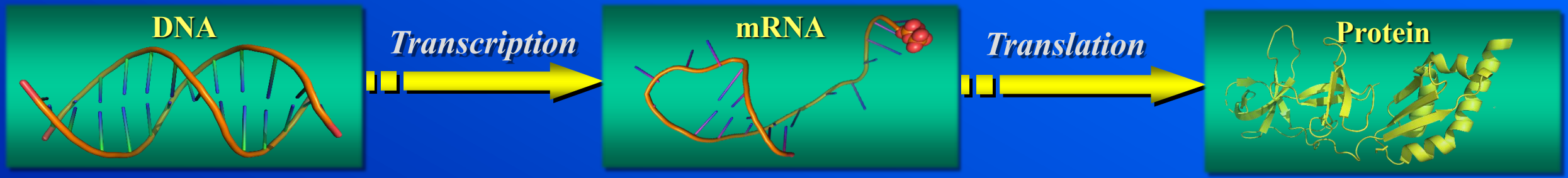
- *Archives merge and access routes change.*



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DNA, RNA and protein data

Central dogma of biology



Linked database records

DNA-level data

Genomic sequences and gene context

NIH GenBank

EMBL's European Bioinformatics Institute EMBL-EBI

DDBJ

Cross-references

RNA-level data

RefSeq

mRNAsequences

RNAcentral

Noncoding RNA sequences

Rfam

RNA families

Cross-references

Protein-level data

Sequence and structure
Function and interactions

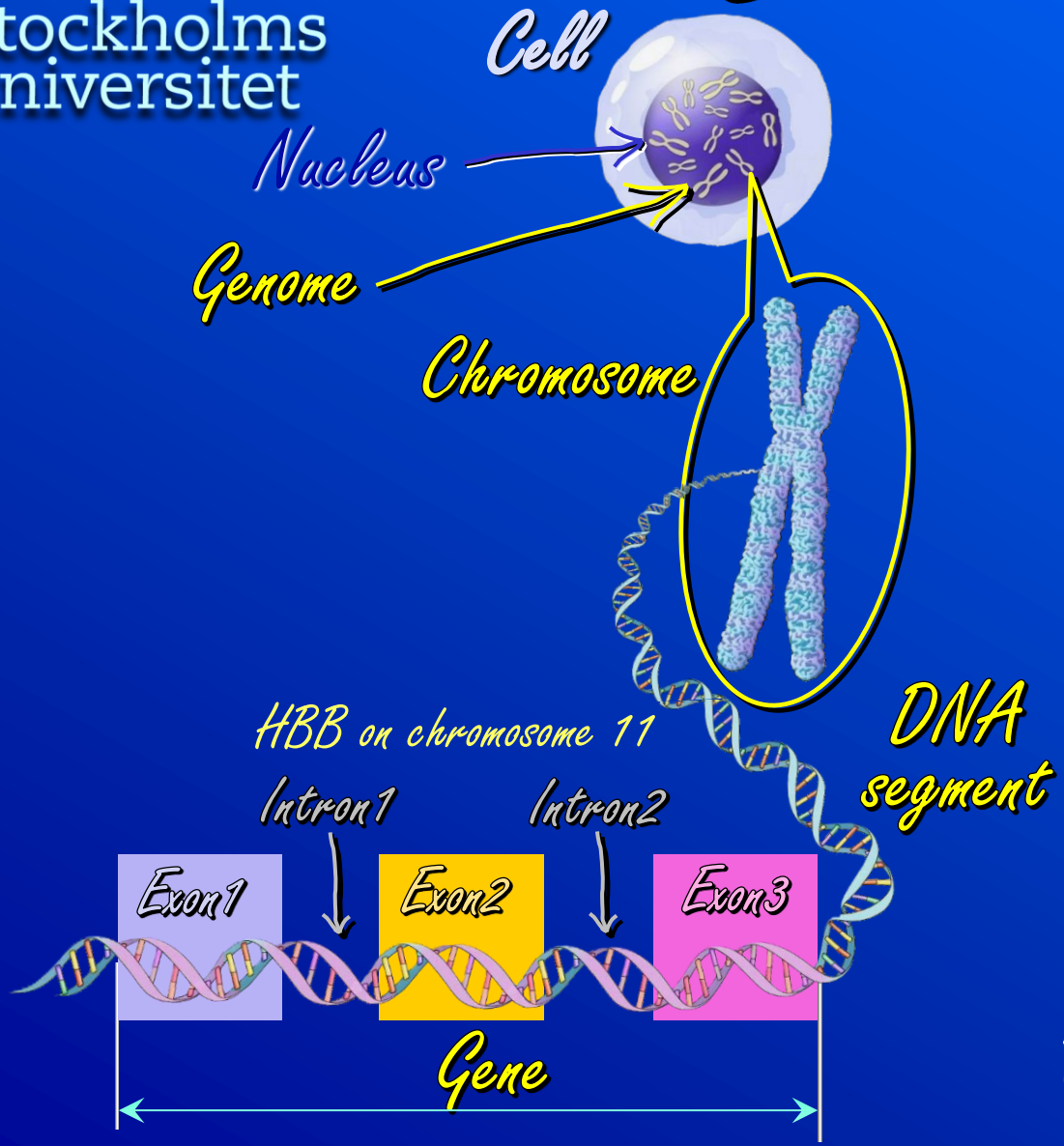
UniProt

RCSB PDB PROTEIN DATA BANK

IntAct

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DNA-level data



- **Genome**
  Genome assemblies and their annotations
- **Chromosome overview**
  Chromosome sequences and mapped information
- **Selected DNA region**
  Genes and non-coding DNA within a region
- **Gene**
  Gene structure and links to RNA and proteins

NCBI Gene

GenBank

ENA

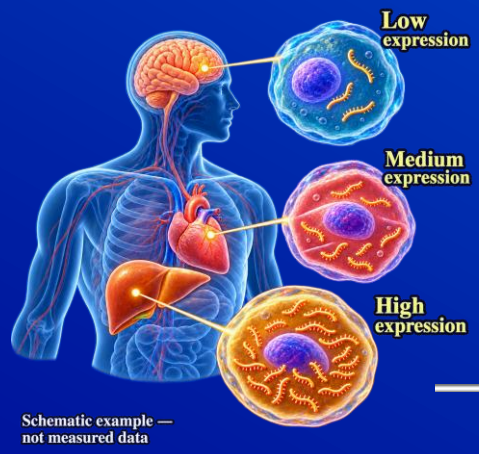
DDBJ Annotated/ Assembled sequences

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RNA-level data







RNA transcripts

Sequences, exon composition and transcript variants

RefSeq

Ensembl

Noncoding RNAs

RNA sequences, families and structure

RNAcentral

Rfam

RNA expression

Where RNA is detected and how abundant it is

Expression Atlas

GEO

GEA

Same RNA sequence, two writing conventions

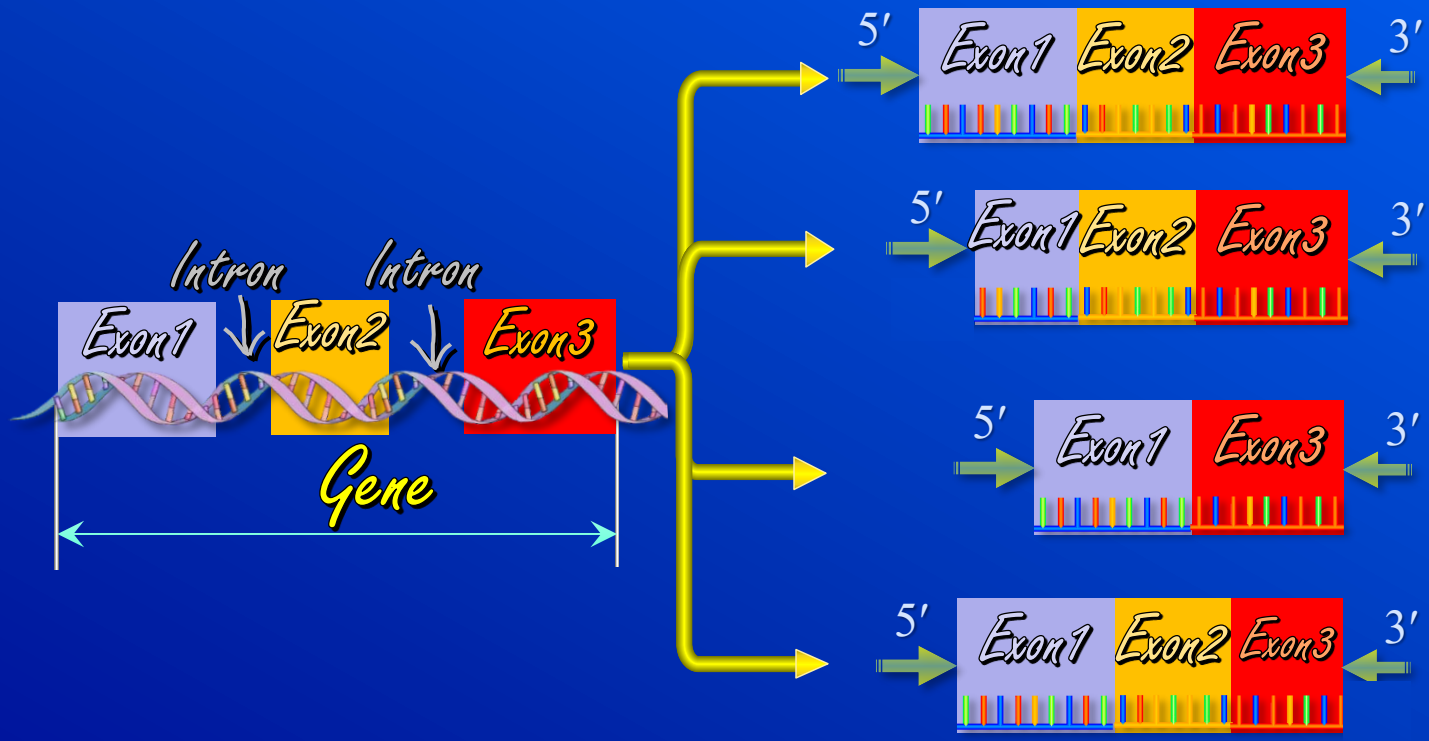
RNA notation 5′ A G C U U C G A 3′

DNA notation 5′ A G C T T C G A 3′

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RNA-level data

One gene, several RNA transcripts



- Baseline transcript
- Transcription start at different sites
- Alternative splicing
- Alternative cleavage/polyadenylation sites

Two transcripts, same gene?

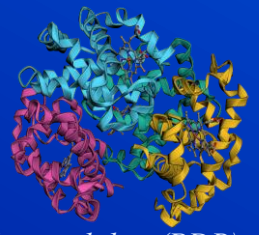
Transcripts originating from the same gene share a linked gene identifier.

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Protein-level data

MVHLTPEEKS
AVTALWGKVN

β-globin sequence



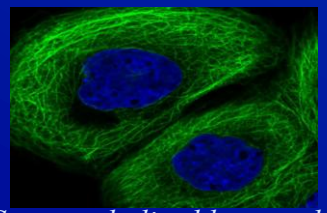
Haemoglobin (PDB)



Domain map: schematic



HBB network (STRING)



Green: tubulin; blue: nuclei

Amino-acid sequence

- Sequences and annotated variants

UniProtKB Protein Knowledgebase

NCBI PROTEIN

Three-dimensional structure

- Experimental (PDB) or predicted (AlphaFold)

RCSB PDB PROTEIN DATA BANK

PDBj Protein Data Bank Japan

PDBe Protein Data Bank in Europe

AlphaFold

Domains and families

- Conserved regions and functional sites

Pfam

InterPro

ECOD

Interactions and associations

- Physical / genetic interactions and functional links

BioGRID

BioGRID

STRING

IntAct

Abundance and location

- Where a protein occurs and how much is detected

THE HUMAN PROTEIN ATLAS

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Behind database records



Experiments generate raw data

- Sequencing signals
- Experimental images
- Instrument readings
- Phenotypic observations



Analysis produces results and predictions

- Assembled sequences
- Translated protein sequences
- Domain assignments
- Three-dimensional structures



Curation evaluates and connects the information

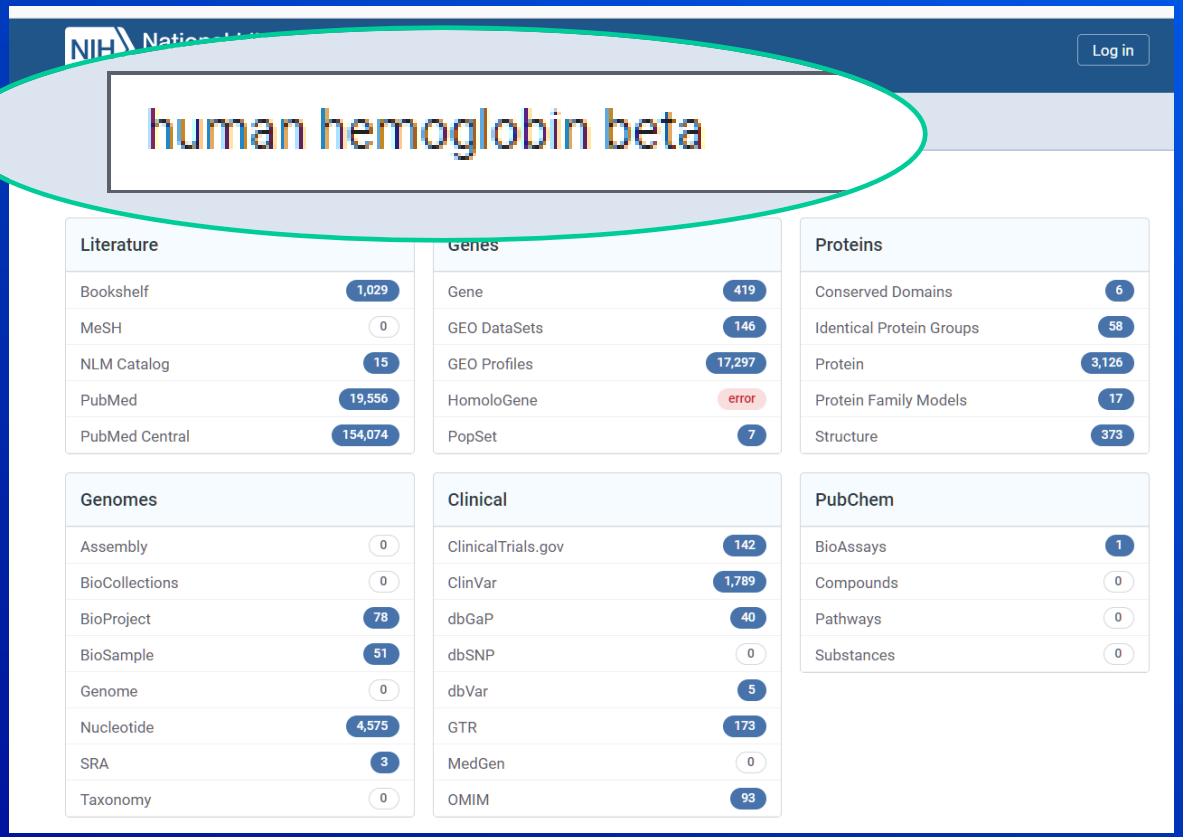
- Assess supporting evidence
- Create and update annotations
- Compare findings across studies
- Link related records

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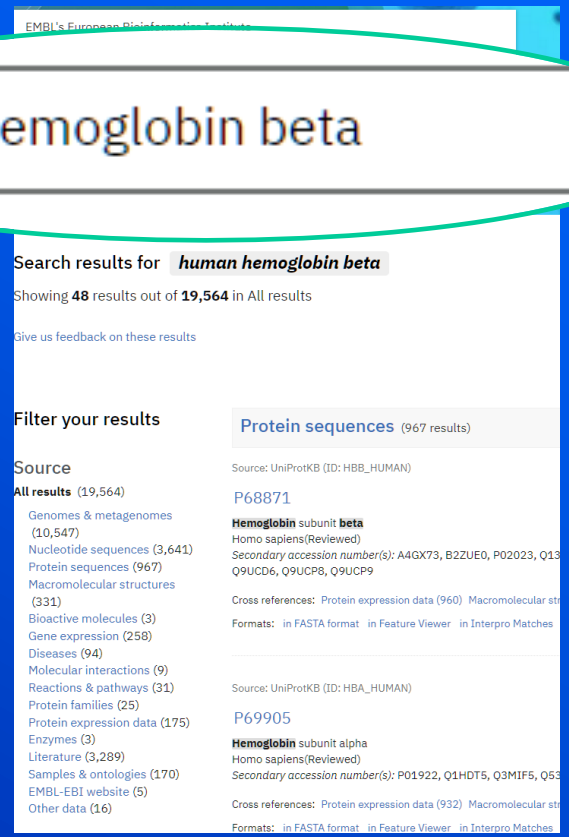
Search by name

- One biological name, different databases

NCBI



EMBL-EBI

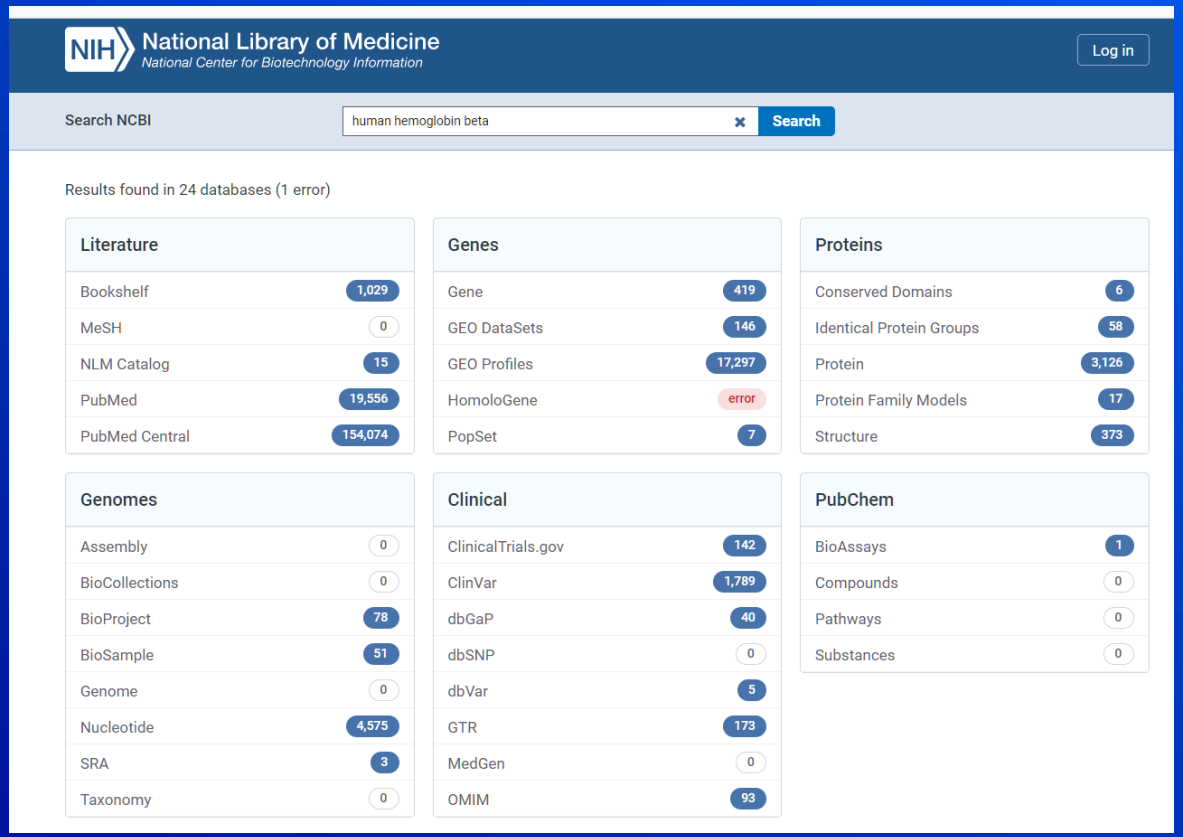


Stockholms universitet

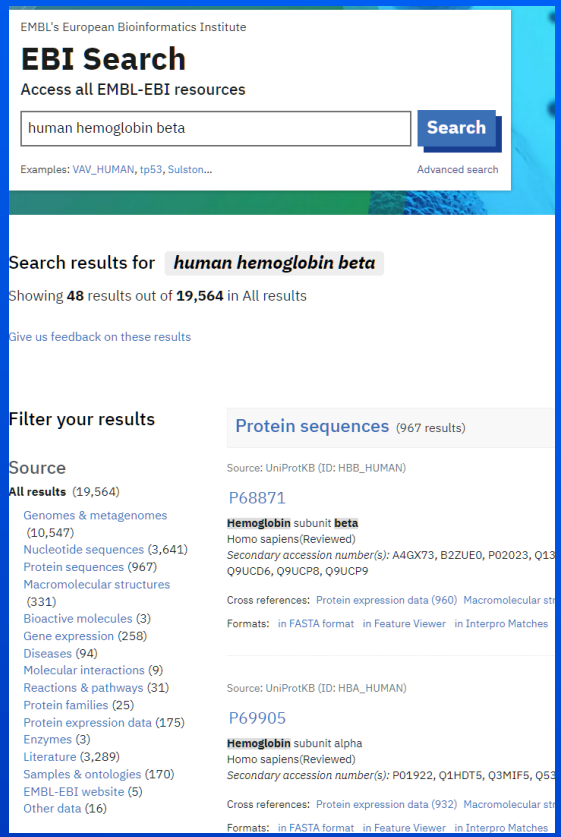
Search by name

- One search, several kinds of records

NCBI



EMBL-EBI

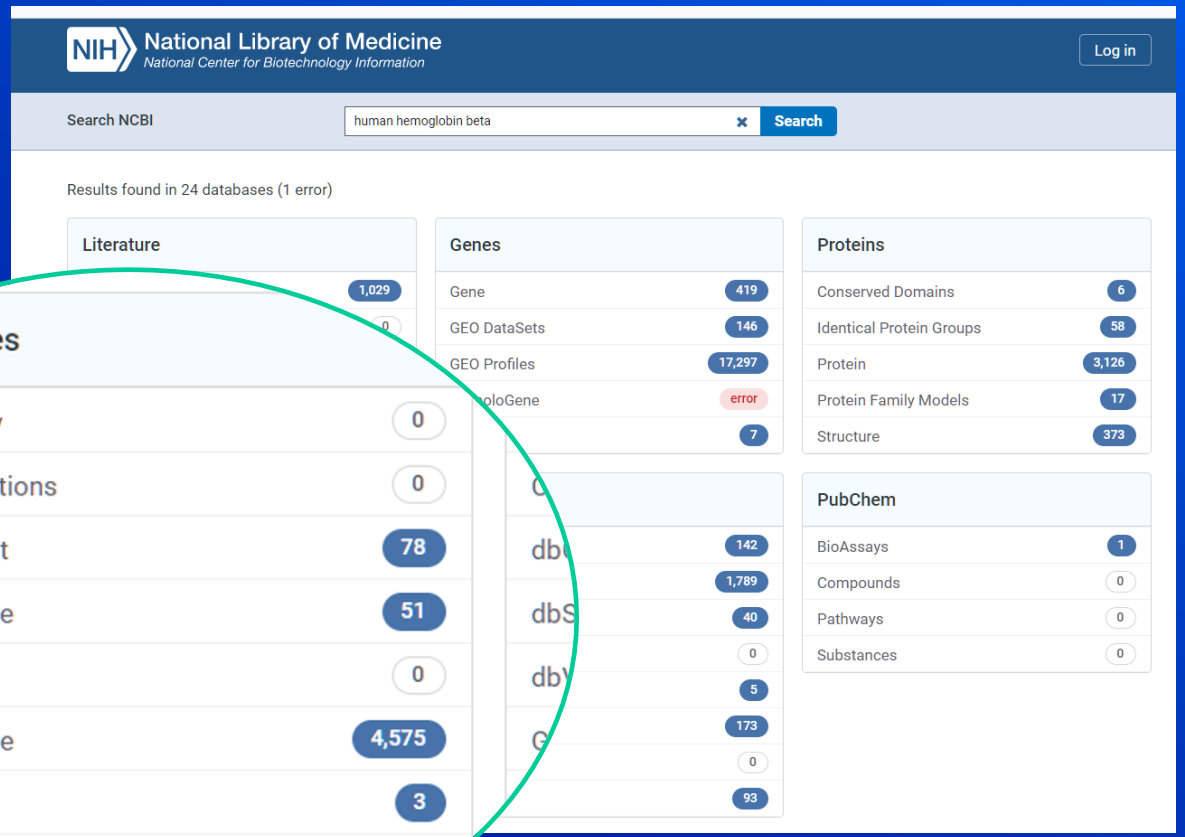


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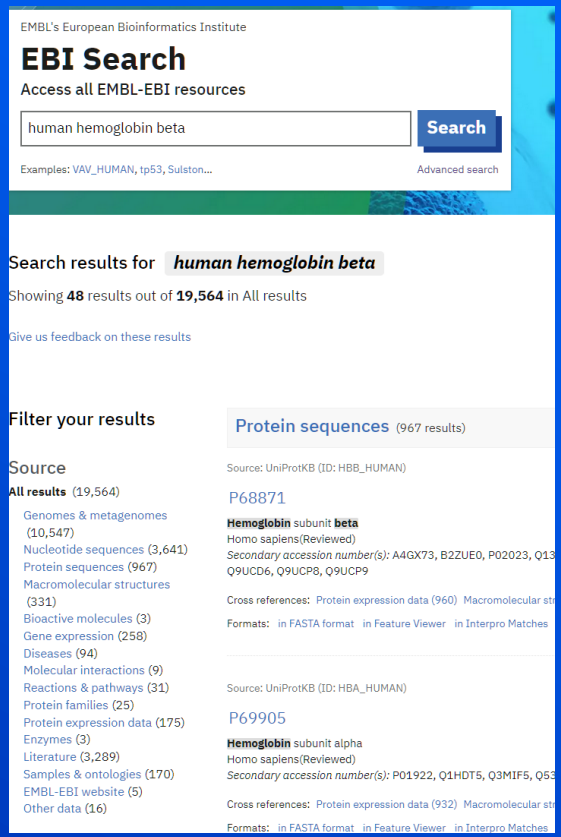
Search by name

Nucleotide sequences and their sources

NCBI



EMBL-EBI

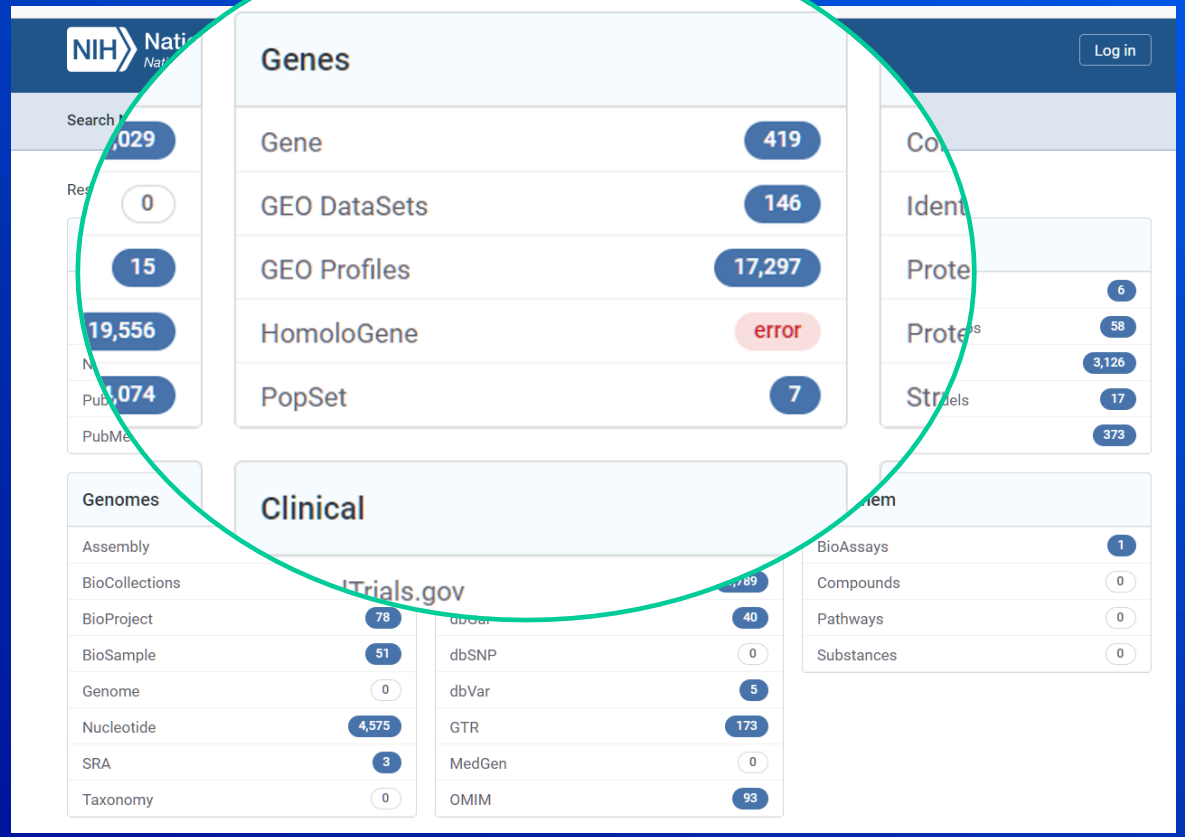


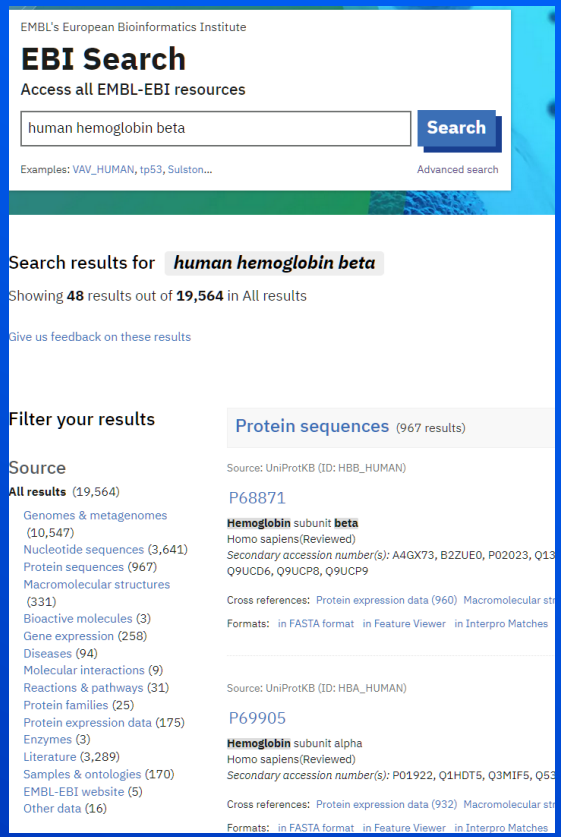
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Search by name

- Gene records and expression studies

- EMBL-EBI



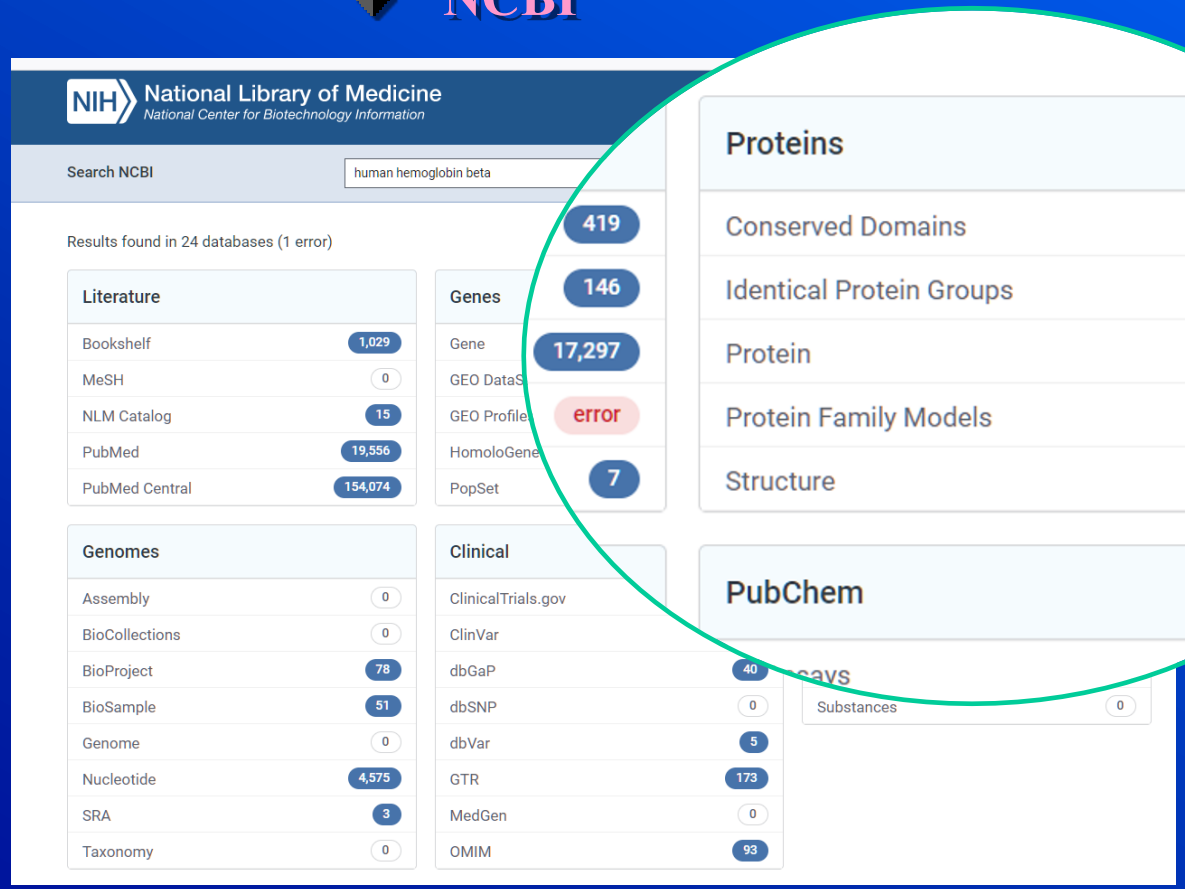


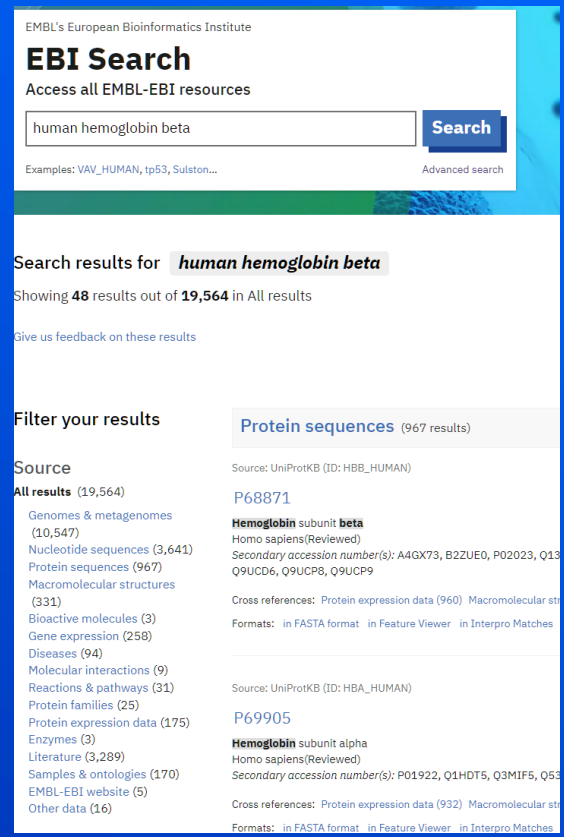
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Search by name

Different questions about a protein

- NCBI
- EMBL-EBI



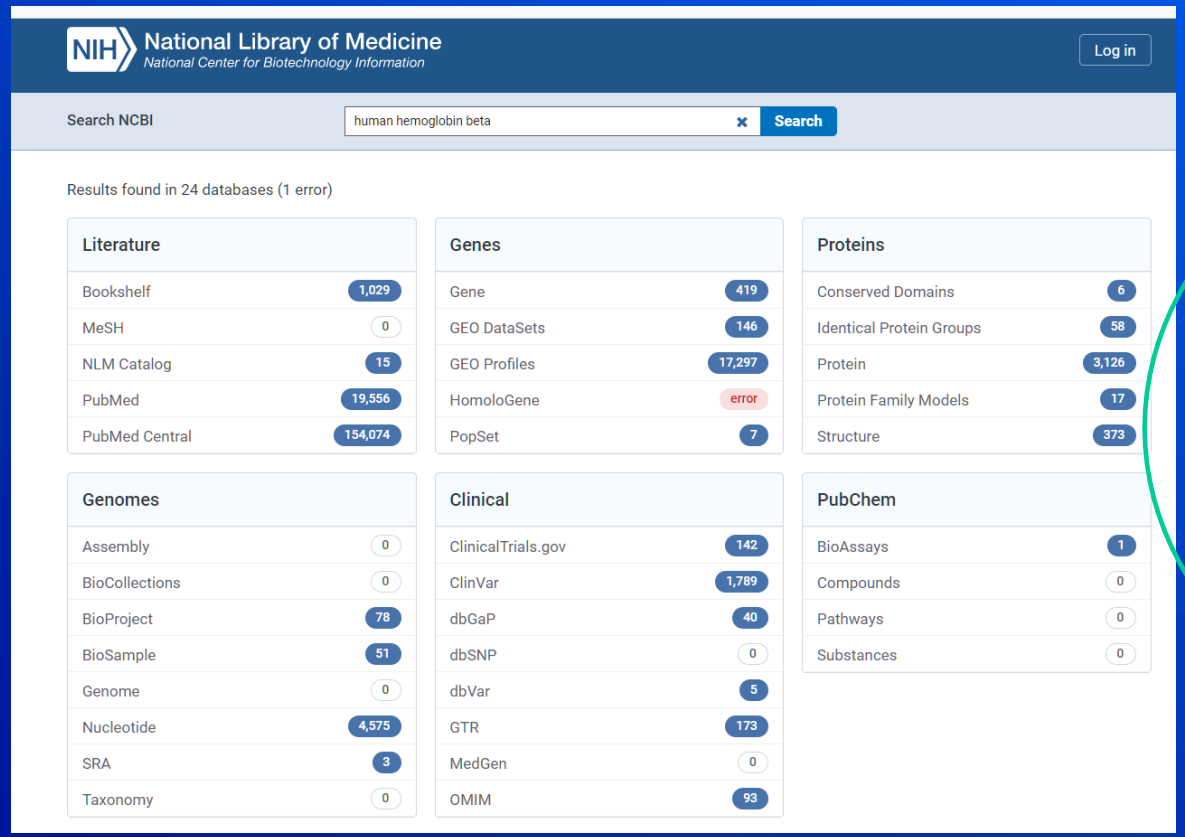


Stockholms universitet

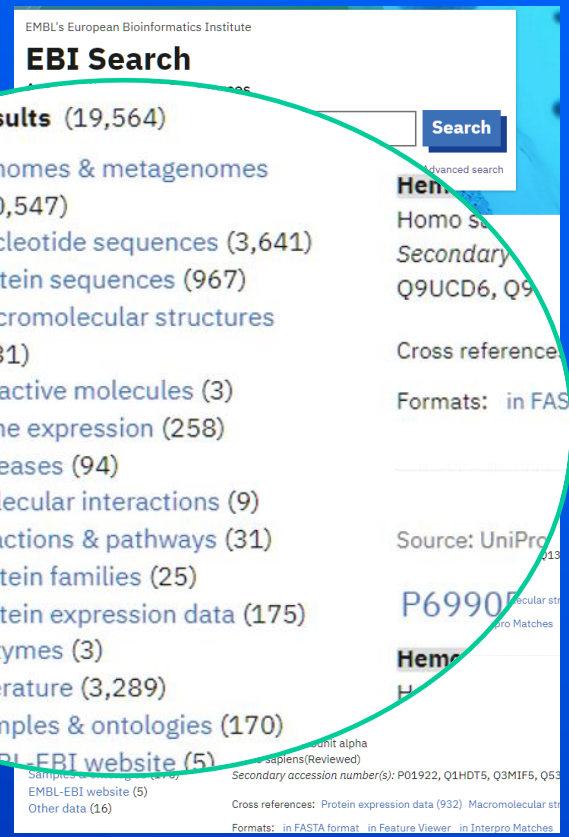
Search by name

Choosing and checking a result

NCBI



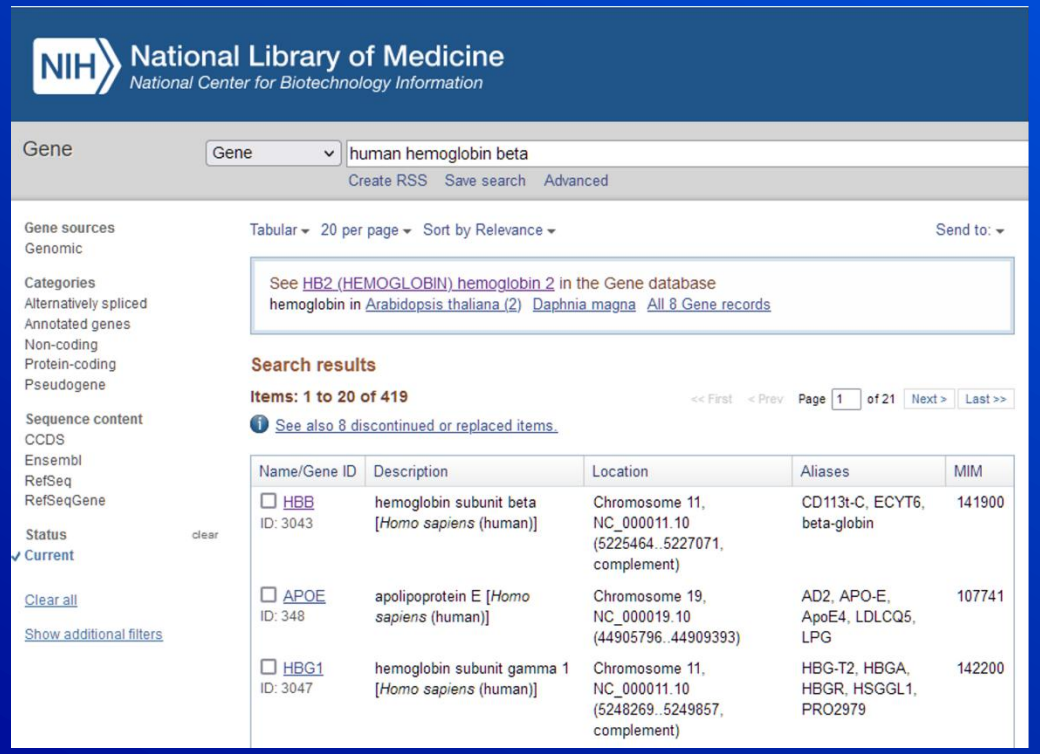
EMBL-EBI

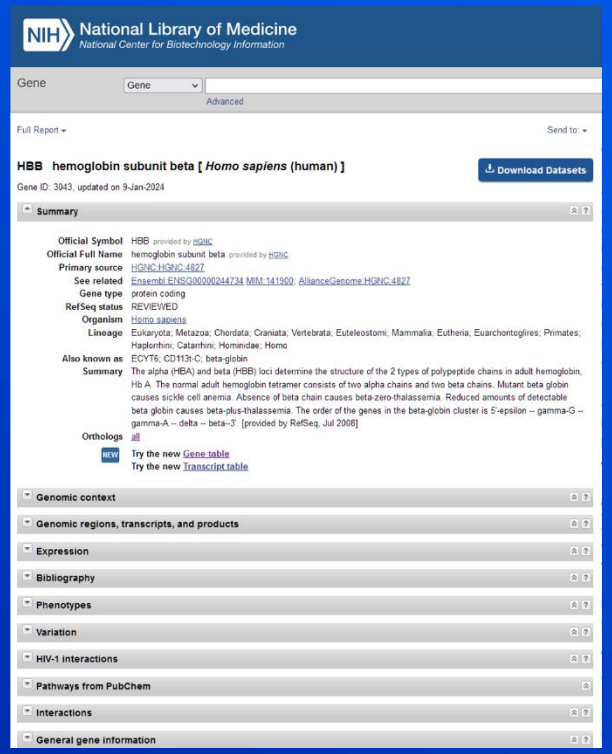


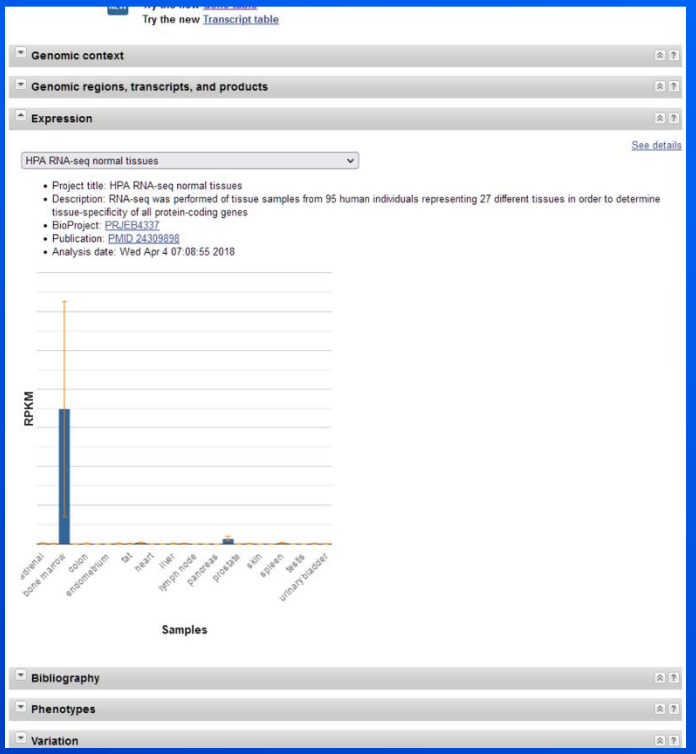
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Search by name

- Search within selected resource
  - NCBI Gene





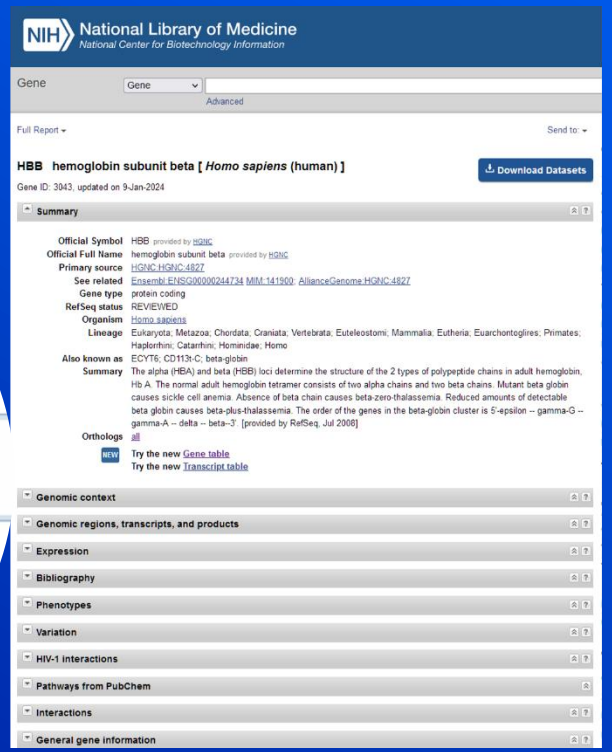


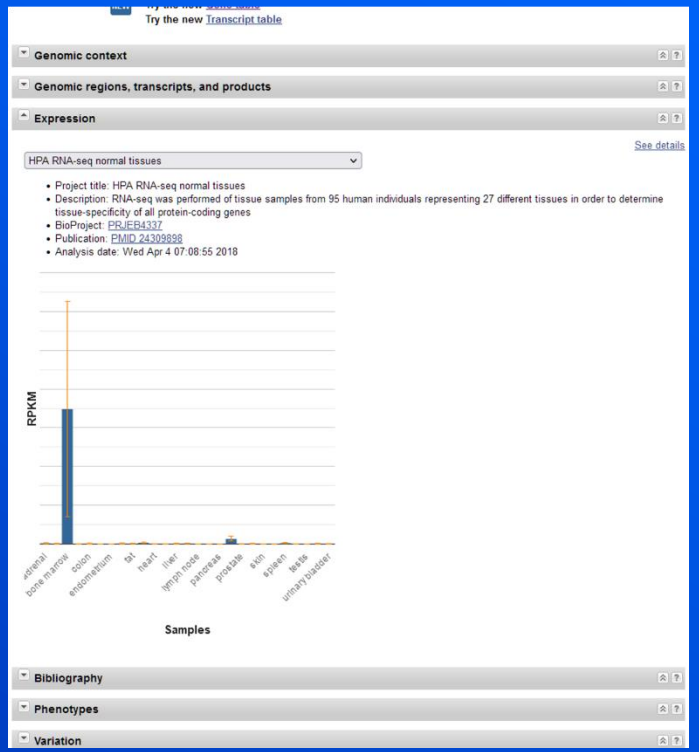
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Search by name

- Check the name and organism
  - NCBI Gene



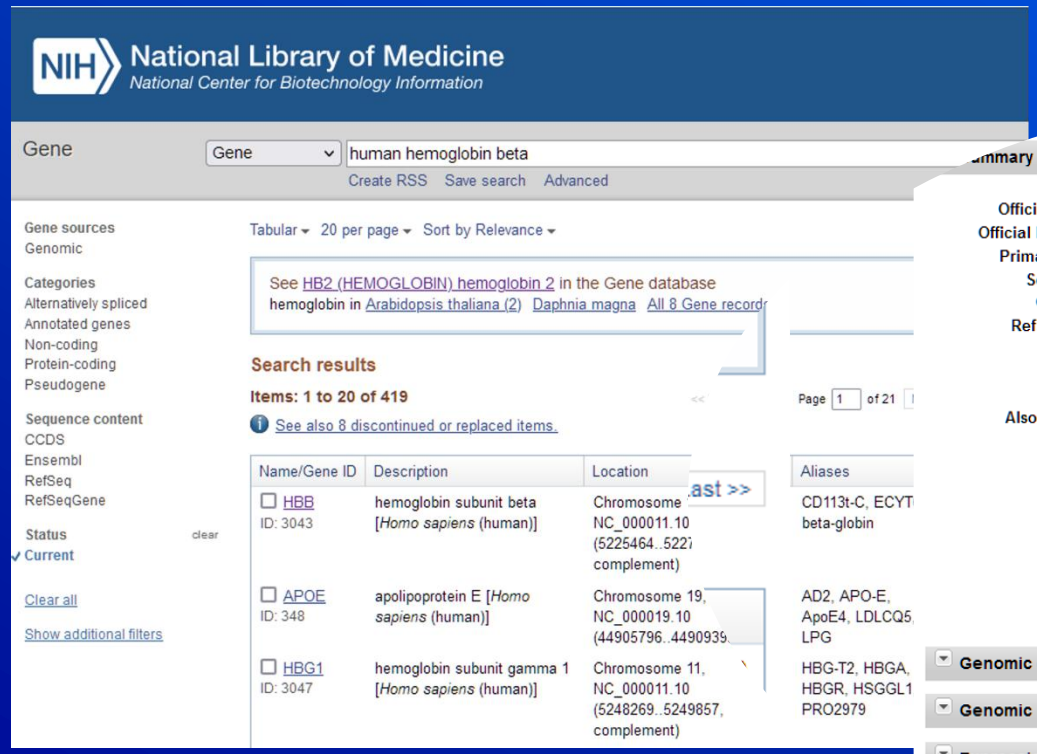




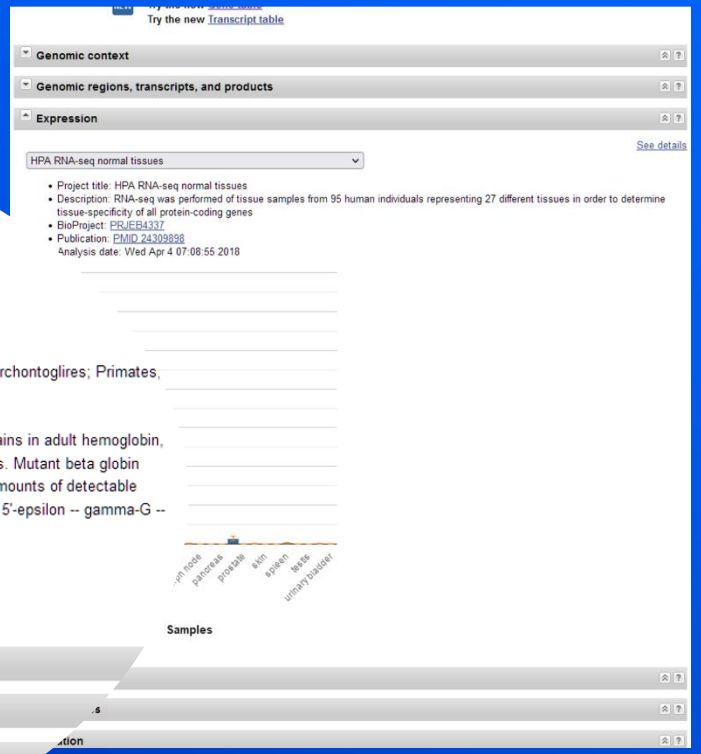
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Search by name

- Explore the gene record
  - NCBI Gene



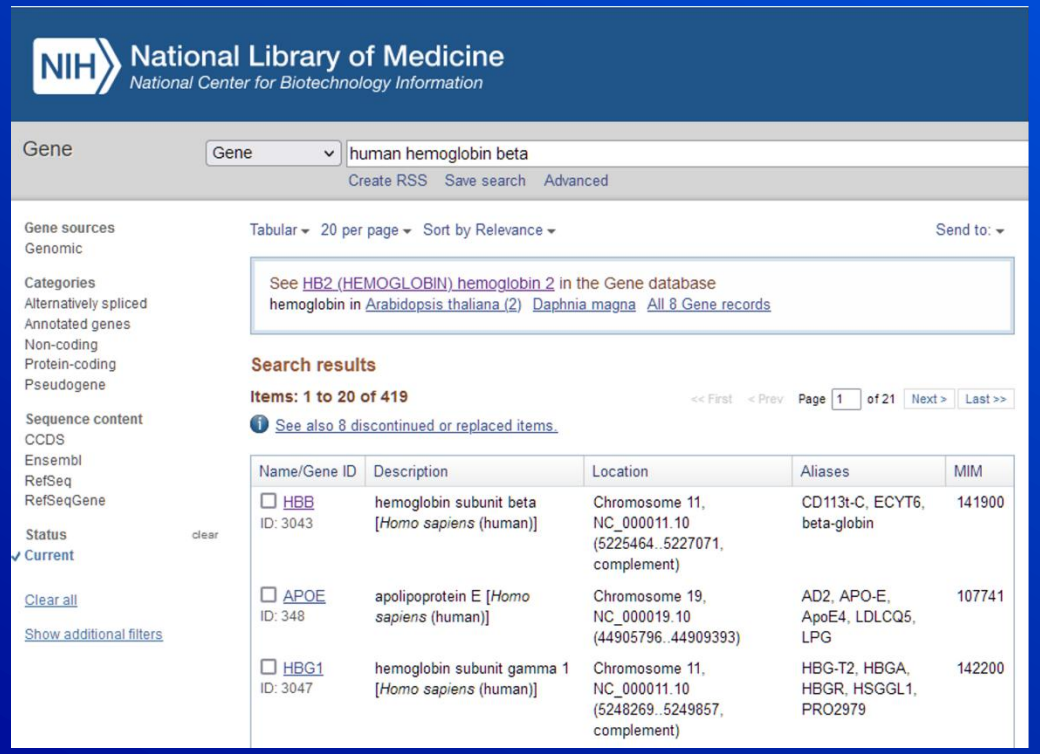


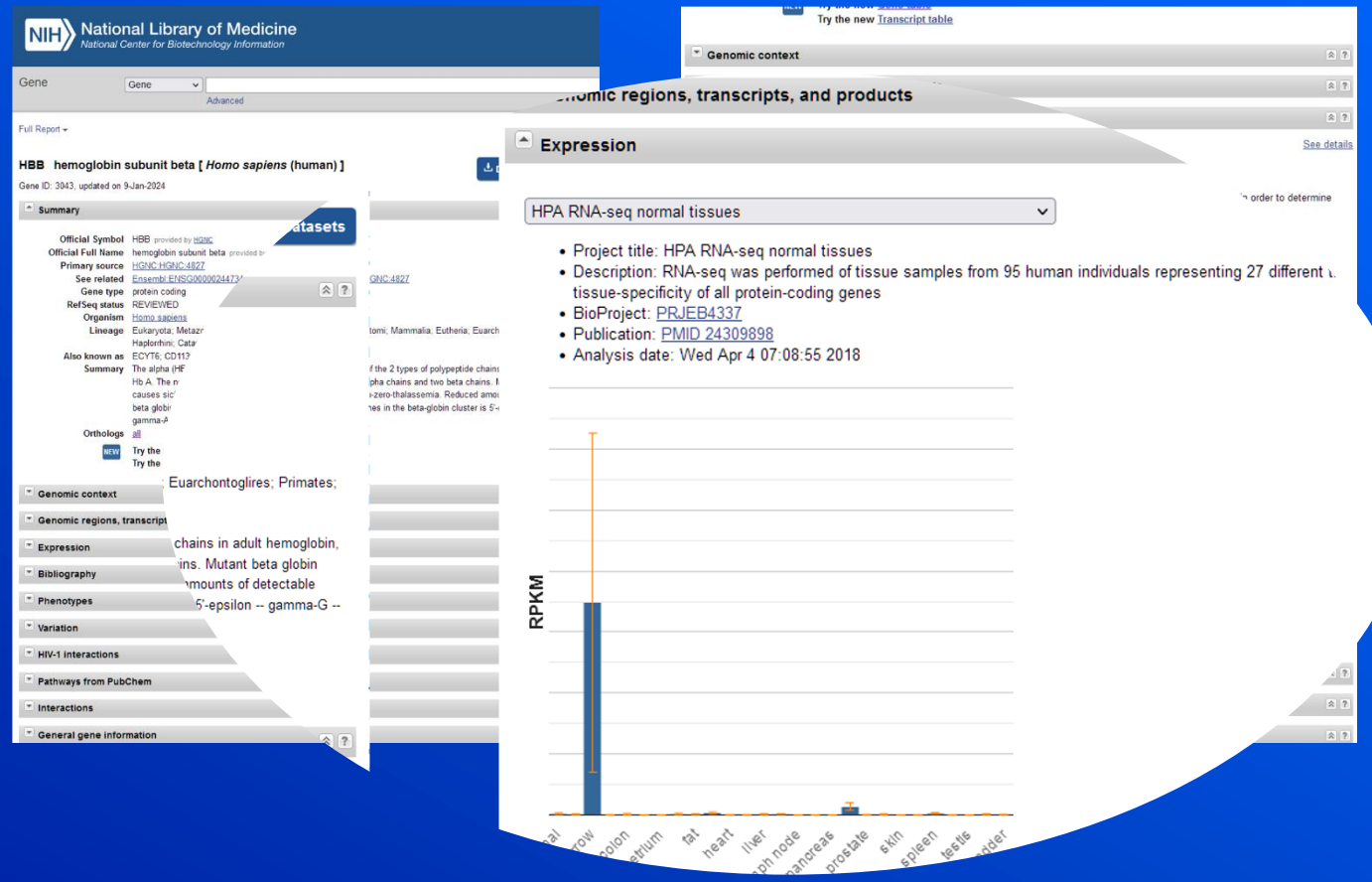


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Search by name

- Trace expression data to the study
  - NCBI Gene

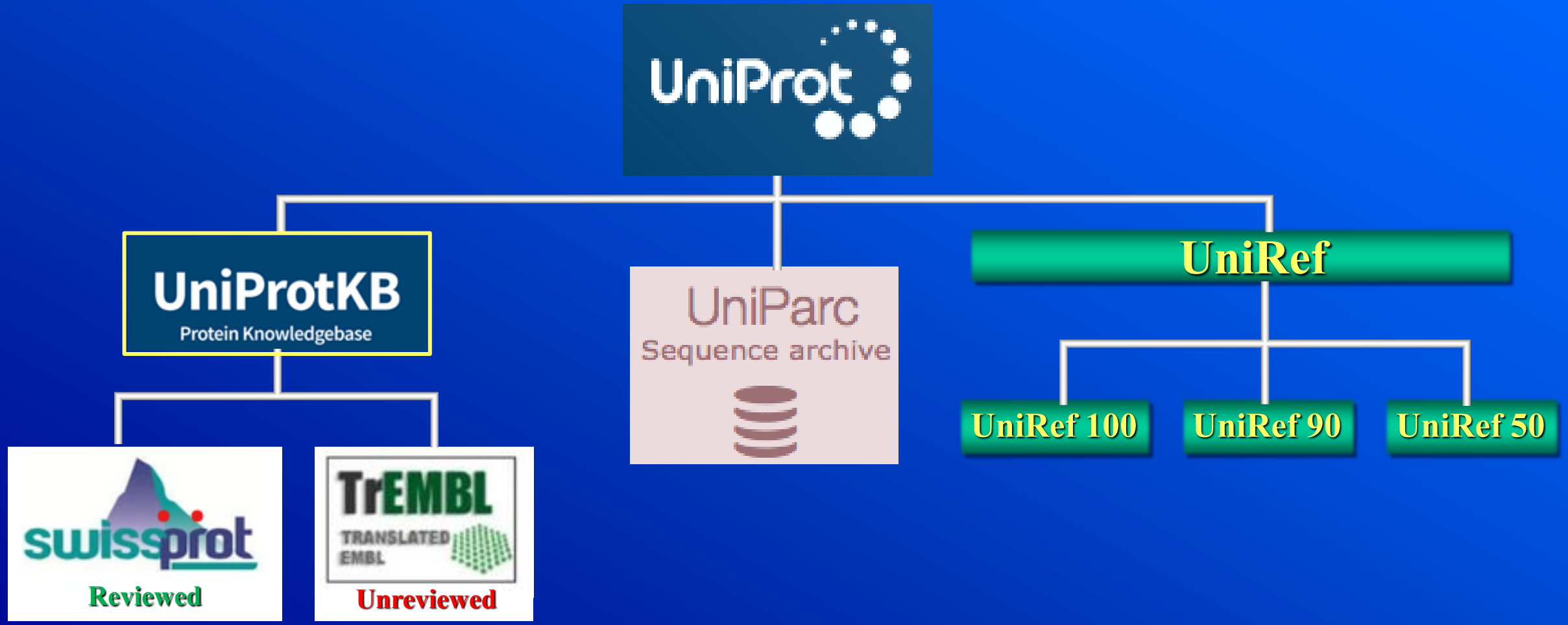




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Universal Protein Resource

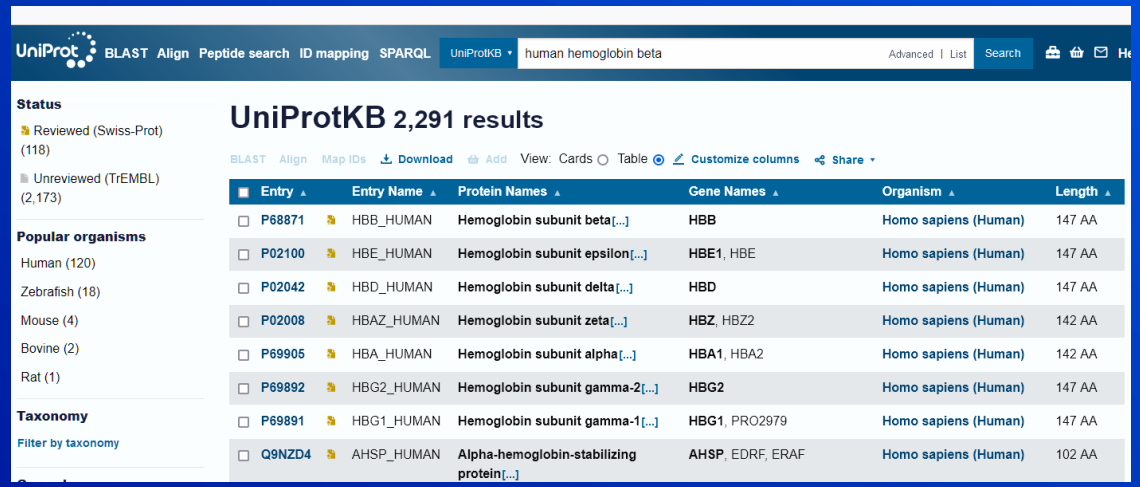
Resource content



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Search by name

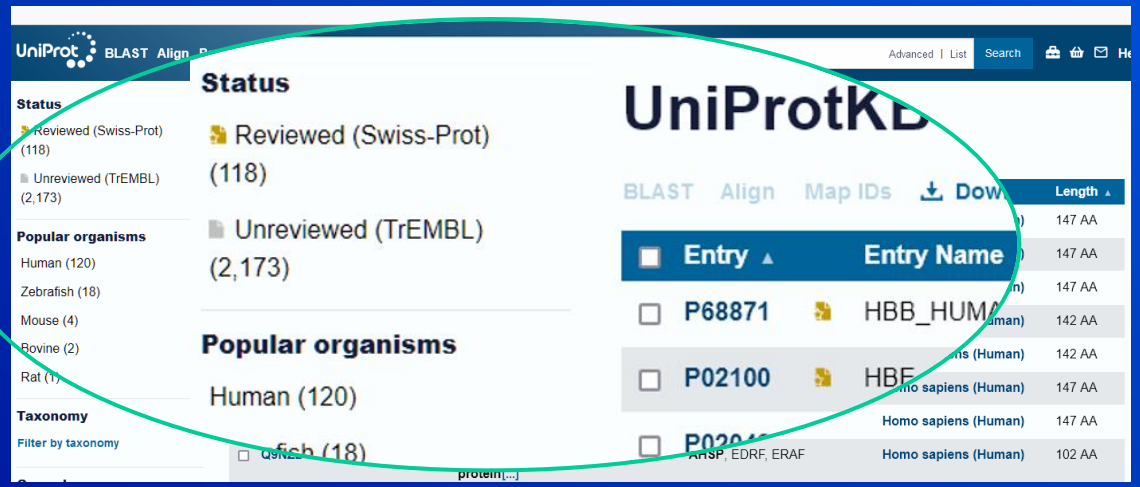
- Search within UniProtKB
  - UniProtKB

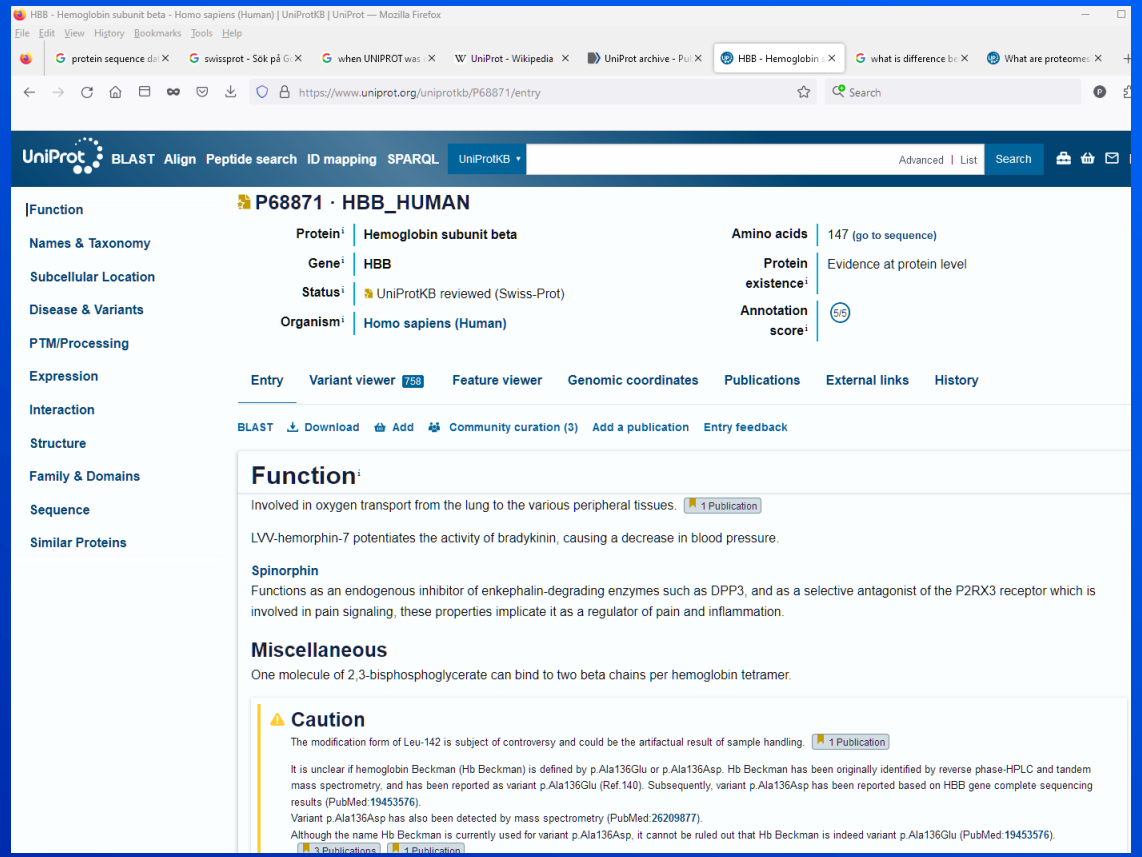


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Search by name

- Filter by organism and review status
  - UniProtKB

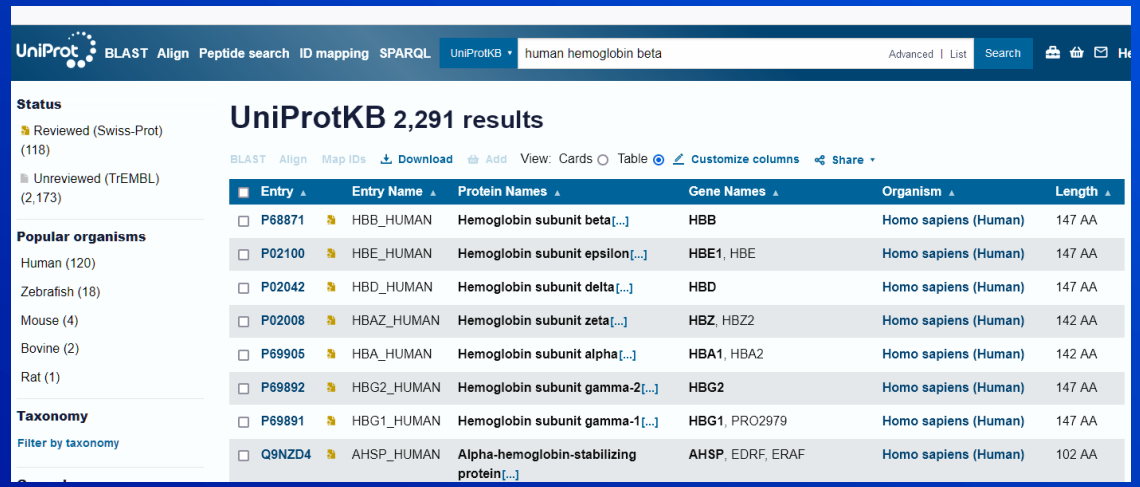


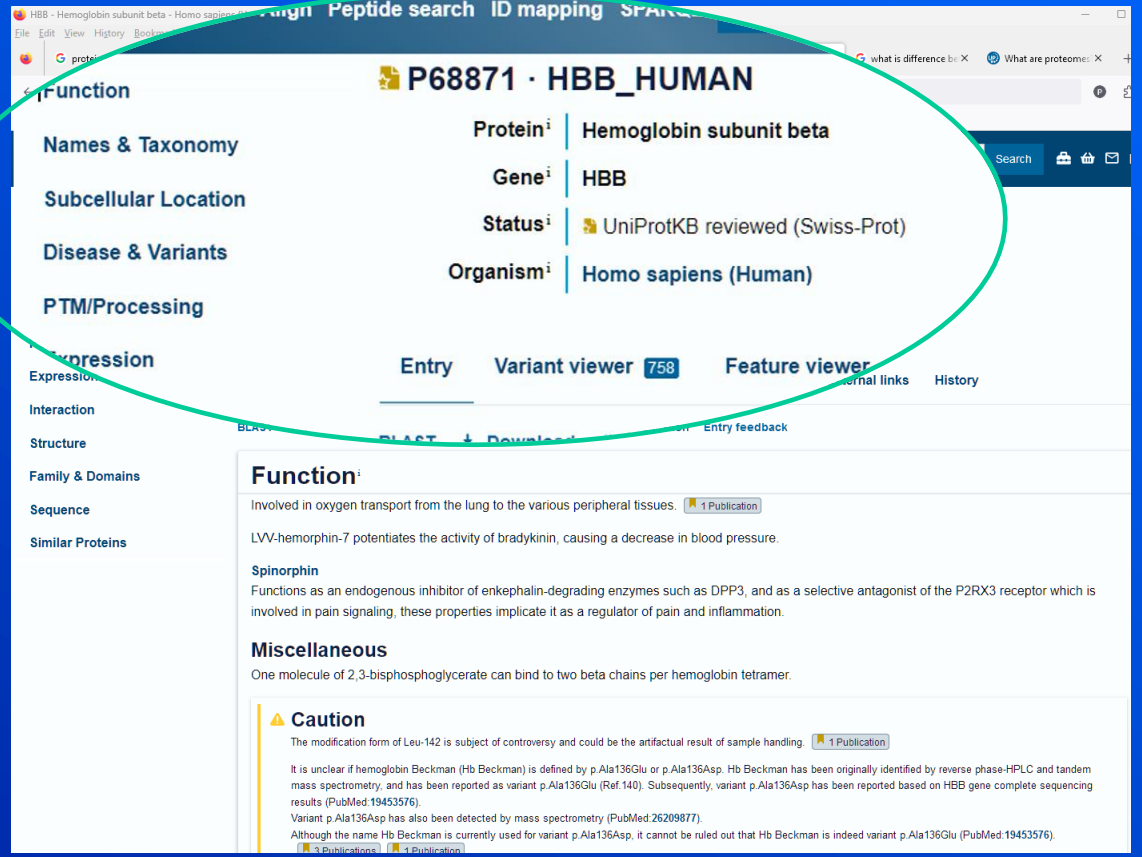


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Search by name

- Check the protein name and organism
  - UniProtKB

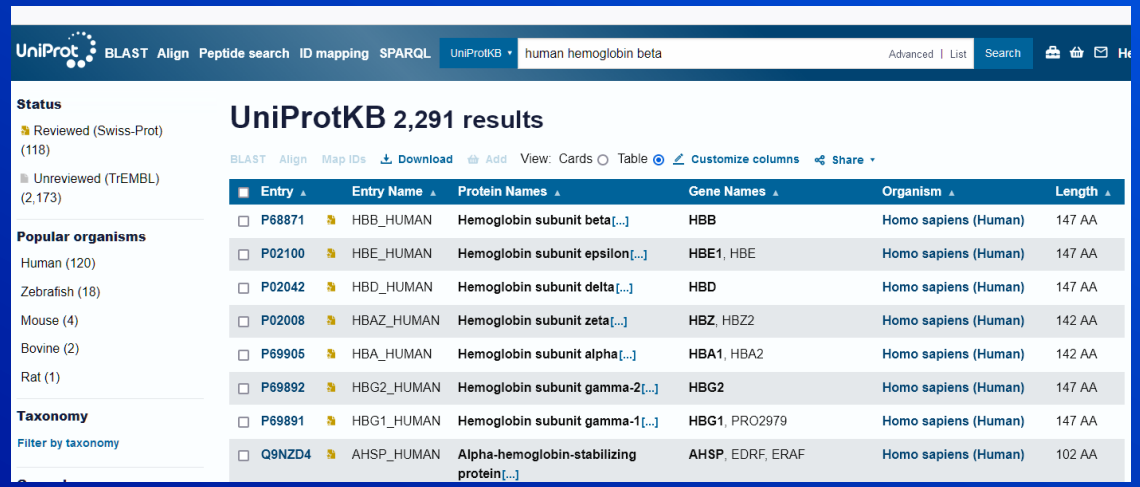


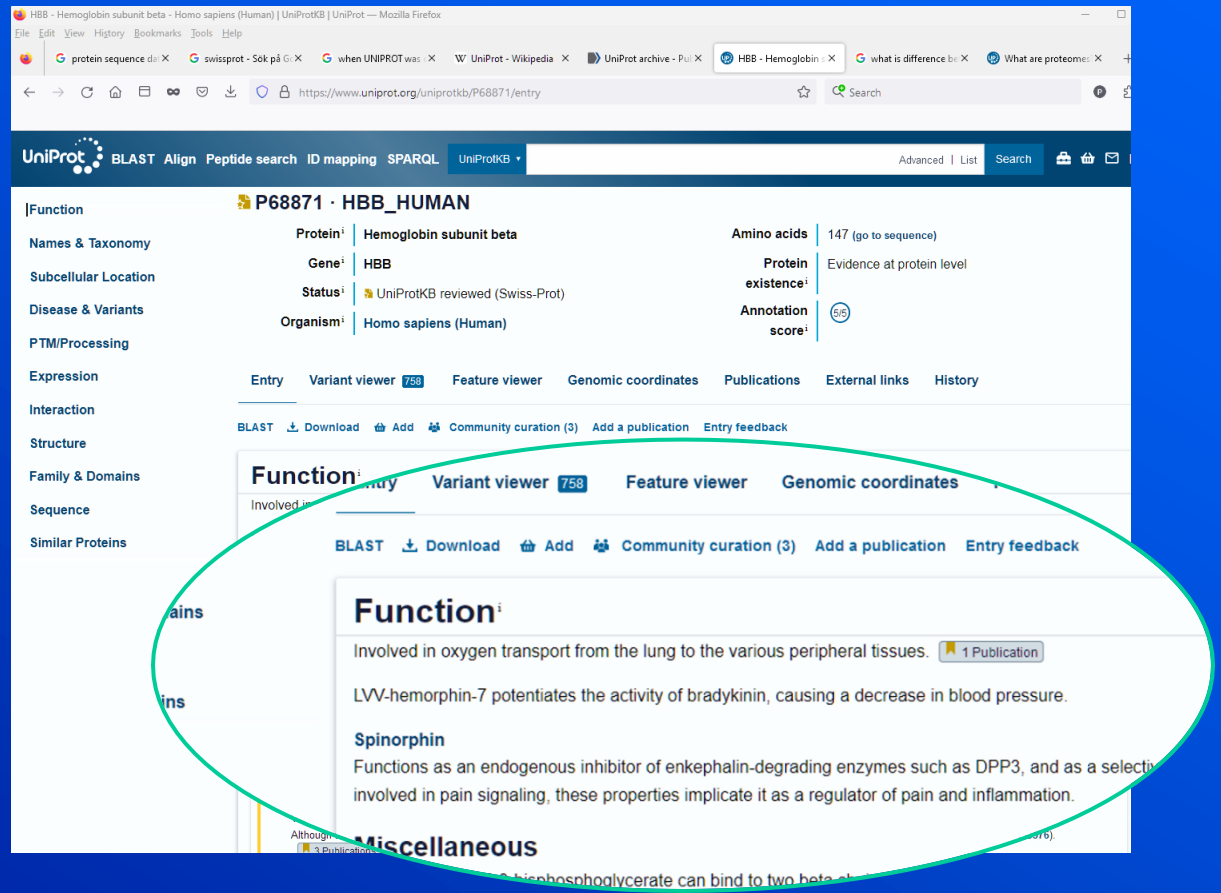


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Search by name

- Check the function and its evidence
- UniProtKB





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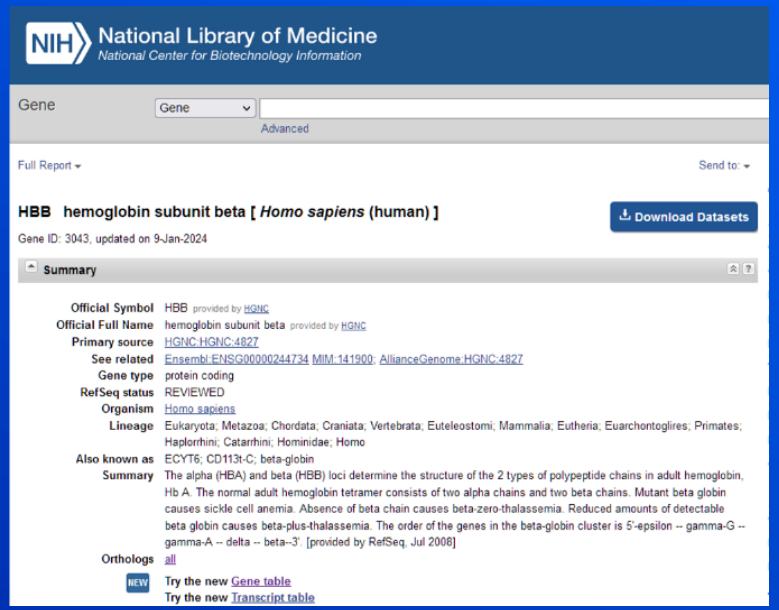
Search by official symbol

One symbol, different record IDs

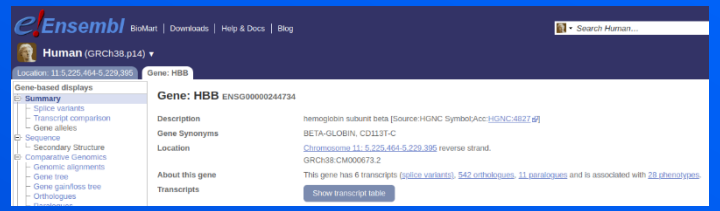
Official gene symbol (HBB)



NCBI



EMBL-EBI



DDBJ

```
LOCUS       V00497                626 bp    mRNA    linear   HUM 07-OCT-2008
DEFINITION  Human messenger RNA for beta-globin.
ACCESSION   V00497
VERSION     V00497.1
KEYWORDS    beta-globin; complementary DNA; globin.
SOURCE      Homo sapiens (human)
  ORGANISM  Homo sapiens
            Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
            Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini;
            Catarrhini; Hominidae; Homo.
REFERENCE   1  (bases 1 to 626)
  AUTHORS   Kafatos F.C., Efstratiadis A., Forget B.G., Weissman S.M.
  TITLE     Molecular evolution of human and rabbit beta-globin mRNAs
  JOURNAL   Proc. Natl. Acad. Sci. U.S.A. 74(12), 5618-5622(1977).
   PUBMED   271989
REFERENCE   2  (bases 51 to 599)
  AUTHORS   Marotta C.A., Wilson J.T., Forget B.G., Weissman S.M.
  TITLE     Human beta-globin messenger RNA. III. Nucleotide sequences derived
            from complementary DNA
  JOURNAL   J Biol Chem 252(14), 5040-5053(1977).
   PUBMED   68958
REFERENCE   3  (bases 480 to 626)
  AUTHORS   Proudfoot N.J.
  TITLE     Complete 3' noncoding region sequences of rabbit and human
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  JOURNAL   Cell 10(4), 559-570(1977).
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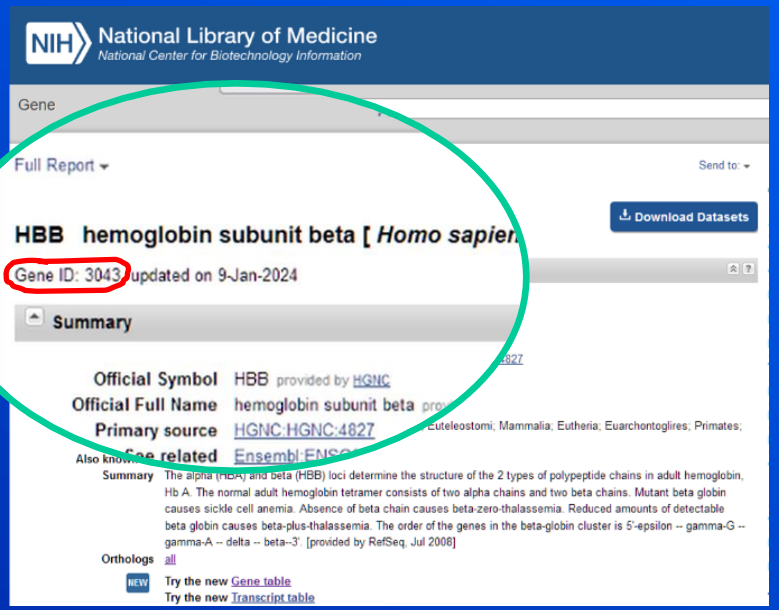
Search by official symbol

One symbol, different record IDs

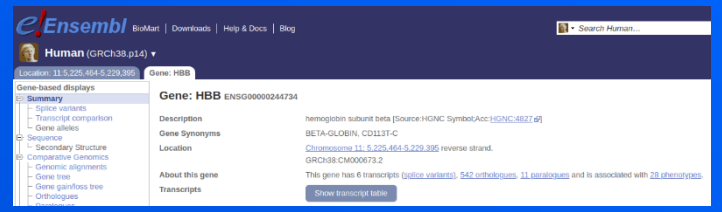
Official gene symbol (HBB)



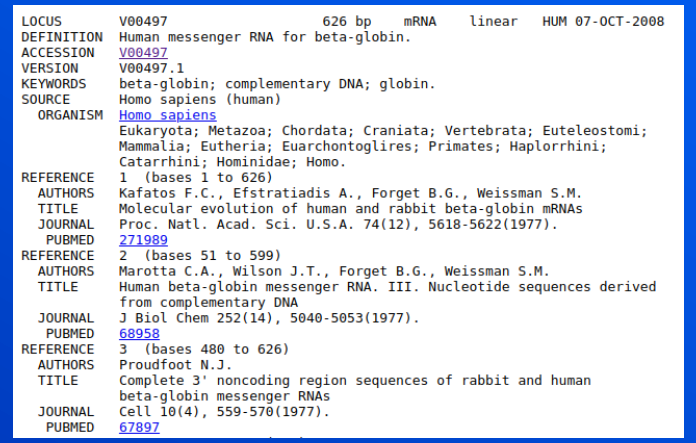
NCBI



EMBL-EBI



DDBJ



```
LOCUS       V00497                626 bp    mRNA    linear   HUM 07-OCT-2008
DEFINITION  Human messenger RNA for beta-globin.
ACCESSION   V00497
VERSION     V00497.1
KEYWORDS    beta-globin; complementary DNA; globin.
SOURCE      Homo sapiens (human)
  ORGANISM  Homo sapiens
            Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
            Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini;
            Catarrhini; Hominidae; Homo.
REFERENCE   1  (bases 1 to 626)
  AUTHORS   Kafatos F.C., Efstratiadis A., Forget B.G., Weissman S.M.
  TITLE     Molecular evolution of human and rabbit beta-globin mRNAs
  JOURNAL   Proc. Natl. Acad. Sci. U.S.A. 74(12), 5618-5622(1977).
   PUBMED   271989
REFERENCE   2  (bases 51 to 599)
  AUTHORS   Marotta C.A., Wilson J.T., Forget B.G., Weissman S.M.
  TITLE     Human beta-globin messenger RNA. III. Nucleotide sequences derived
            from complementary DNA
  JOURNAL   J Biol Chem 252(14), 5040-5053(1977).
   PUBMED   68958
REFERENCE   3  (bases 480 to 626)
  AUTHORS   Proudfoot N.J.
  TITLE     Complete 3' noncoding region sequences of rabbit and human
            beta-globin messenger RNAs
  JOURNAL   Cell 10(4), 559-570(1977).
   PUBMED   67897
```

Stockholms universitet

Search by official symbol

One symbol, different record IDs

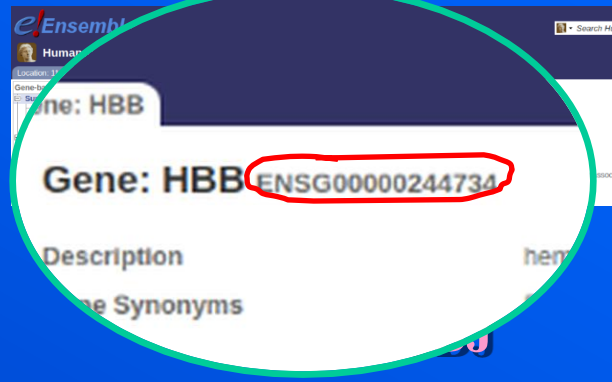
Official gene symbol (HBB)



NCBI



EMBL-EBI




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   PUBMED   271989
REFERENCE   2  (bases 51 to 599)
  AUTHORS   Marotta C.A., Wilson J.T., Forget B.G., Weissman S.M.
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  JOURNAL   Cell 10(4), 559-570(1977).
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```

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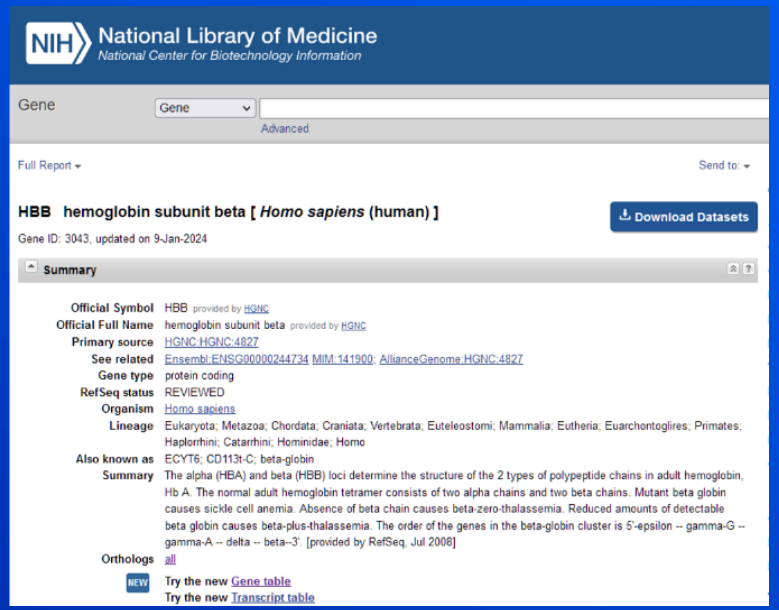
Search by official symbol

One symbol, different record IDs

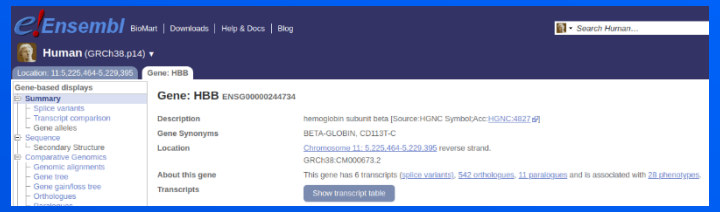
Official gene symbol (HBB)



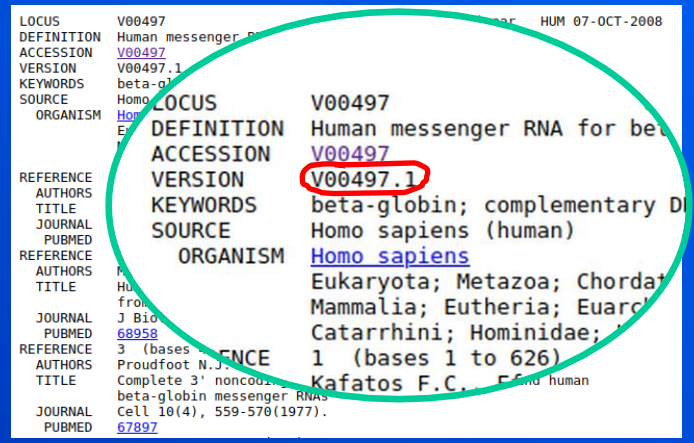
NCBI



EMBL-EBI



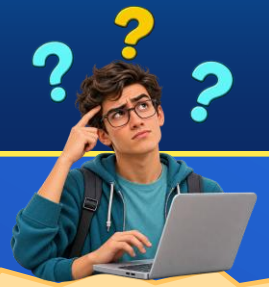
DDBJ



From a symbol to a record

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Approved human gene symbol
HBB



HGNC
Nomenclature
HGNC:4827

NCBI
Gene record
3043

Ensembl
Gene record
ENSG00000244734

NCBI
Genomic sequence
NG_059281.1

NCBI
ClinVar variant
VCV000015495.3

NCBI
Reference mRNA
NM_000518.5

DDBJ
Submitted mRNA
V00497.1

Ensembl
Ensembl transcript
ENST00000335295.4

Ensembl
Ensembl protein
ENSP00000333994.3

NCBI PROTEIN
RefSeq protein
NP_000509.1

UniProtKB
Protein function
P68871

PDB
Haemoglobin complex
2HHB

InterPro
Globin family
IPR000971

Pfam
Globin domain
PF00042

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From a symbol to a record

What kind of sequence?

Which version did we use?

RefSeq mRNA

NM_000518.5

RefSeq protein

NP_000509.1

Save the full accession and version to retrieve the same sequence.

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Clues in the code

Prefixes and suffixes carry clues

Resource	Pattern or example	What the pattern tells us
RefSeq	NM_000518.5 NP_000509.1	NM_ mRNA, NP_ protein .5 sequence version 5
Ensembl (human)	ENSG… / ENST… / ENSP… ENST00000335295.4	Gene / transcript / protein .4 version of this transcript
INSDC read runs	SRR… / ERR… / DRR… Example SRR390728	Sequencing run Prefix shows the originating archive
GEO	GSE… / GSM… GSE10072 / GSM254625	GSE study series GSM sample
UniProtKB isoforms	P04150-2	-2 identifies a protein isoform
UniProtKB features	VAR_002863 PRO_0000052976	VAR_ natural sequence variant PRO_ processed chain or peptide
Rfam / Pfam	RF00005 / PF00042	RF RNA family PF protein family or domain



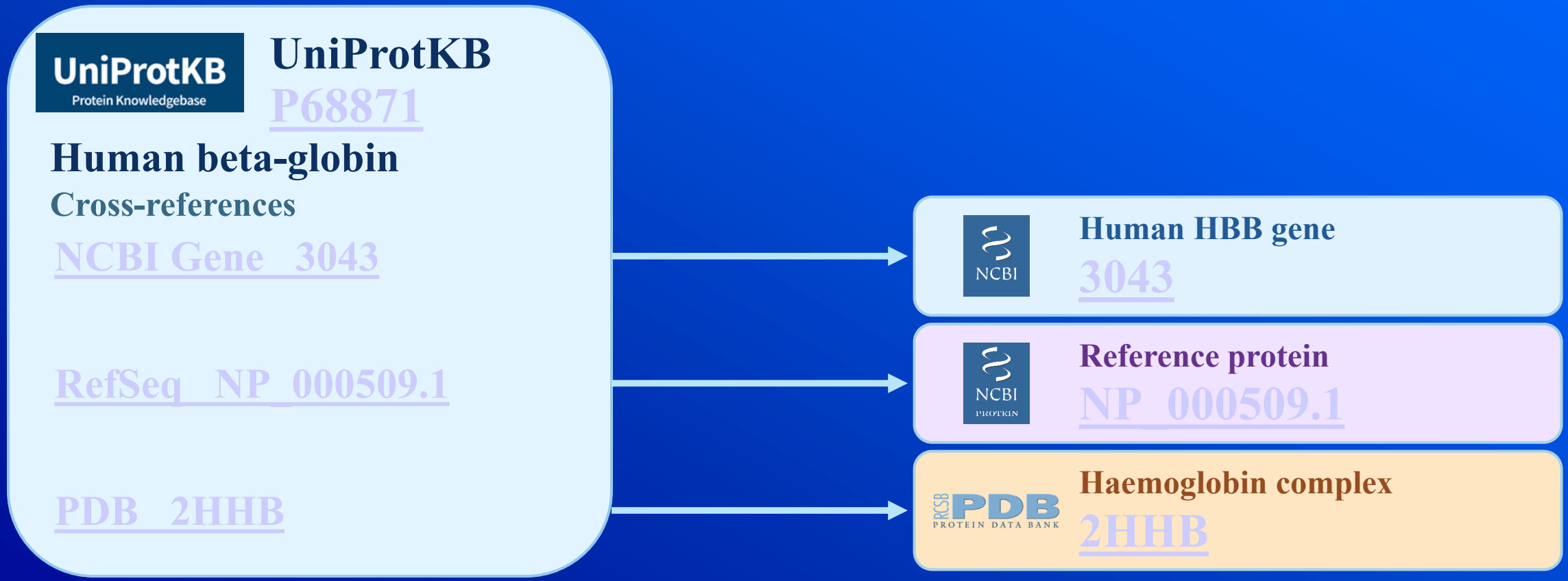
Check the resource's rules

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Links across resources

Linked identifiers are cross-references

They lead to relevant information in another resource.



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Routes to database info

Database record pages



Inspect records and check their evidence.

Programmatic access



Automate queries using code or command-line tools.

Genome browsers

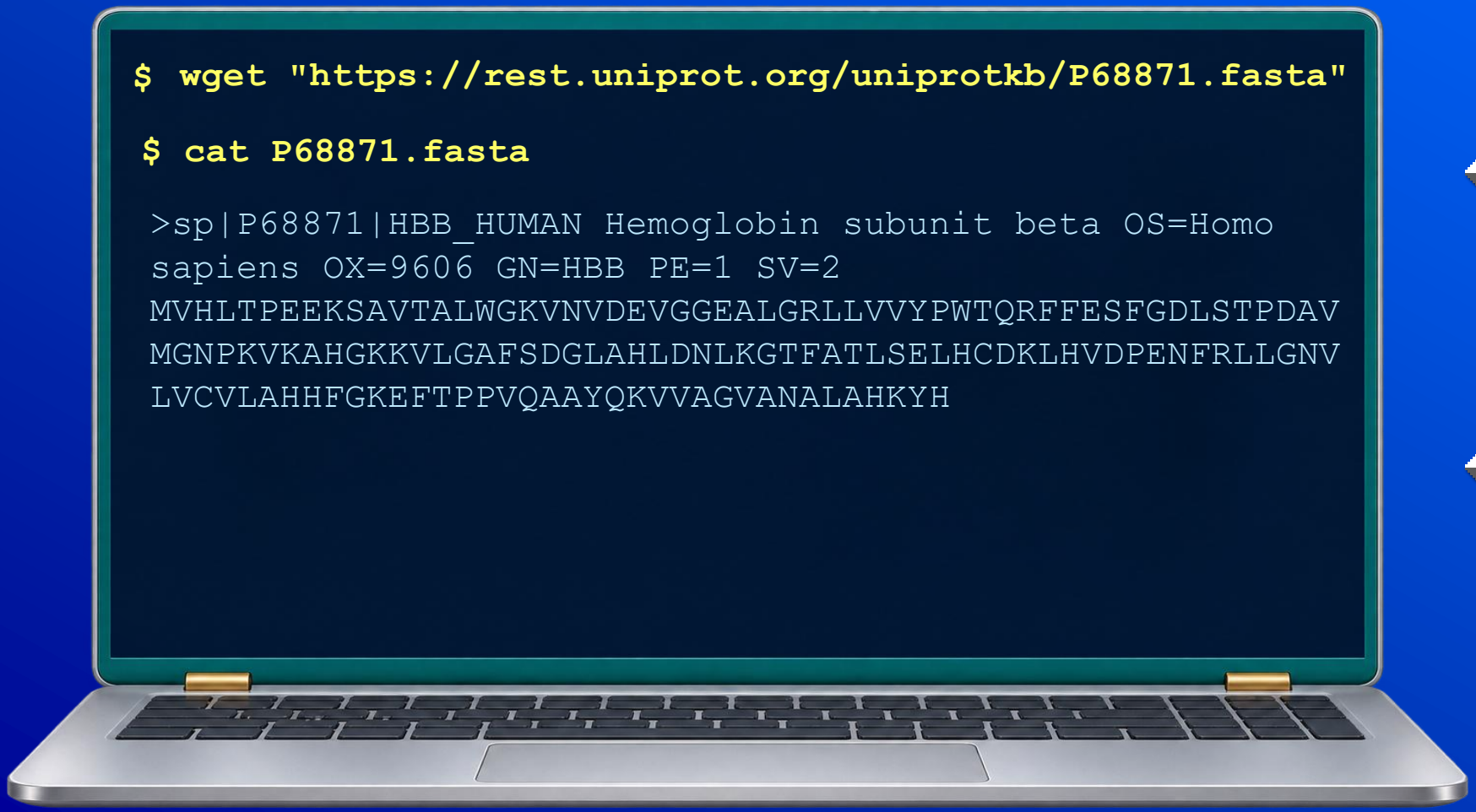


Explore DNA, transcripts and proteins through one web interface

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Programmatic access

One command, one file



- File may contain many records
- File processing needs additional tools or code

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Programmatic access

Download first, process next

Python script

```
import requests
url = ("https://rest.uniprot.org/uniprotkb/"
       "P68871.json")

r = requests.get(url, timeout=30)
r.raise_for_status()

record = r.json()

print(record["primaryAccession"],
      record["organism"]["scientificName"],
      record["sequence"]["length"])
```



Output

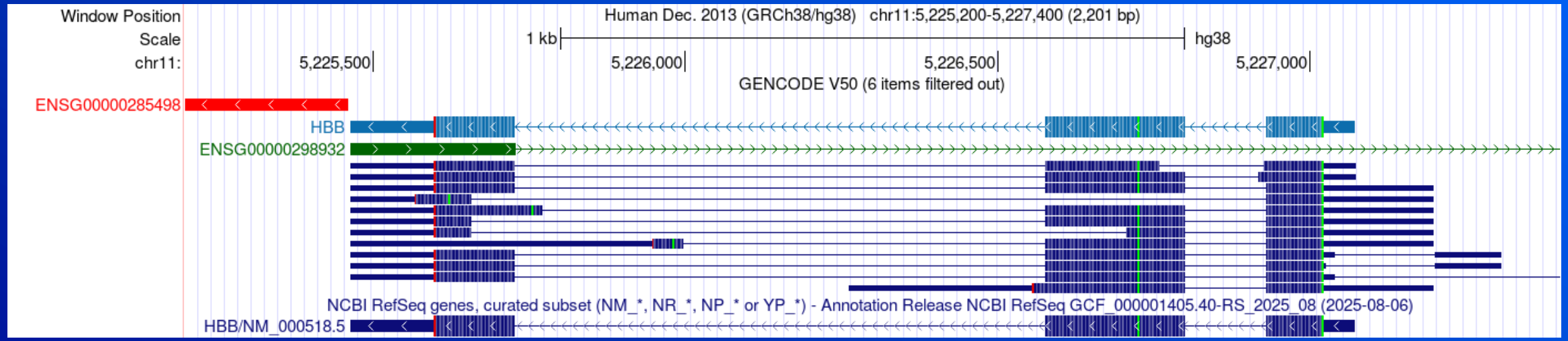
```
P68871 Homo sapiens 147
```

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Genome browsers

Everything in one interface

HBB in UCSC Genome Browser: https://genome.ucsc.edu



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Genome browsers

Reference assembly and displayed region

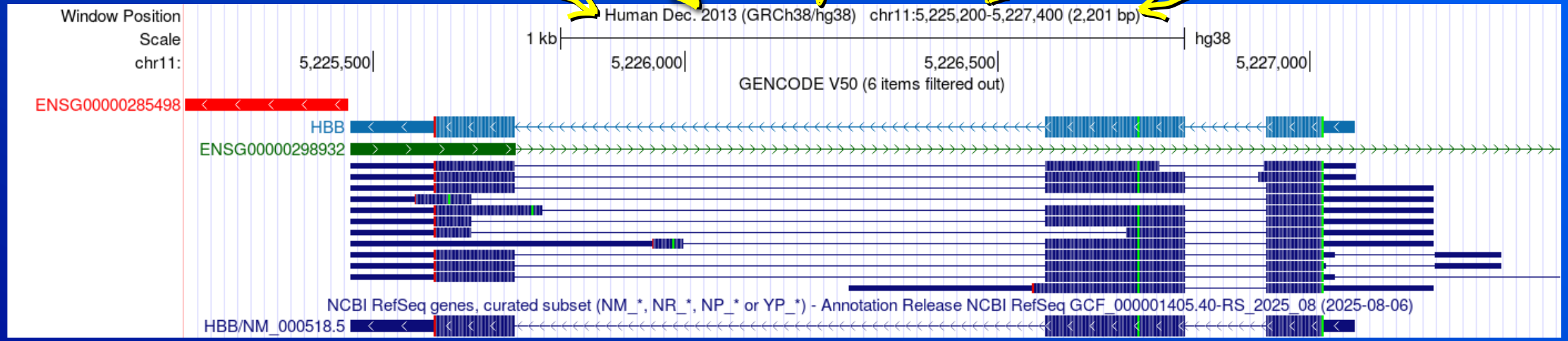
Assembly organism

Assembly release date

Assembly name

Chromosome interval

Window length



HBB in UCSC Genome Browser: https://genome.ucsc.edu

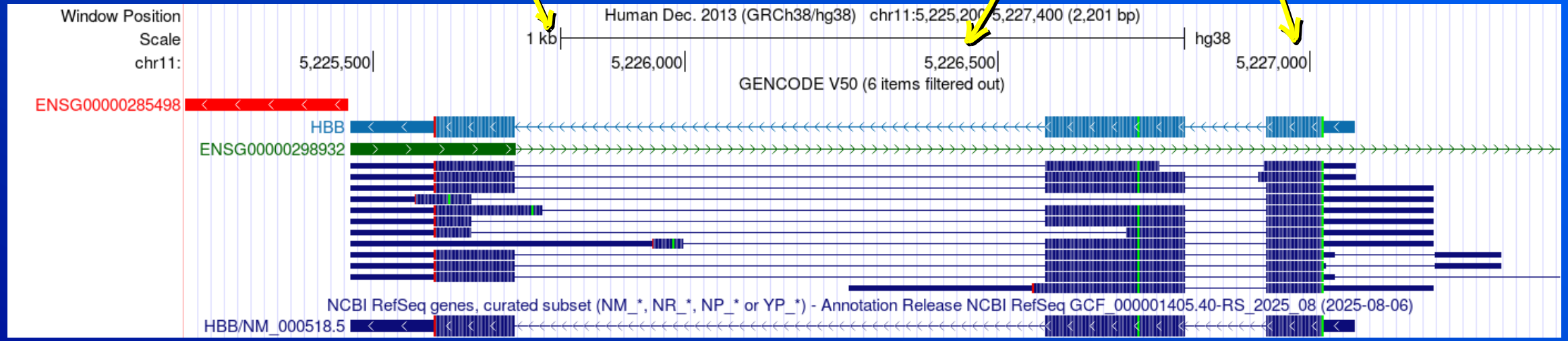
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Genome browsers

Distances along chromosome 11

Scale bar (1,000 bases)

Positions along the chromosome 11



HBB in UCSC Genome Browser: https://genome.ucsc.edu

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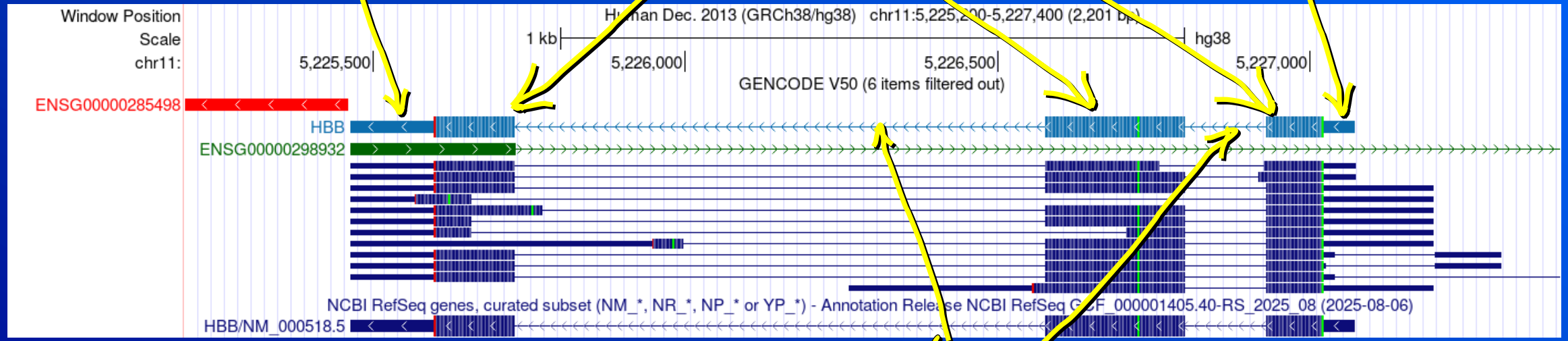
Genome browsers

Exons and Introns

Non-coding exon part (UTR)

Coding exon parts

UTR



Introns

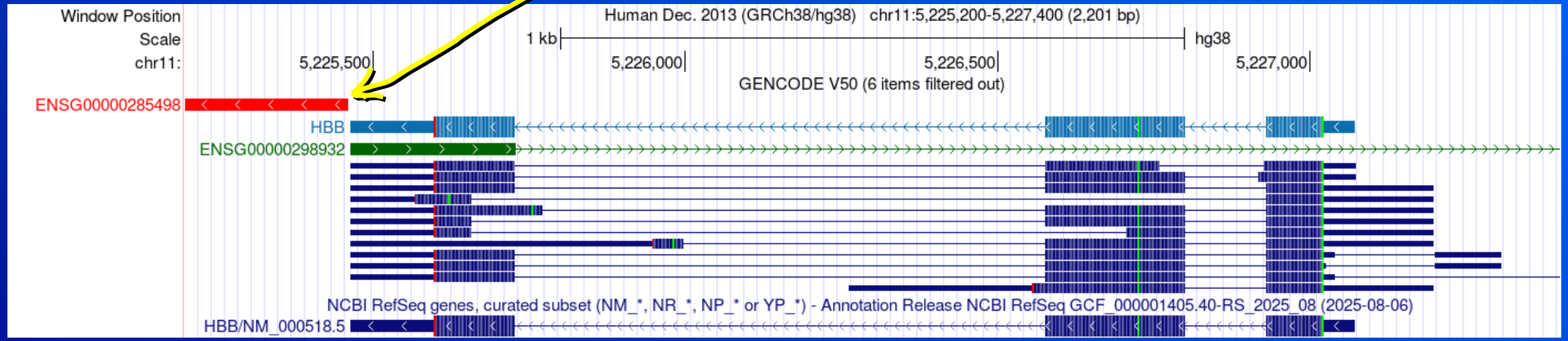
HBB in UCSC Genome Browser: https://genome.ucsc.edu

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Genome browsers

- Some annotations still need confirmation

Possible transcript



HBB in UCSC Genome Browser: https://genome.ucsc.edu

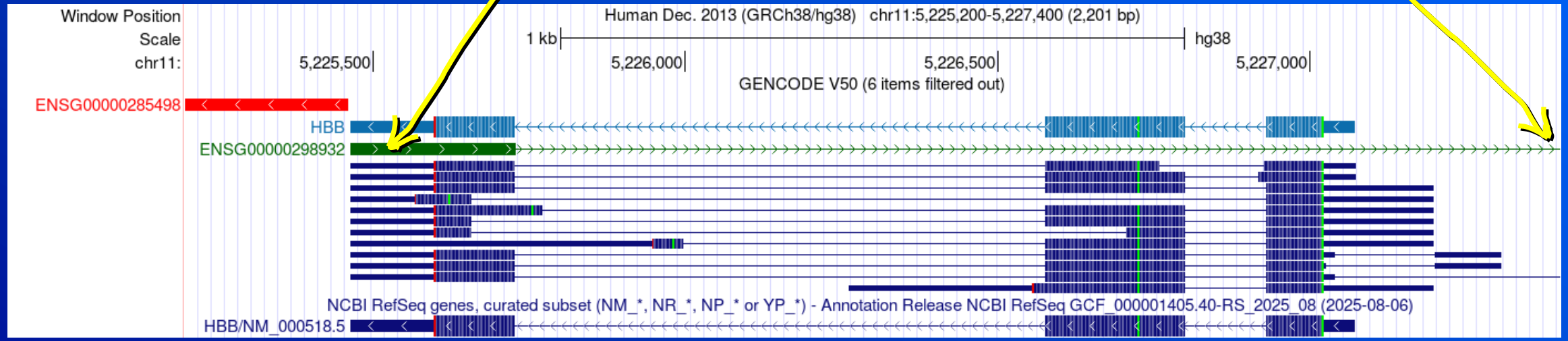
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Genome browsers

A neighbouring long non-coding RNA

Next exon outside this window

Non-coding RNA exon on opposite DNA strand



HBB in UCSC Genome Browser: https://genome.ucsc.edu

Stockholms universitet

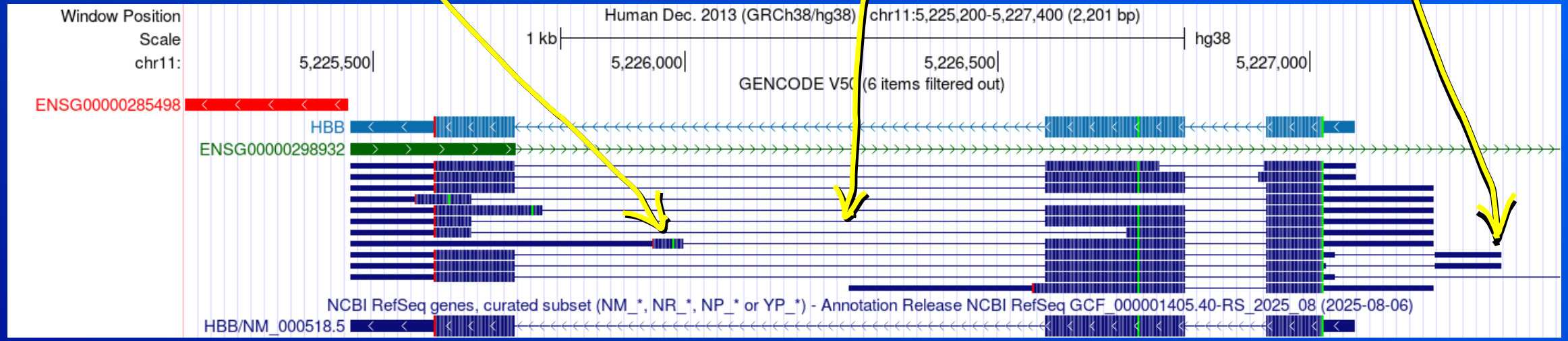
Genome browsers

One gene, several transcripts

Different exon boundaries

Alternative transcripts

Different transcript ends



HBB in UCSC Genome Browser: https://genome.ucsc.edu

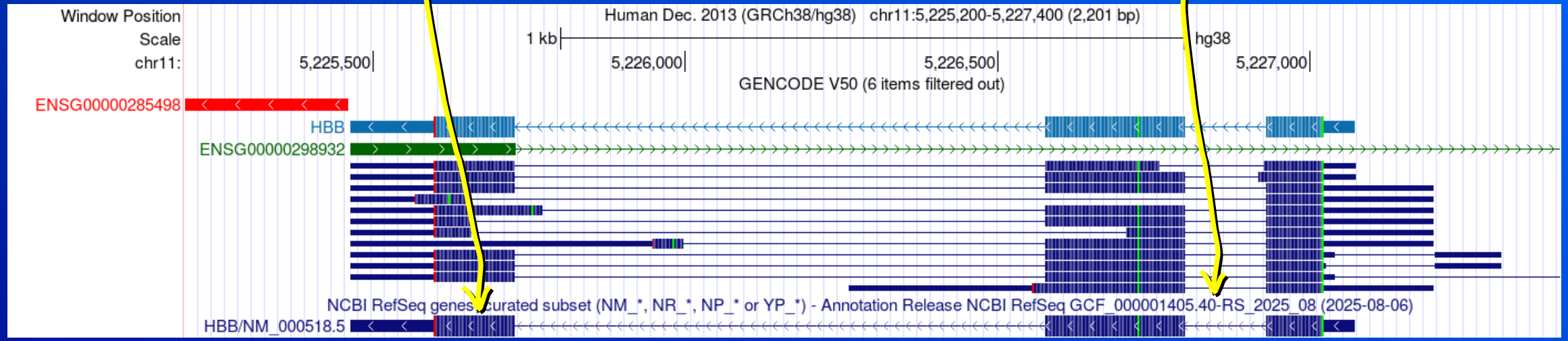
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Genome browsers

One curated HBB reference transcript

Curated RefSeq annotation

Annotation release identification



HBB in UCSC Genome Browser: https://genome.ucsc.edu

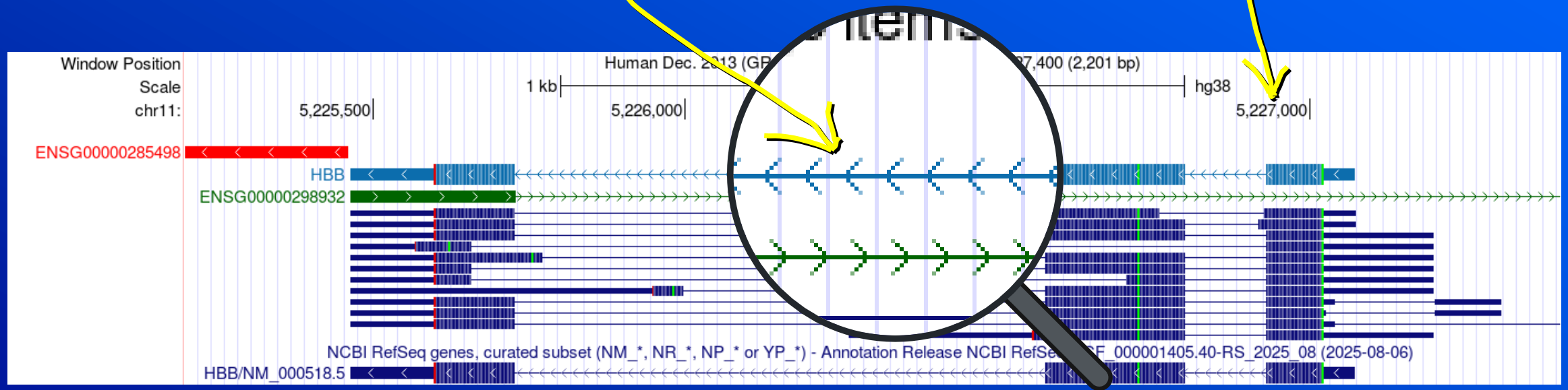
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Genome browsers

Transcription direction

HBB transcription runs to the left

Coordinates increase to the right



HBB in UCSC Genome Browser: https://genome.ucsc.edu

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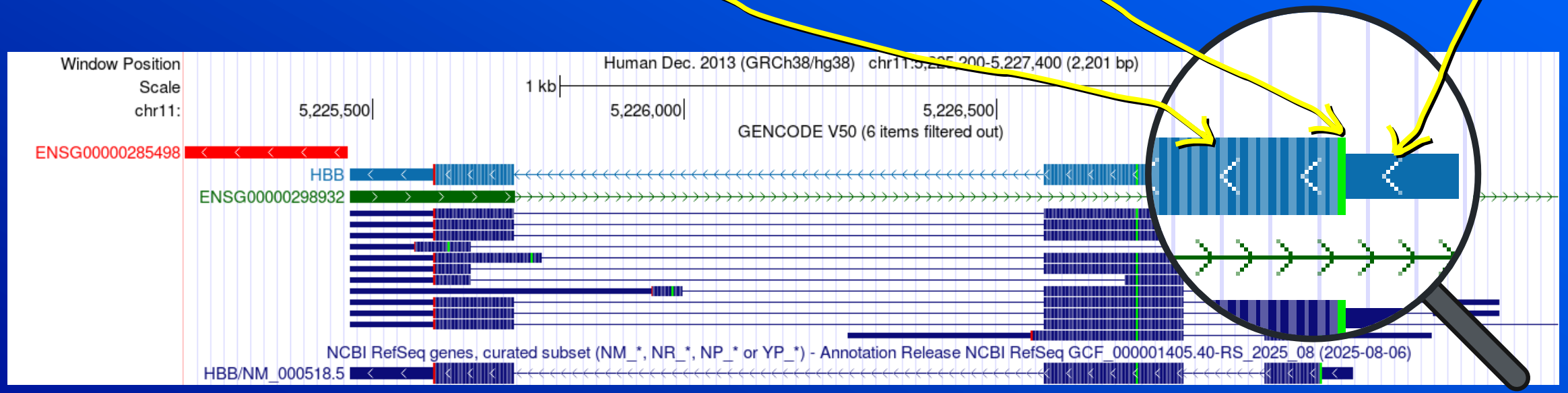
Genome browsers

Exon structure and codon colours

Blue shades: successive codons

Green: start codon (methionine)

Thin part: untranslated region



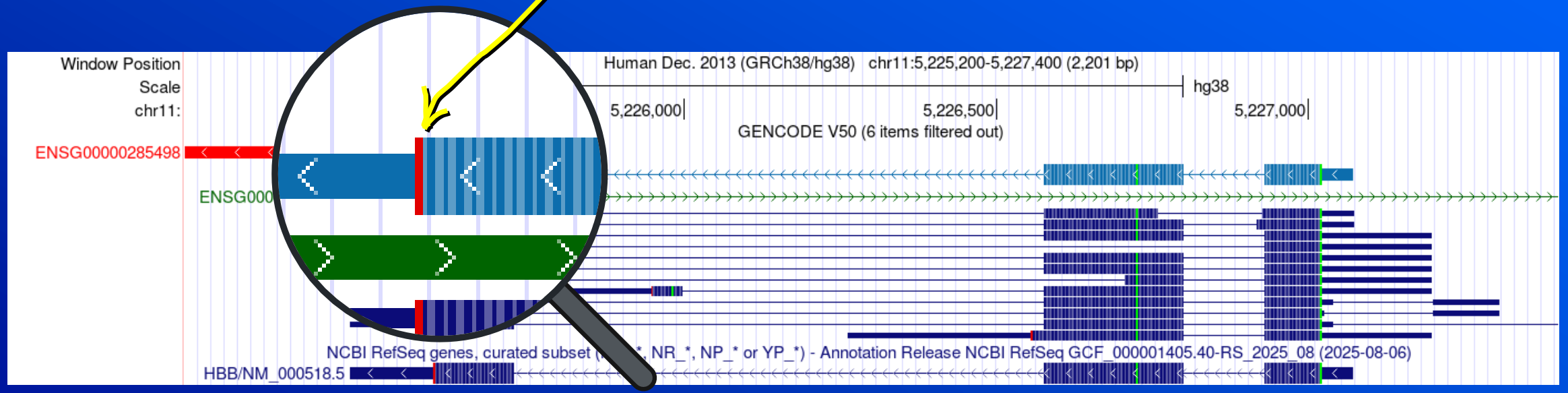
HBB in UCSC Genome Browser: https://genome.ucsc.edu

Stockholms universitet

Genome browsers

Exon structure and codon colours

Red: stop codon



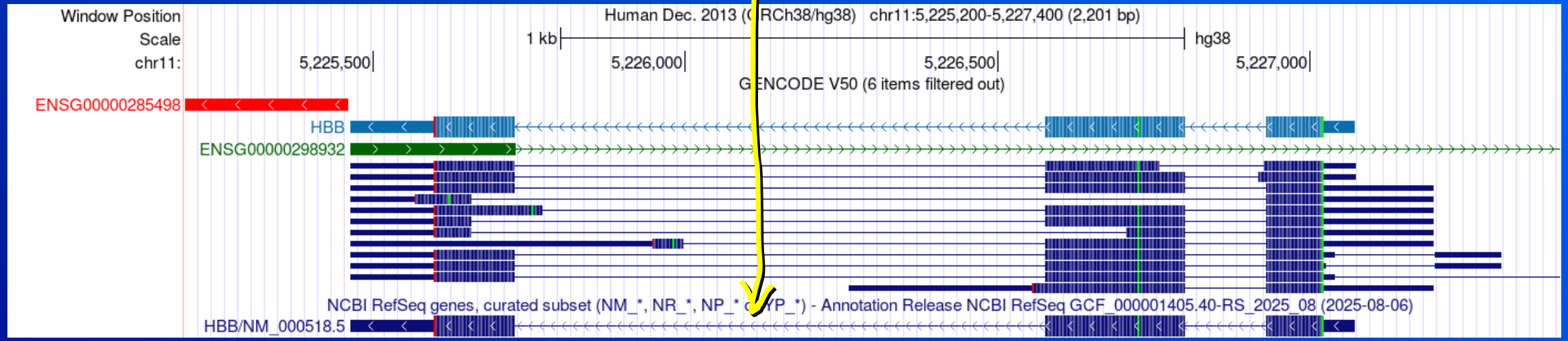
HBB in UCSC Genome Browser: https://genome.ucsc.edu

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Genome browsers

- One transcript, different sequences

Click this transcript



Stockholms universitet

Genome browsers

One transcript, different sequences

RefSeq Gene HBB

RefSeq: NM_000518.5 **Status:** Reviewed
Description: hemoglobin subunit beta
Molecule type: mRNA
Source: BestRefSeq
Biotype: protein_coding
Synonyms: beta-globin,CD113t-C,ECYT6
OMIM: 141900
Protein: NP_000509.1

Links to sequence:

- Predicted Protein
- Predicted mRNA may be different from the genon
- Genomic Sequence from assembly

Protein (147 amino acids)

>NP_000509.1 length=147

MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLS
TPDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVD
PENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH

Full mRNA (628 nucleotides)

>NM_000518.5

acatttgcttctgacacaactgtgttcactagcaacctcaaacagacaccatggtgcatctgac
tcctgaggagaagtctgccgttactgccctgtggggcaaggtgaacgtggatgaagttggtggt
gaggccctgggcaggctgctggtggtctacccttggacccagaggttctttgagtcctttgggg
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caatttctattaaaggttcctttgttccctaagtccaactactaaactgggggatattatgaag
ggccttgagcatctggattctgcctaataaaaaacatttattttcattgcaa

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Genome browsers

One transcript, different sequences

RefSeq Gene HBB

RefSeq: NM_000518.5 **Status:** Reviewed
Description: hemoglobin subunit beta
Molecule type: mRNA
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Biotype: protein_coding
Synonyms: beta-globin,CD113t-C,ECYT6
OMIM: 141900
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Links to sequence:

- Predicted Protein
- Predicted mRNA may be different from the genomic sequence.
- Genomic Sequence from assembly

Get DNA for

Position chr11:5,225,464-5,227,071

Note: if you would prefer to get DNA for more than one feature of this tra

Sequence Retrieval Region Options:

- ☐ Promoter/Upstream by 1000 bases
- ☑ 5' UTR Exons
- ☑ CDS Exons
- ☑ 3' UTR Exons
- ☑ Introns
- ☐ Downstream by 1000 bases
- ◉ One FASTA record per gene.
- ○ One FASTA record per region (exon, intron, etc.) with 0 extra
  - ☐ Split UTR and CDS parts of an exon into separate FASTA records

Note: if a feature is close to the beginning or end of a chromosome and edge of the chromosome.

Sequence Formatting Options:

- ◉ Exons in upper case, everything else in lower case.
- ○ CDS in upper case, UTR in lower case.
- ○ All upper case.
- ○ All lower case.
- ☐ Mask repeats: ◉ to lower case ○ to N

Get DNA | Extended case/color options

Stockholms universitet

Genome browsers

One transcript, different sequences

Get DNA for

Position chr11:5,225,464-5,227,071

Note: if you would prefer to get DNA for more than one feature of this tra

Sequence Retrieval Region Options:

- ☐ Promoter/Upstream by 1000 bases
- ☑ 5' UTR Exons
- ☑ CDS Exons
- ☑ 3' UTR Exons
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- ○ CDS in upper case, UTR in lower case.
- ○ All upper case.
- ○ All lower case.
- ☐ Mask repeats: ◉ to lower case ○ to N

Get DNA | Extended case/color options

Genomic DNA: 1,608 nt

>hg38_ncbiRefSeqCurated_NM_000518.5
range=chr11:5225464-5227071 5'pad=0 3'pad=0 strand=- repeatMasking=none

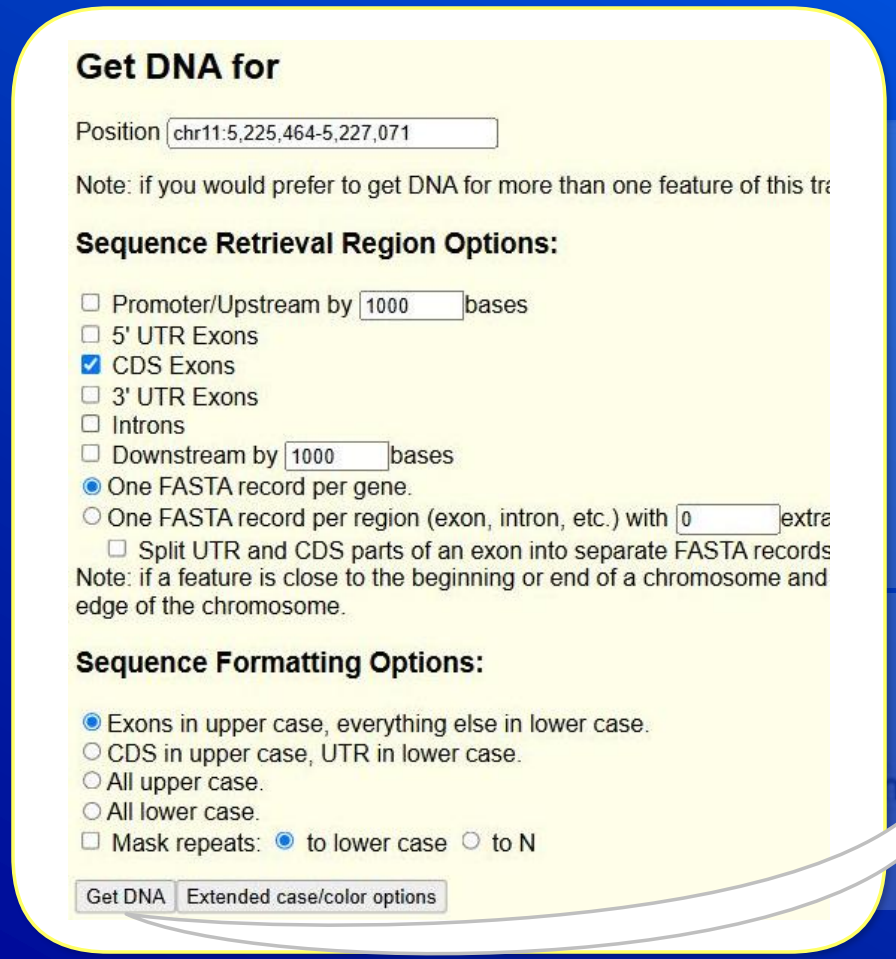
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gtctatttcccacccttagGCTGCTGGTGGTCTACCCTTGGACCCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCT
GATGCTGTTATGGGCAACCCTAAGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTCACCTGGAC
AACCTCAAGGGCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCTGCACGTGGATCCTGAGAACTTCAGGgtgagtcta
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AGTGGTGGCTGGTGTGGCTAATGCCCTGGCCCACAAGTATCACTAAGCTCGCTTTCTTGCTGTCCAATTTCTATTAAAGGTTCC
TTTGTTCCCTAAGTCCAACTACTAAACTGGGGGATATTATGAAGGGCCTTGAGCATCTGGATTCTGCCTAATAAAAAACATTTA
TTTTCATTGCAA

Uppercase: exons Lowercase: introns

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Genome browsers

One transcript, different sequences



Coding sequence: 444 nt

>hg38_ncbiRefSeqCurated_NM_000518.5
range=chr11:5225598-5227021 5'pad=0 3'pad=0 strand=- repeatMasking=none

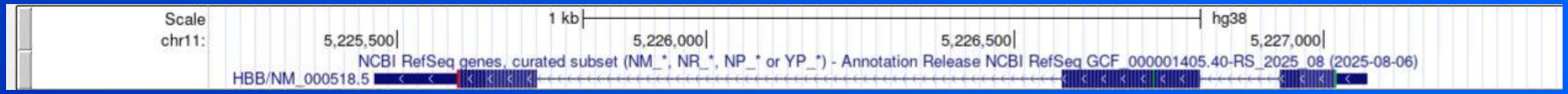
ATGGTGCATCTGACTCCTGAGGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGA
ACGTGGATGAAGTTGGTGGTGAGGCCCTGGGCAGGCTGCTGGTGGTCTACCCTTGGAC
CCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCTGATGCTGTTATGGGCAAC
CCTAAGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTC
ACCTGGACAACCTCAAGGGCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCT
GCACGTGGATCCTGAGAACTTCAGGCTCCTGGGCAACGTGCTGGTCTGTGTGCTGGCC
CATCACTTTGGCAAAGAATTCACCCCACCAGTGCAGGCTGCCTATCAGAAAGTGGTGG
CTGGTGTGGCTAATGCCCTGGCCCACAAGTATCACTAA

Joined CDS exons, including terminal stop codon

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Genome browsers

One transcript, different sequences



Genomic DNA, 1,608 nt

```
ACATTTGCTTCTGACACAACTGTGTTCACTAGCAACCTCAAACAGACACCATGGTGCATCTGACTCCTGAGGAGAAGTCTGCCGT
TACTGCCCTGTGGGGCAAGGTGAACGTGGATGAAGTTGGTGGTGAGGCCCTGGGCAGgttggtatcaaggttacaagacaggttt
aaggagaccaatagaaactgggcatgtggagacagagaagactcttgggtttctgataggcactgactctctctgcctattggtc
tattttcccaccctagGCTGCTGGTGGTCTACCCTTGGACCCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCTGATG
CTGTTATGGGCAACCCTAAGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTCACCTGGACAACCT
CAAGGGCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCTGCACGTGGATCCTGAGAACTTCAGGgtgagtctatgggac
gcttgatgttttctttcccttctttttctatggttaagttcatgtcataggaaggggataagtaacagggtacagtttagaatgg
gaaacagacgaatgattgcatcagtgtggaagtctcaggatcgttttagtttcttttatttgctgttcataacaattgttttctt
ttgtttaattcttgctttctttttttttcttctccgcaatttttactattatacttaatgccttaacattgtgtataacaaaagg
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CTGGTCTGTGTGCTGGCCCATCACTTTGGCAAAGAATTCACCCCACCAGTGCAGGCTGCCTATCAGAAAGTGGTGGCTGGTGTGG
CTAATGCCCTGGCCCACAAGTATCACTAAGCTCGCTTTCTTGCTGTCCAATTTCTATTAAAGGTTCCTTTGTTCCCTAAGTCCAA
CTACTAAACTGGGGGATATTATGAAGGGCCTTGAGCATCTGGATTCTGCCTAATAAAAAACATTTATTTTCATTGCAA
```

Join exons

Full transcript, 628 nt

```
acatttgcttctgacacaactgtgttcactagcaacctcaaacagacaccatggtgcatctgac
tcctgaggagaagtctgccgttactgccctgtggggcaaggtgaacgtggatgaagttggtggt
gaggccctgggcaggctgctggtggtctacccttggacccagaggttctttgagtcctttgggg
atctgtccactcctgatgctgttatgggcaaccctaaggtgaaggctcatggcaagaaagtgct
cggtgcctttagtgatggcctggctcacctggacaacctcaagggcacctttgccacactgagt
gagctgcactgtgacaagctgcacgtggatcctgagaacttcaggctcctgggcaacgtgctgg
tctgtgtgctggcccatcactttggcaaagaattcaccccaccagtgcaggctgcctatcagaa
agtggtggctggtgtggctaatgccctggcccacaagtatcactaagctcgctttcttgctgtc
caatttctattaaaggttcctttgttccctaagtccaactactaaactgggggatattatgaag
ggccttgagcatctggattctgcctaataaaaaacatttatttttcattgcaa
```

Select CDS

Coding sequence, 444 nt

```
ATGGTGCATCTGACTCCTGAGGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGA
ACGTGGATGAAGTTGGTGGTGAGGCCCTGGGCAGGCTGCTGGTGGTCTACCCTTGGAC
CCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCTGATGCTGTTATGGGCAAC
CCTAAGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTC
ACCTGGACAACCTCAAGGGCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCT
GCACGTGGATCCTGAGAACTTCAGGCTCCTGGGCAACGTGCTGGTCTGTGTGCTGGCC
CATCACTTTGGCAAAGAATTCACCCCACCAGTGCAGGCTGCCTATCAGAAAGTGGTGG
CTGGTGTGGCTAATGCCCTGGCCCACAAGTATCACTAA
```

Translate

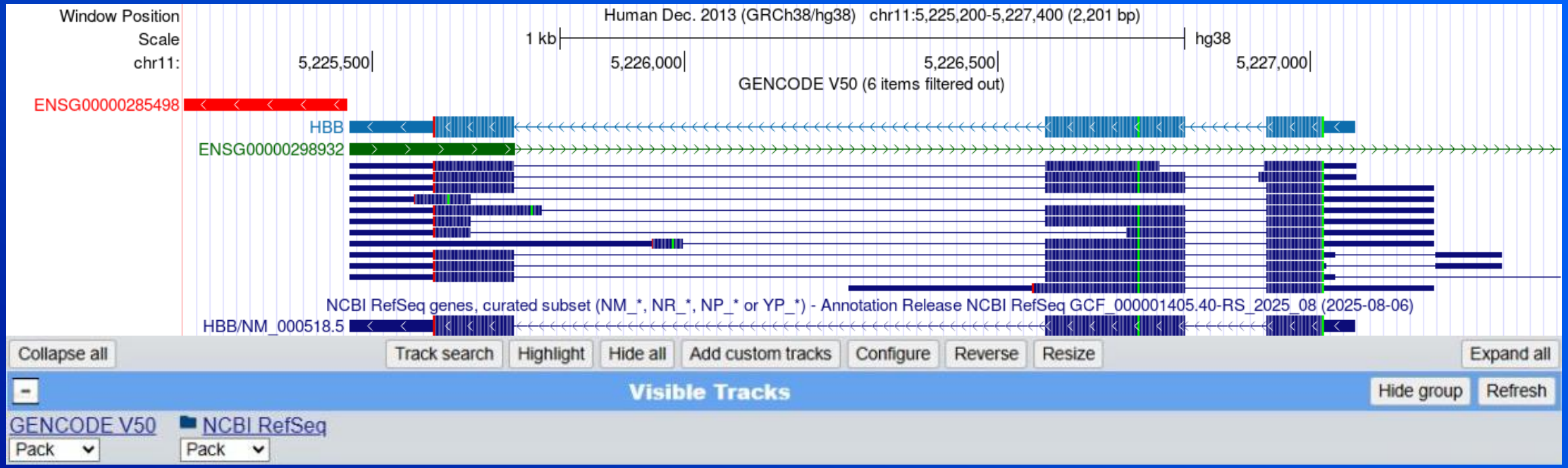
Protein, 147 aa

```
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLS
TPDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVD
PENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH
```

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Genome browsers

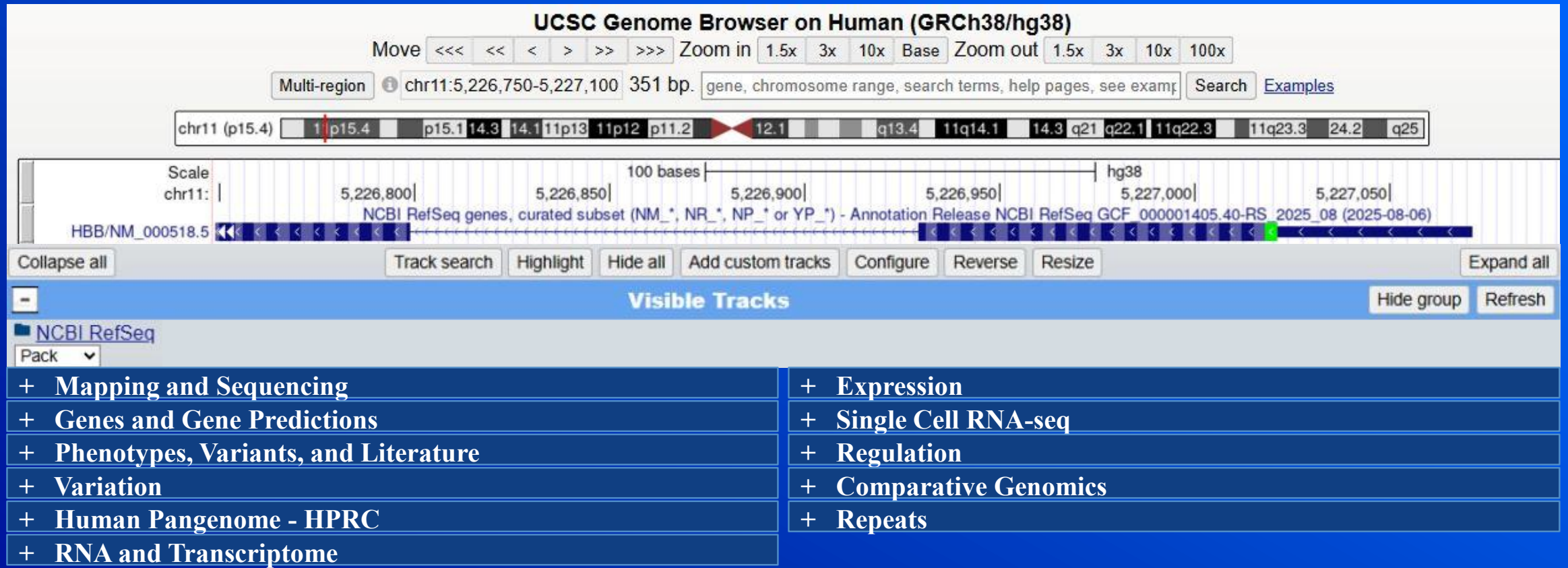
What is visible?



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Genome browsers

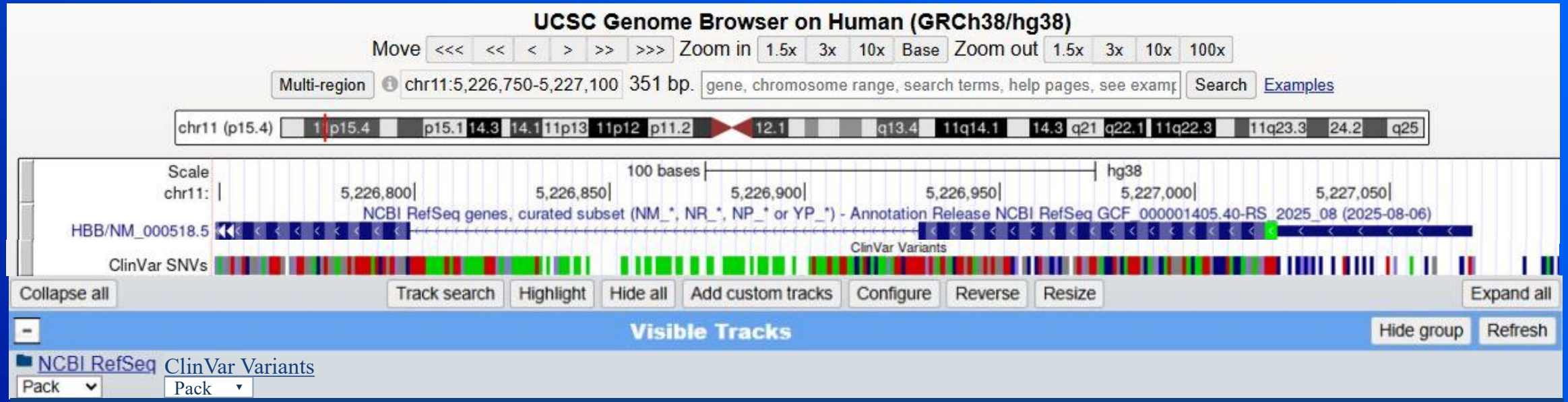
What is visible?



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Genome browsers

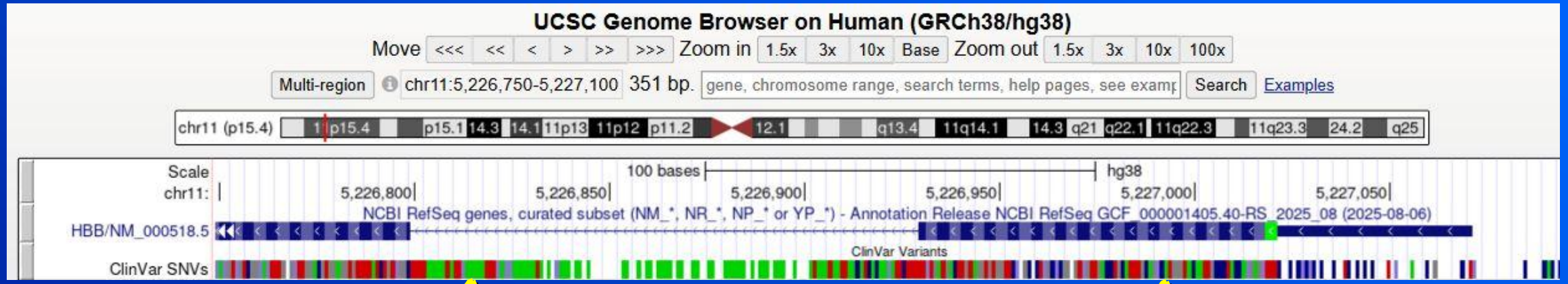
What is visible?



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Genome browsers

Where does the variant lie?



An intronic variant:

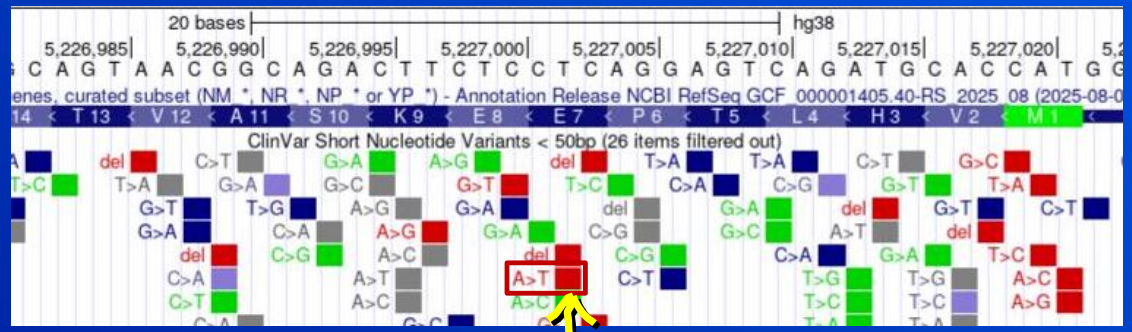
A coding variant:

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Genome browsers

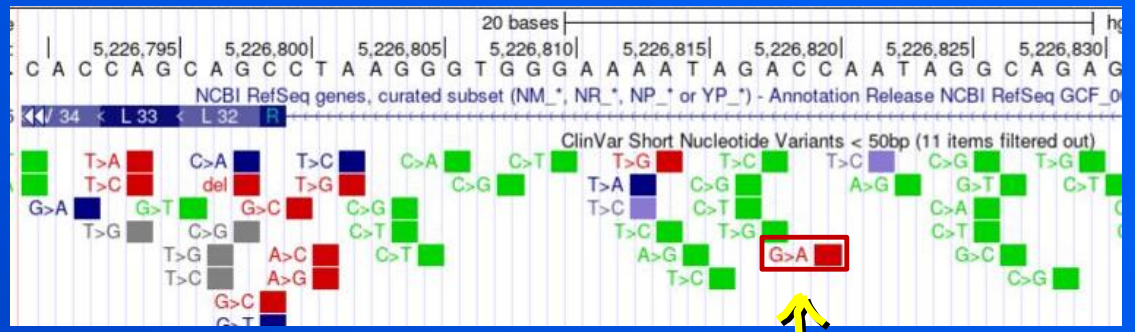
What is consequence of a variant?

Coding variant



at DNA position 5,227,002
A changes to T
GAG becomes GTG
GLU7 in beta-globin becomes Val

Intronic variant



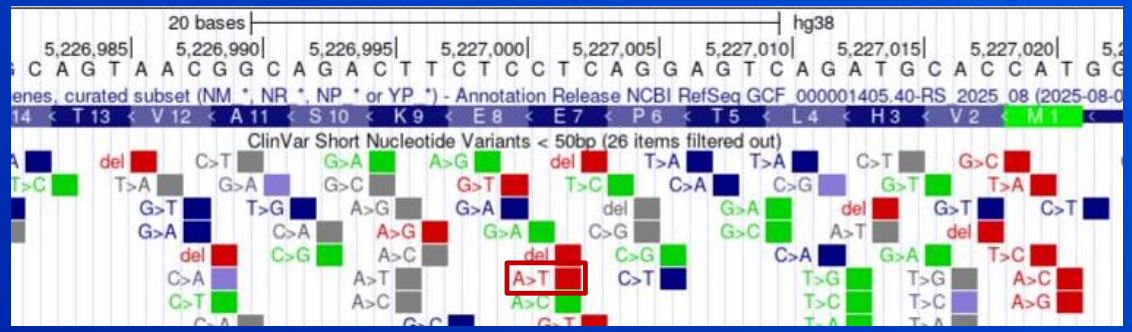
at DNA position 5,226,820
G changes to A
Leaves 19 intronic bases in mRNA
Premature stop codon

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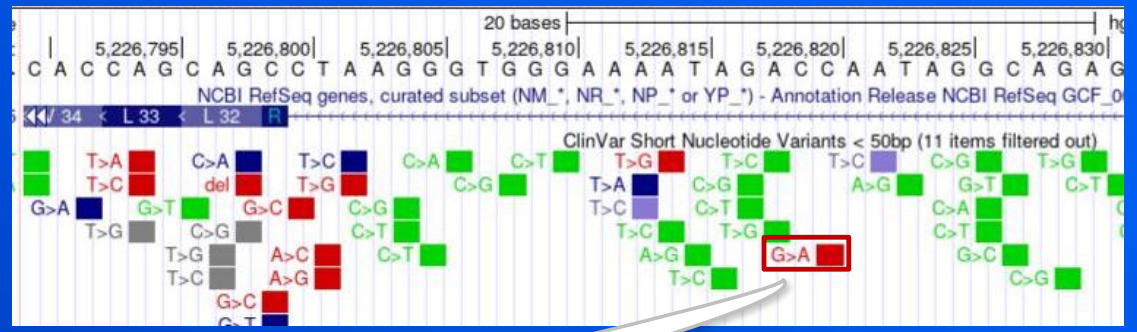
Genome browsers

Follow the evidence

- Coding variant



- Intronic variant



UCSC feature details

Link to ClinVar with Variant ID	VCV000015454 : NM_000518.5(HBB):c.93-21G>A
Clinical significance	Pathogenic
Review Status	★★★☆ based on: reviewed by expert panel

ClinVar record

Supporting submissions and cited studies

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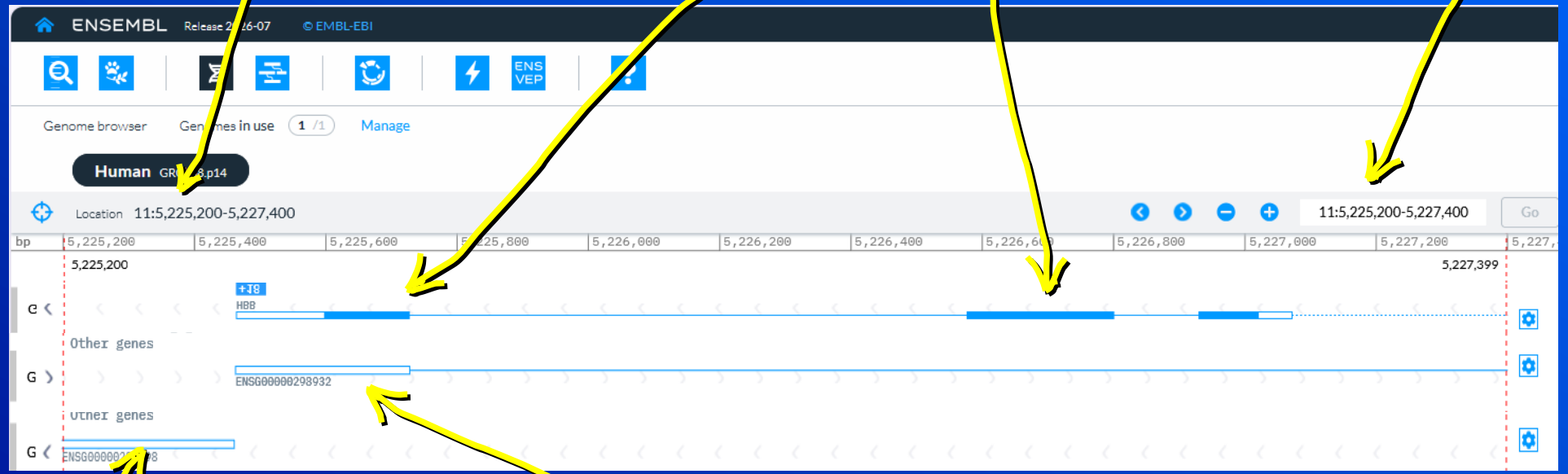
Genome browsers

What can we find in Ensembl?

Genome assembly

Exons separated by introns

Chromosome region



Possible transcript

Non-coding RNA exon on opposite DNA strand

https://www.ensembl.org/genome-browser/GCA_000001405.29?focus=location:11:5225200-5227400&location=11:5225200-5227400

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Genome browsers

One gene, many sequences

Three separate exons
142, 223 and 263 bases

Genomic DNA for
that transcript
1,608 bases

Spliced transcript
628 bases

Coding sequence
444 bases

Whole-gene DNA
3,932 bases



Beta-globin protein
147 amino acids

https://www.ensembl.org/feature-explorer/GCA_000001405.29/gene:ENSG00000244734?view=transcripts

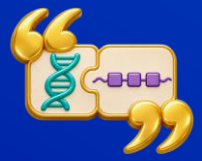
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Finding evidence online

We located an HBB splice variant. Which study measured its effect?



Literature databases



- **Build efficient queries**



- **Filter and open papers**



- **Follow subject headings**



AI-based discovery



- **Find candidate studies**



- **Check claims in the paper**

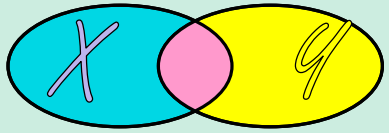
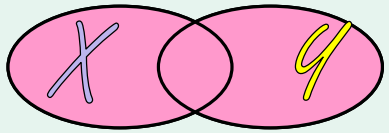
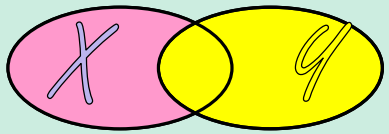


- **Spot AI search pitfalls**

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Building efficient queries

Logical operators

Operator	Schematic matches	X query	Y query	PubMed hits for		
				X search	Y search	Combined
AND		HBB	review	2,899	5,111,910	241
OR		HBB	"beta-globin"	2,899	8,884	11,067
NOT		(HBB AND thalassemia)	review	939	5,111,910	863

PubMed counts: 11 Nov 2026

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Building efficient queries

Restricting search space

Example	Unrestricted query	Hits	Restricted query	Hits
Gene	HBB	2,899	HBB[tiab]	2,701
Article type	review	5,111,910	review[pt]	4,100,845
AND	HBB AND review	241	HBB[tiab] AND review[pt]	148
OR	HBB OR "beta-globin"	11,067	HBB[tiab] OR "beta-globin"[tiab]	10,869
NOT	(HBB AND thalassemia) NOT review	863	(HBB[tiab] AND thalassemia[tiab]) NOT review[pt]	791

[tiab] Title, abstract and author keywords

[pt] Publication type

PubMed counts: 11 Nov 2026

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Building efficient queries

Quotation marks

Query

beta AND globin

beta AND globin

24,779 PubMed records

"beta globin"

8,885 PubMed records

Schematic text matches



Quotation marks keep the words together, in this order

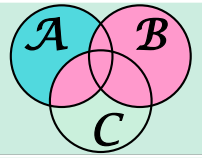
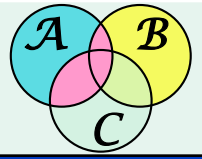
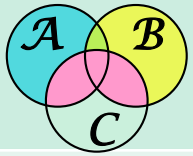
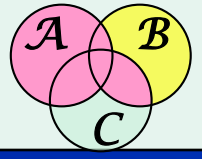


PubMed, 11 Nov 2026. Counts change as the database updates.

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Building efficient queries

Use parentheses with care

Query	Schematic matches	Interpretation	PubMed hits
HBB AND thalassemia OR "beta globin"		“beta globin” alone is sufficient; HBB must be combined with thalassemia	9,357
HBB AND (thalassemia OR "beta globin")		requires HBB and at least one of the other two term	1,189
HBB OR "beta globin" AND thalassemia		Requires thalassemia, together with at least one of the two names.	3,636
HBB OR ("beta globin" AND thalassemia)		HBB alone is sufficient; “beta globin” must be combined with thalassemia	5,596

A: HBB B: "beta globin" C: thalassemia Pink: matching records

PubMed, 11 Nov 2026. Counts change as the database updates.

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Building efficient queries

Wildcards for word variants

- An asterisk (*) can stand for zero or more characters

Search	Query	Matching word examples	PubMed hits
Exact word	HBB AND "thalassemia"	thalassemia	919
Wildcard	HBB AND thalass*	thalassemia, thalassaemia thalassemic, thalassaemic	941

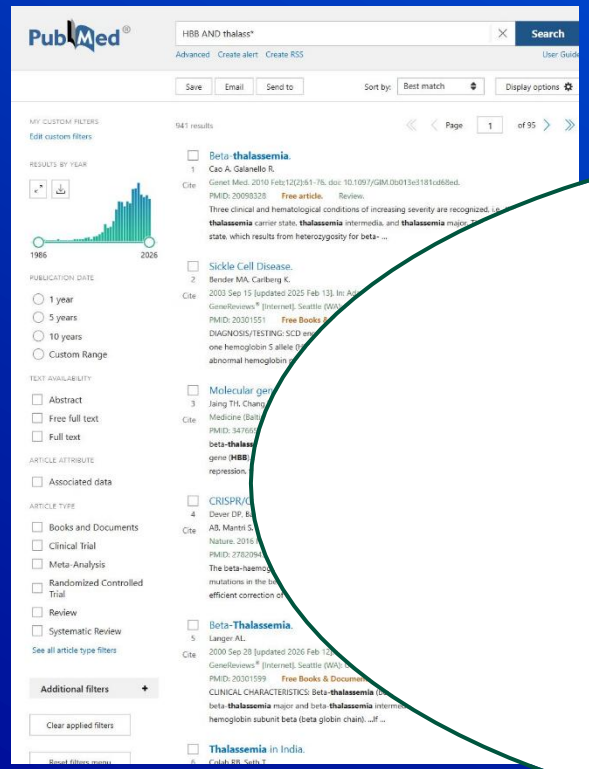
PubMed, 11 Nov 2026. Counts change as the database updates.

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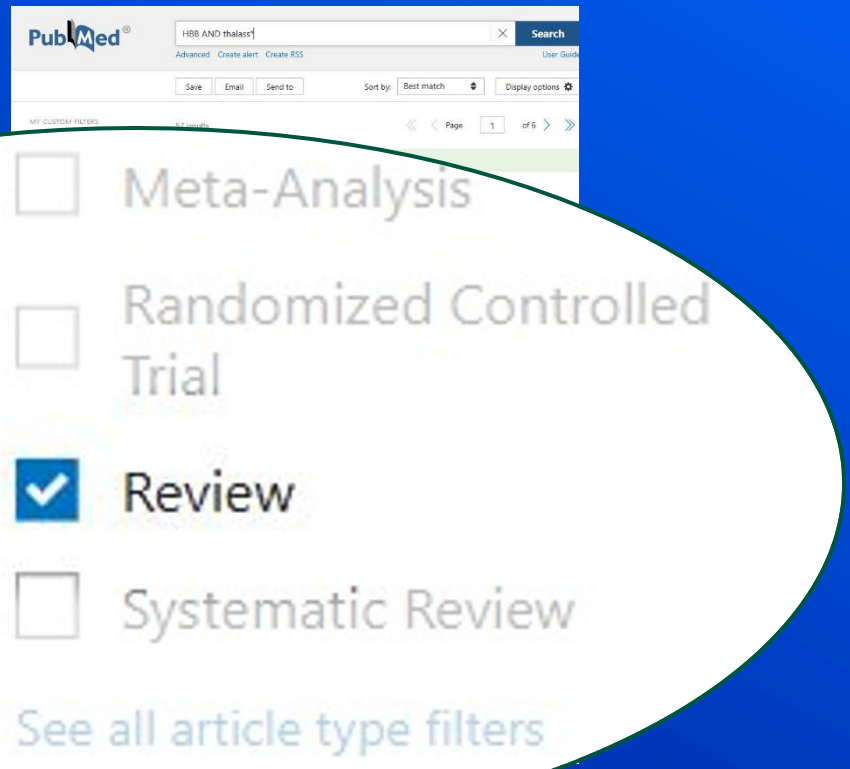
Literature databases

Filter and open papers

- **Query**



- **Filtered results**

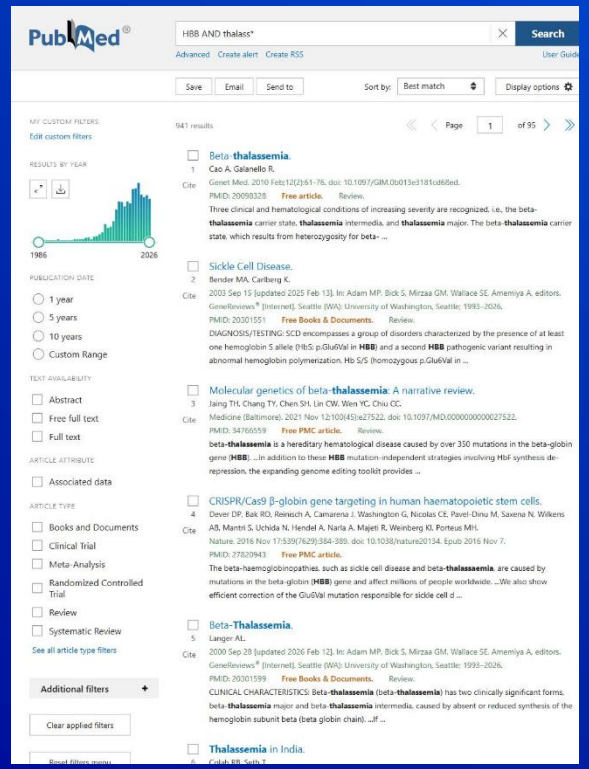


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Literature databases

Filter and open papers

- **Query**



- **Filtered results**

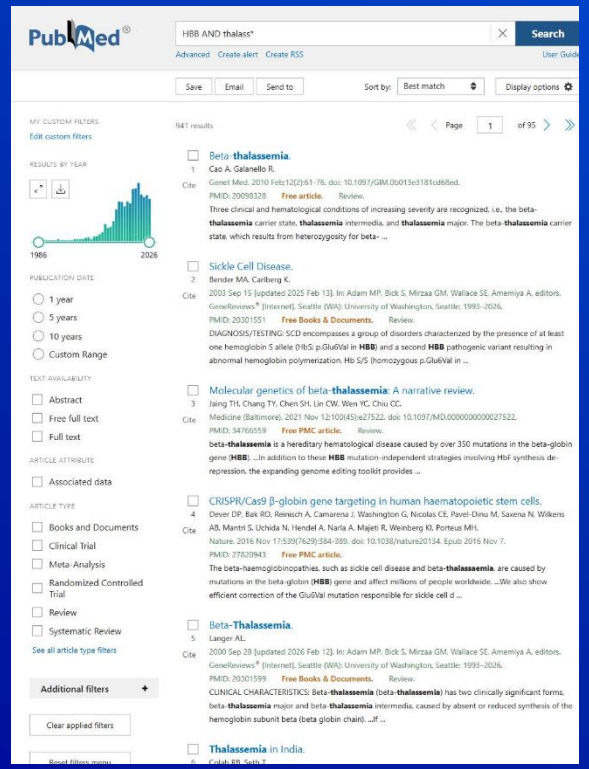


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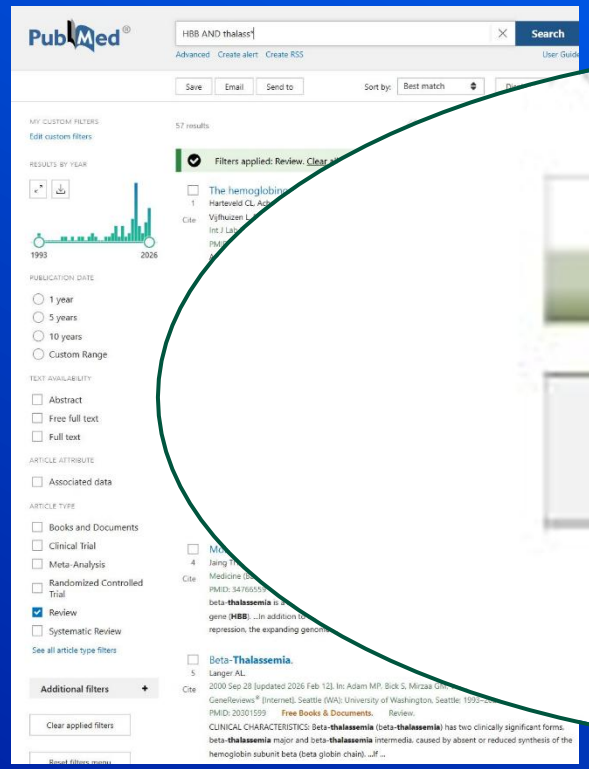
Literature databases

Filter and open papers

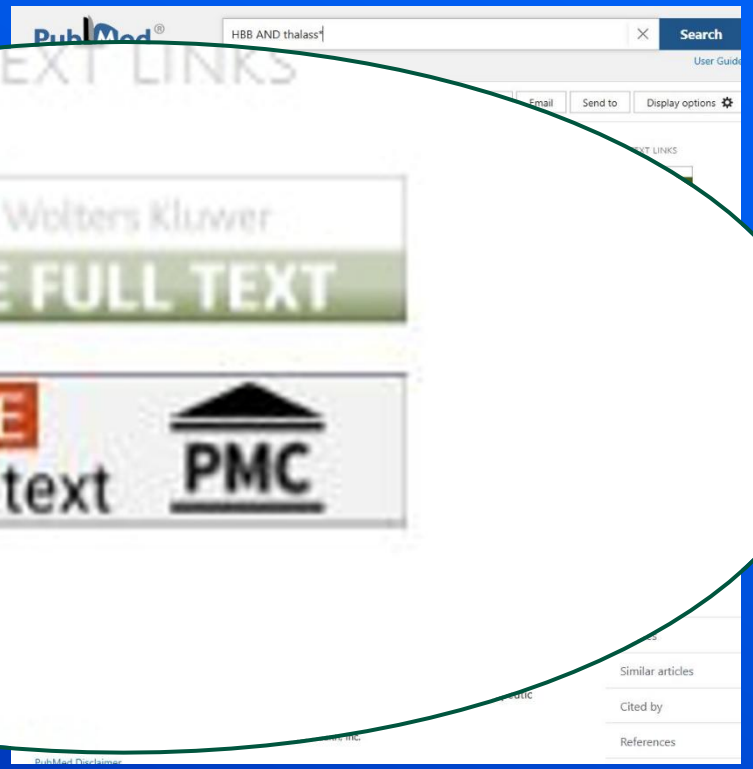
- Query



- Filtered results



- Article record

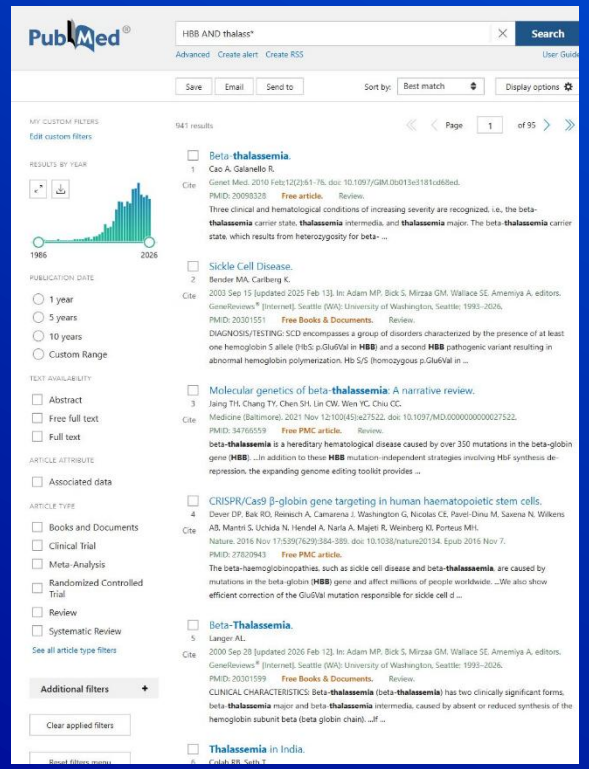


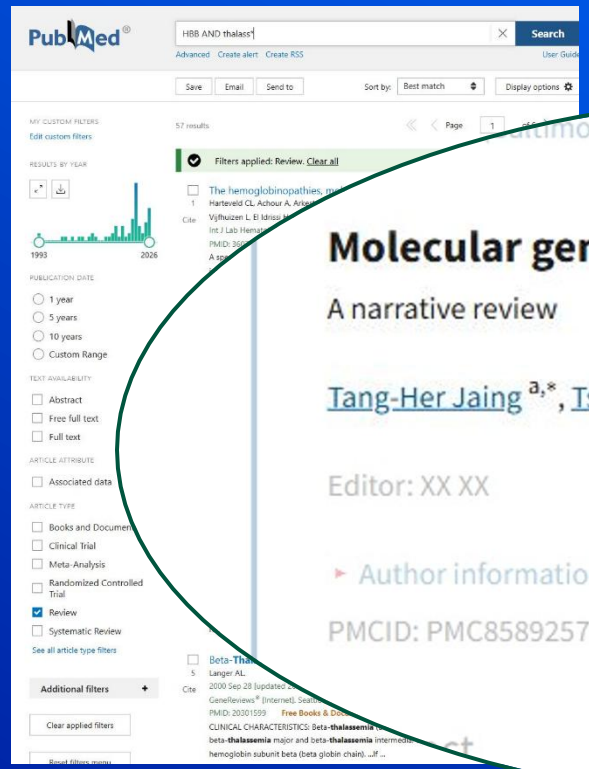
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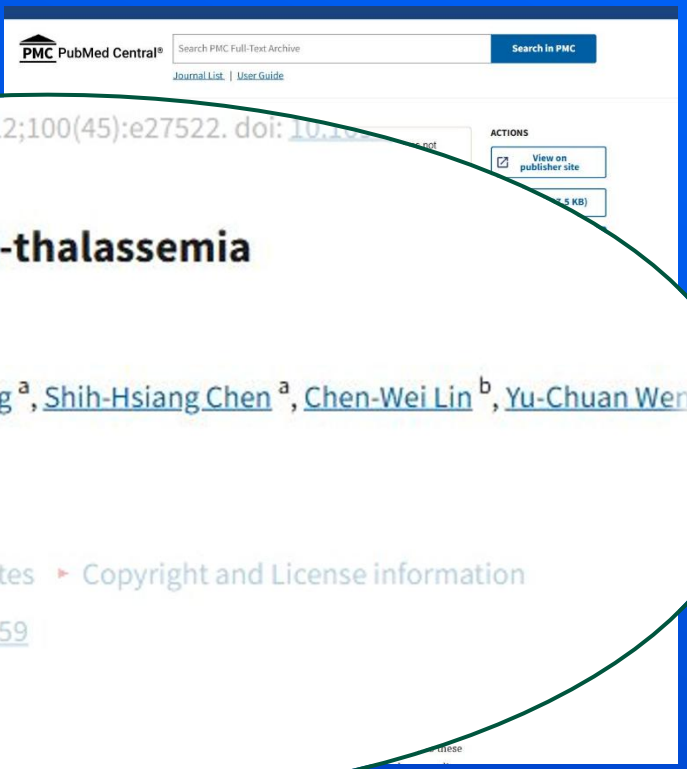
Literature databases

Filter and open papers

- Query
- Filtered results
- Full articles





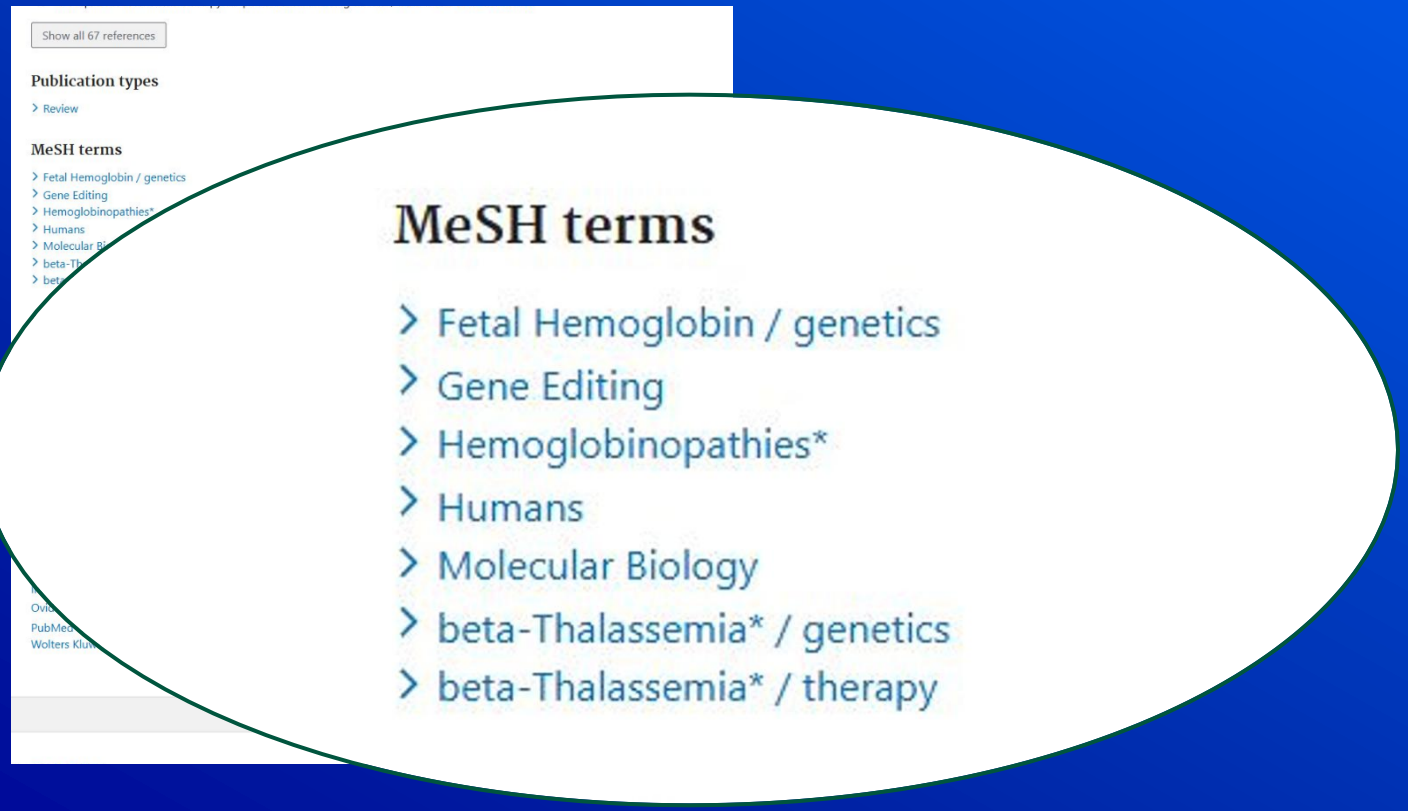


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Literature databases

Follow subject headings

- **Article record**

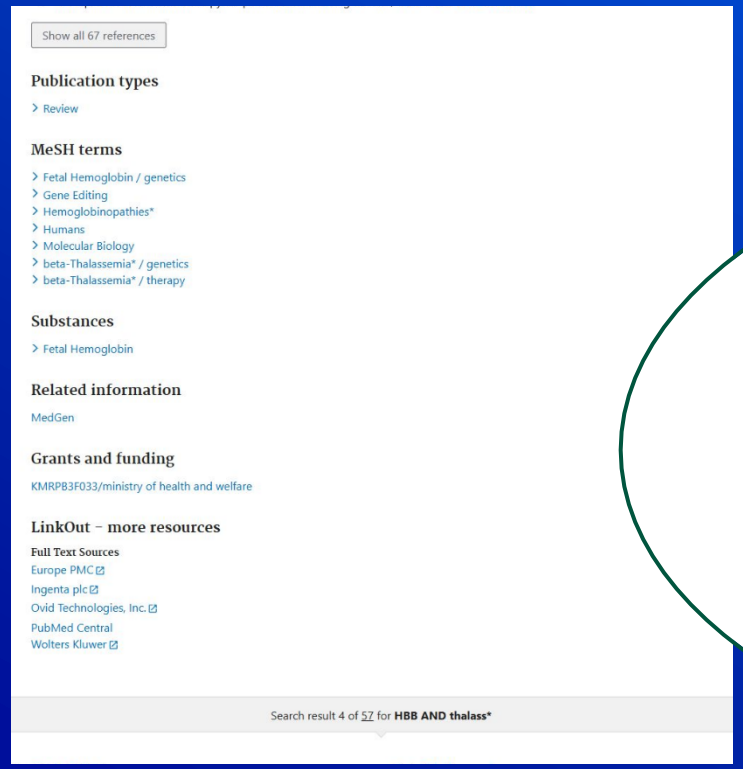


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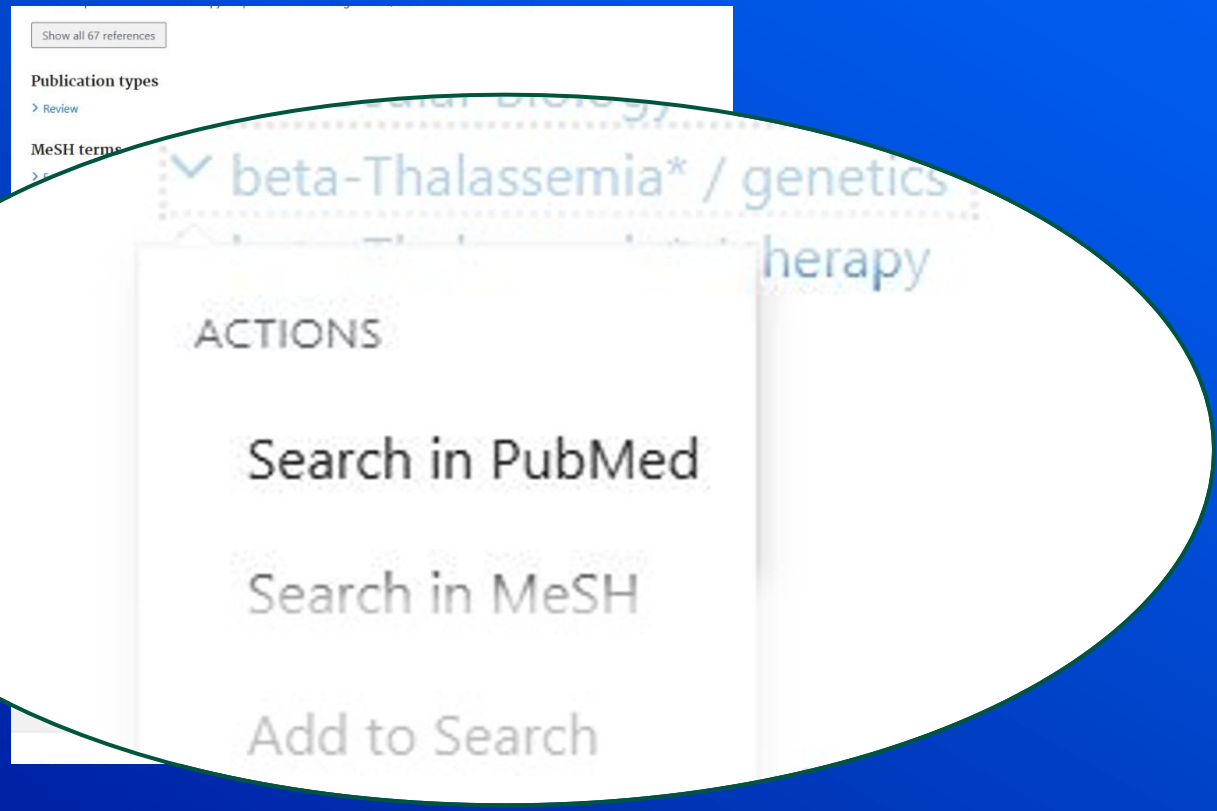
Literature databases

Follow subject headings

- **Article record**



- **Subject heading**

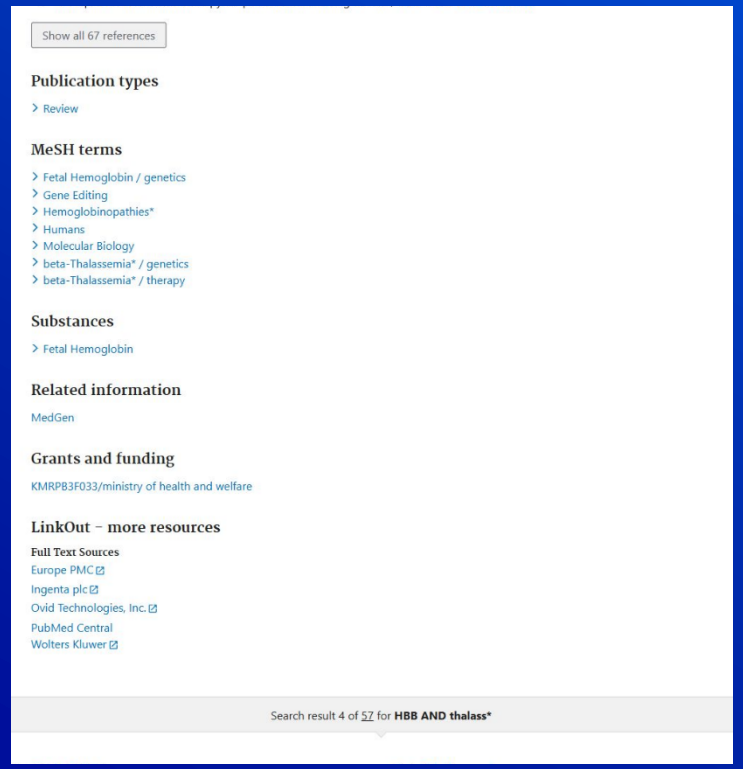


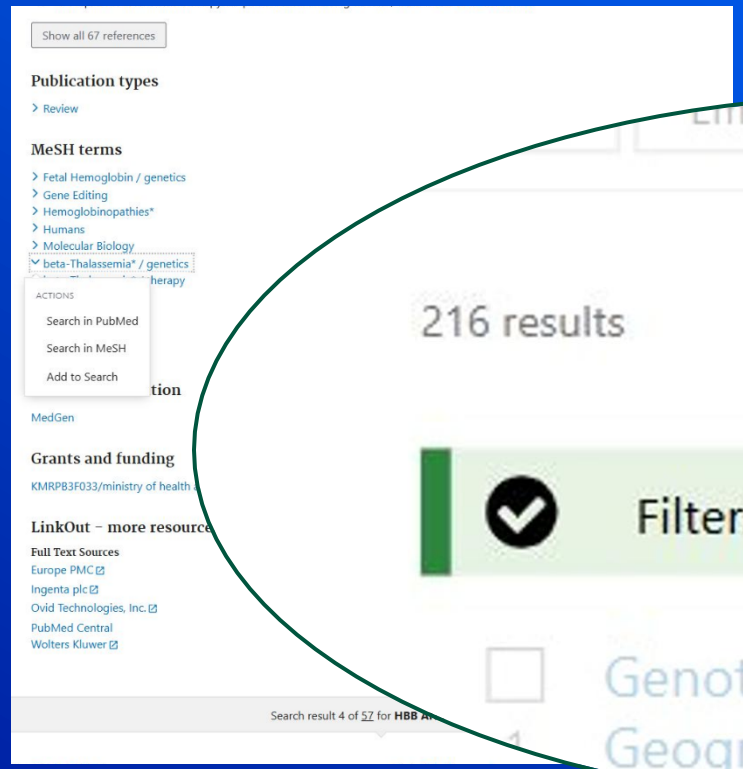
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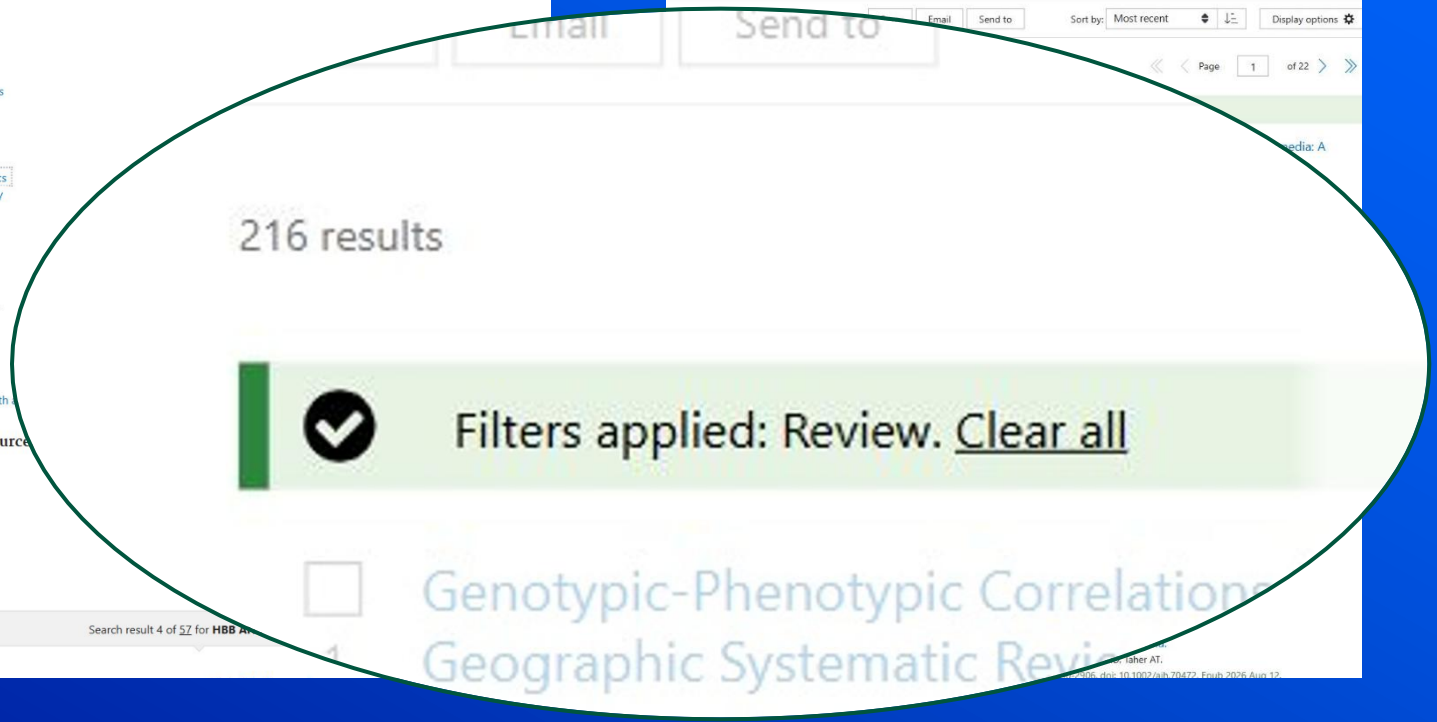
Literature databases

Follow subject headings

- **Article record**
- **Subject heading**
- **New results**





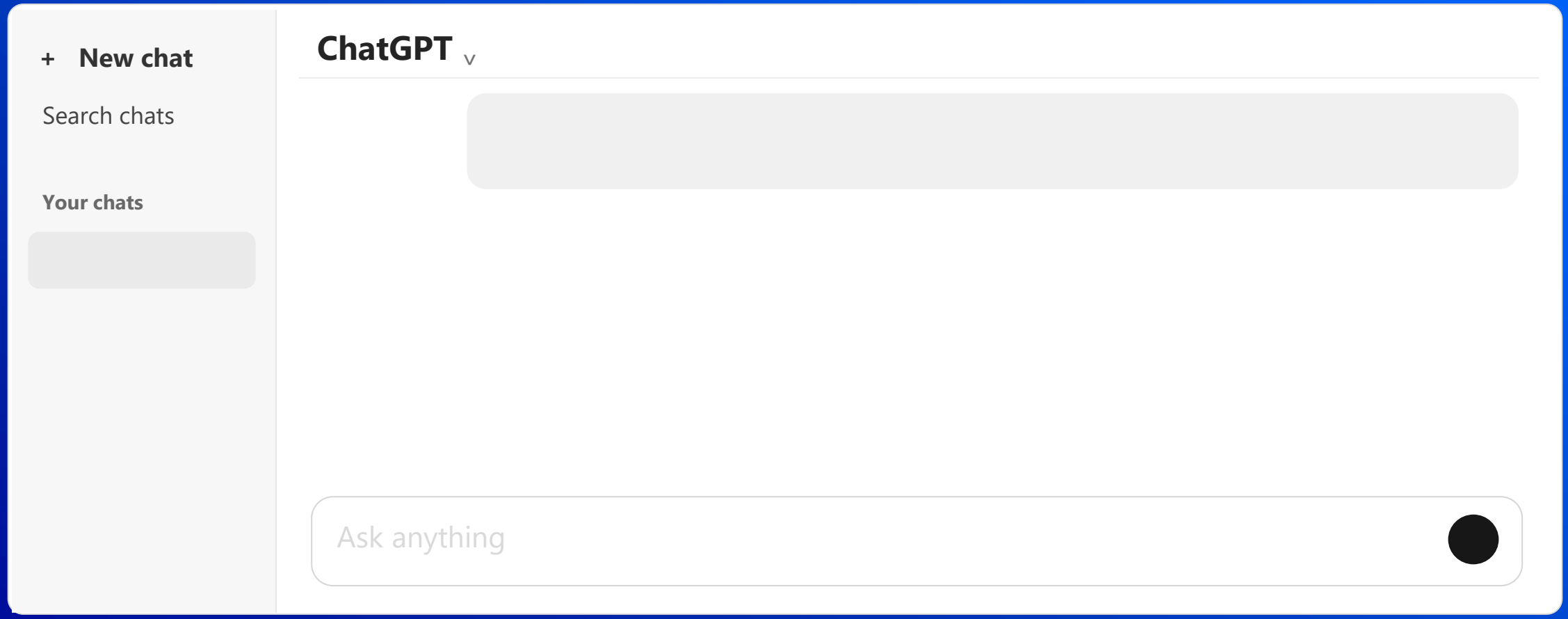


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AI-based discovery

Find candidate studies

- Generic biological question



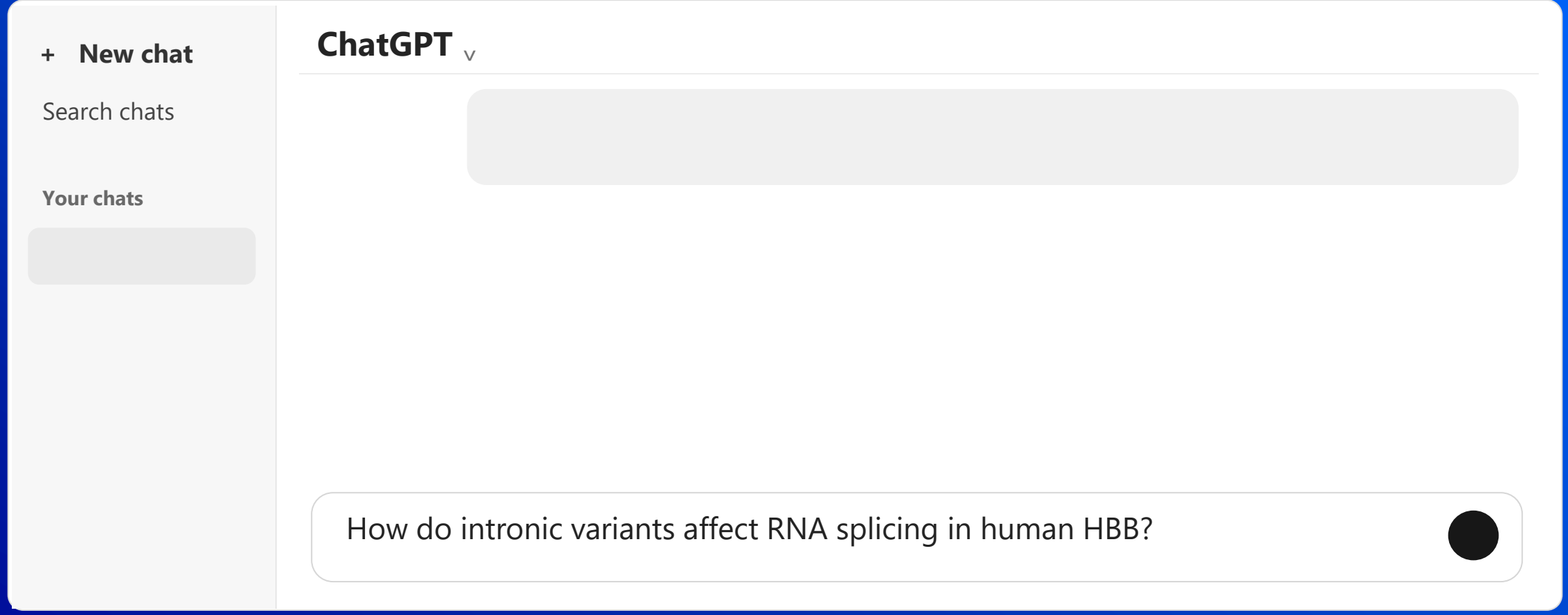
Abridged ChatGPT example

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AI-based discovery

Find candidate studies

- Generic biological question



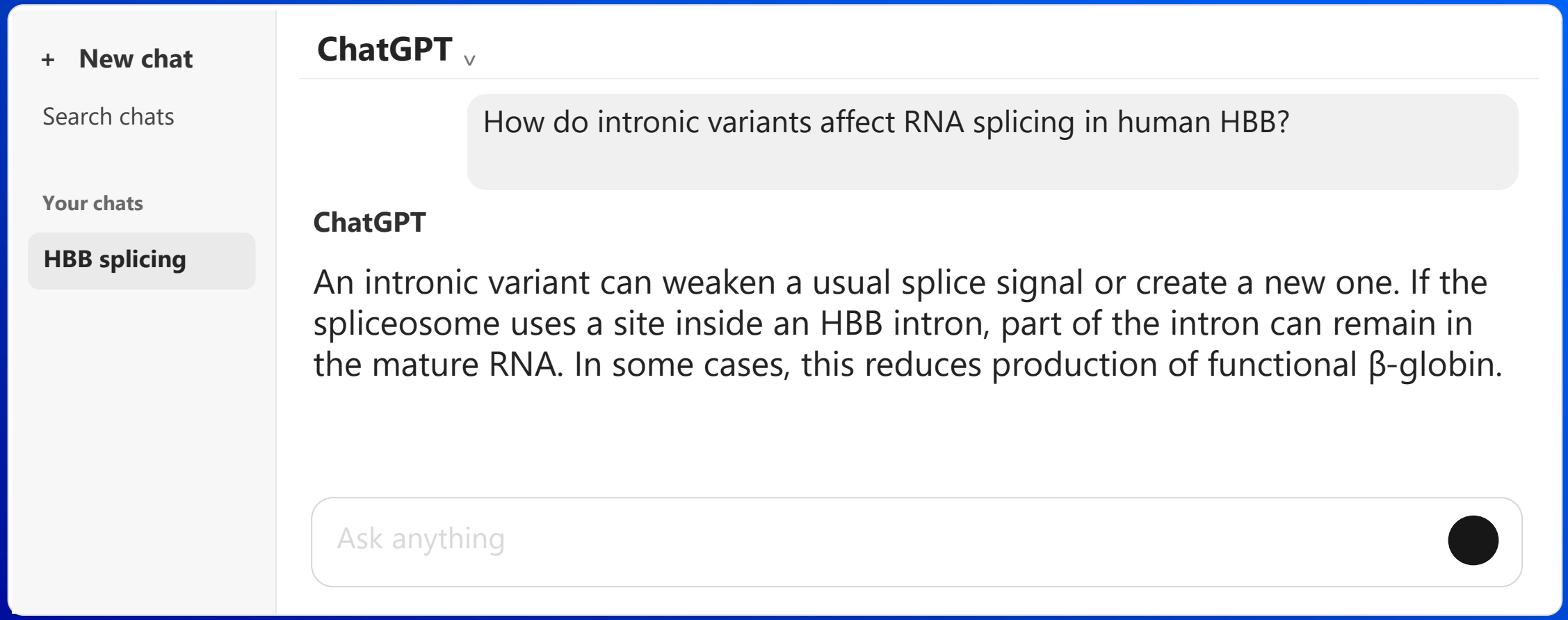
Abridged ChatGPT example

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AI-based discovery

Find candidate studies

- Generic biological question



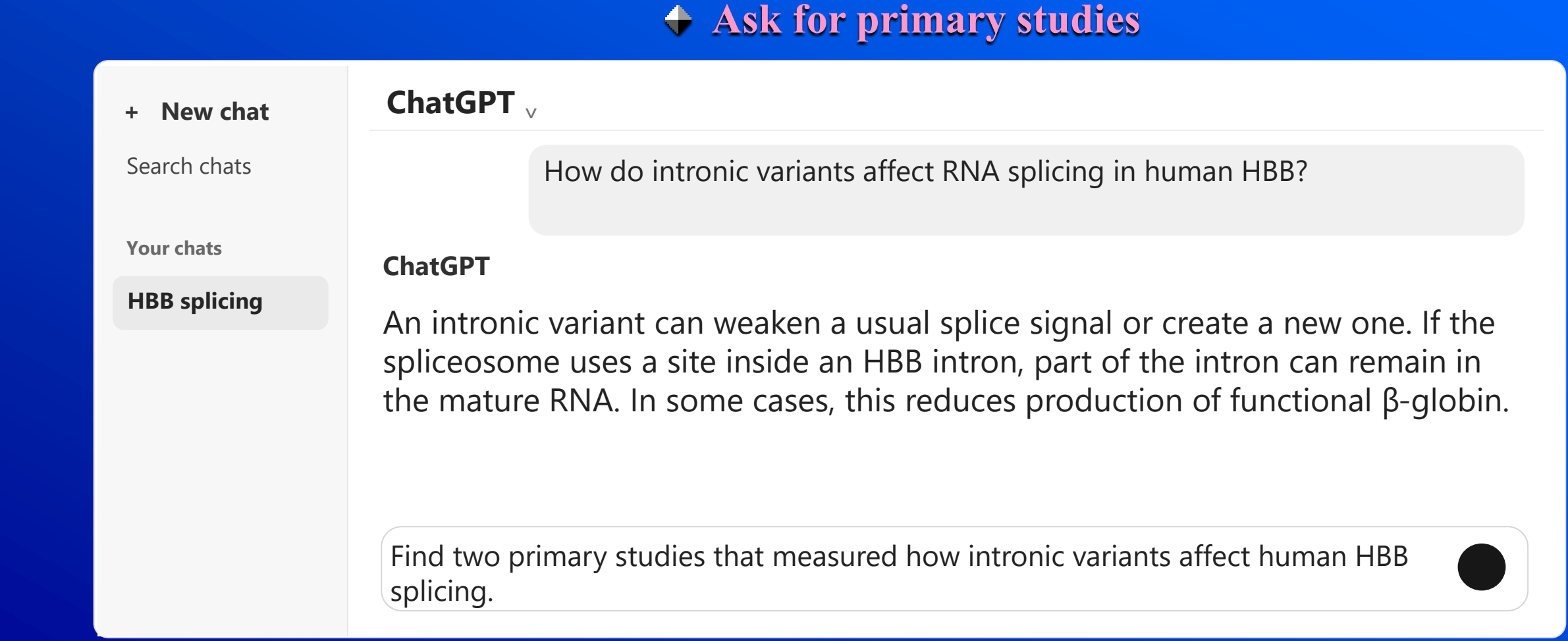
Abridged ChatGPT example

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AI-based discovery

Find candidate studies

Ask for primary studies



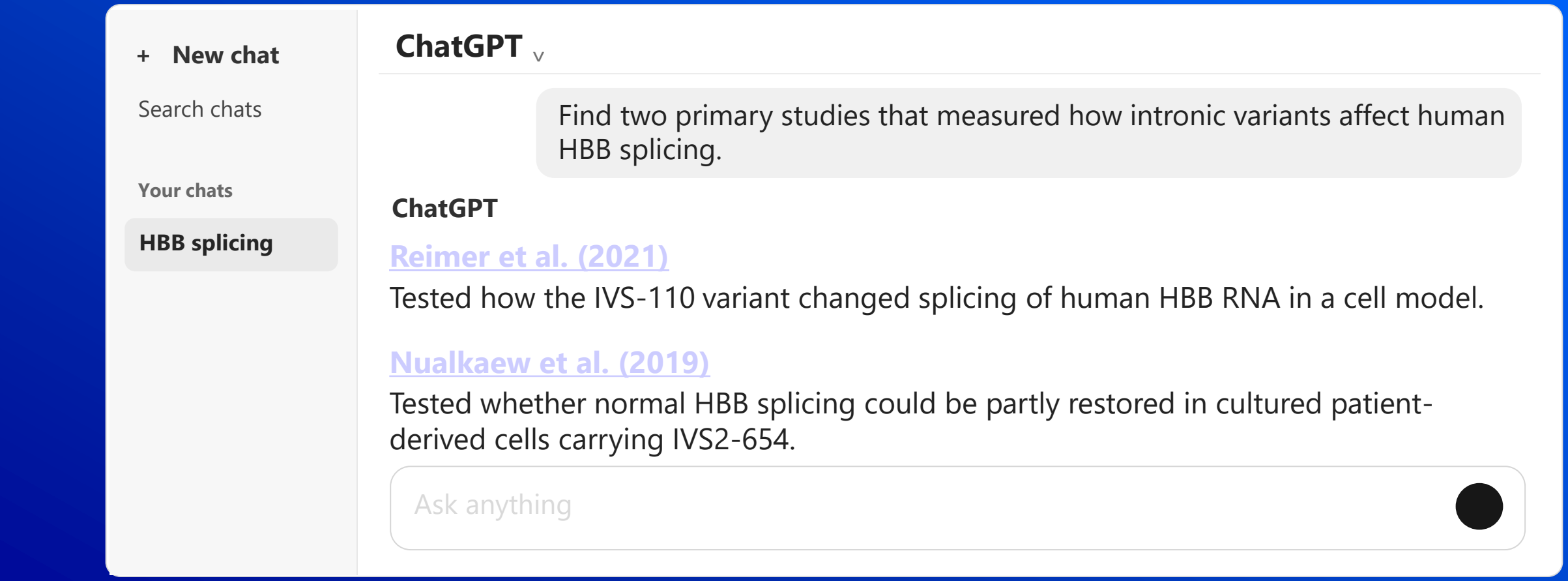
Abridged ChatGPT example

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Find candidate studies

Ask for primary studies

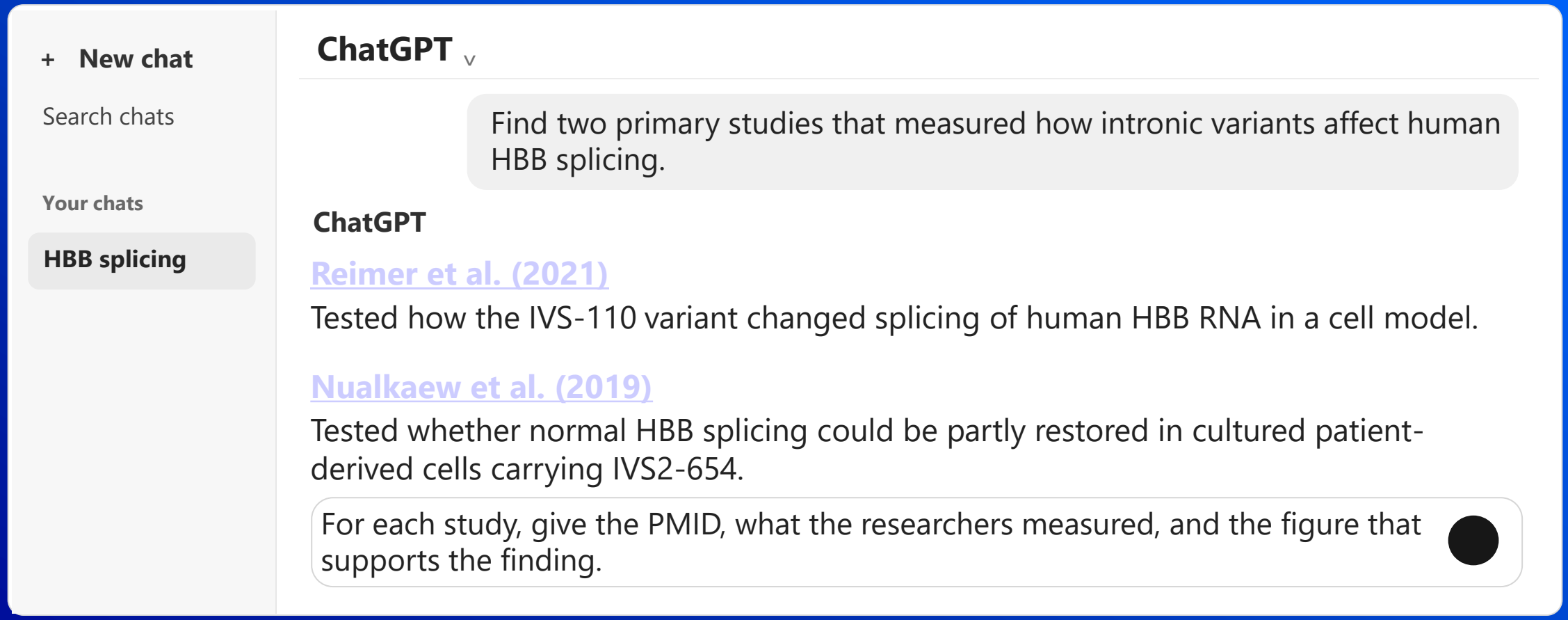


Abridged ChatGPT example

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- Find candidate studies
- Ask for supporting evidence

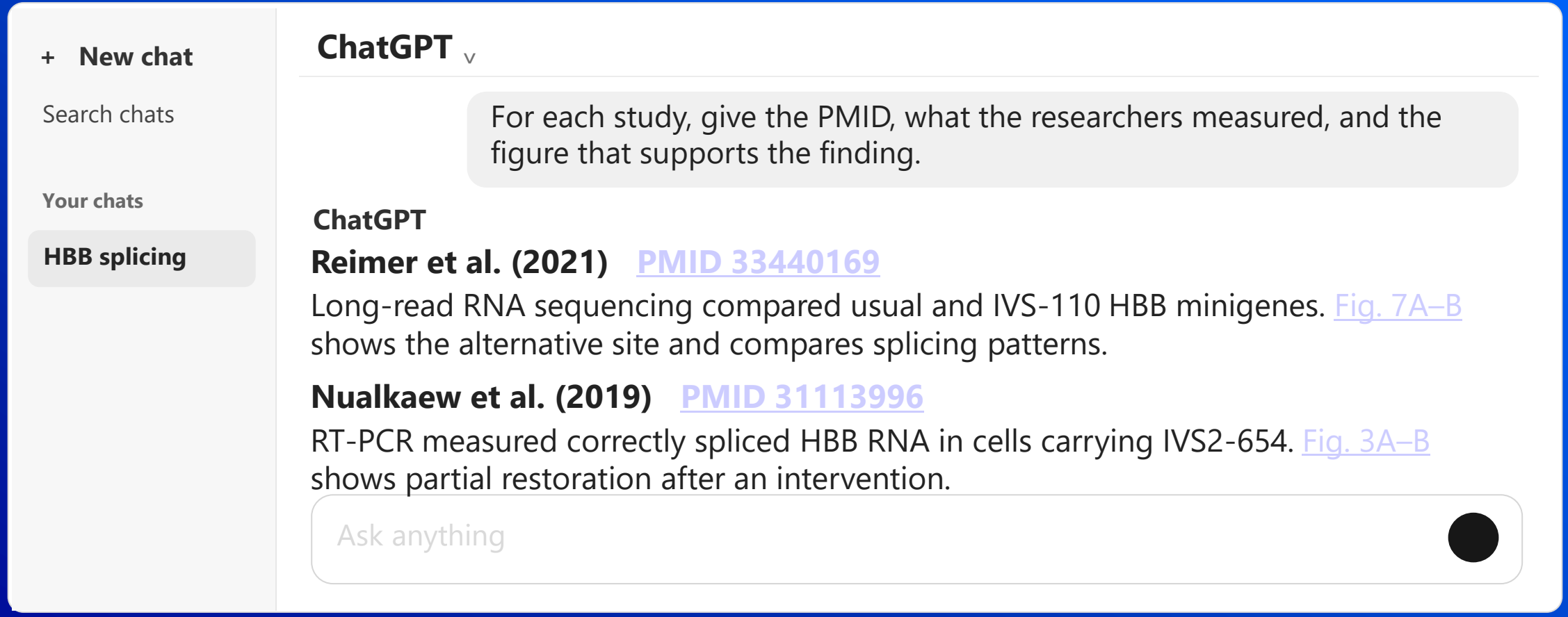


Abridged ChatGPT example

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- Find candidate studies
- Ask for supporting evidence



Abridged ChatGPT example

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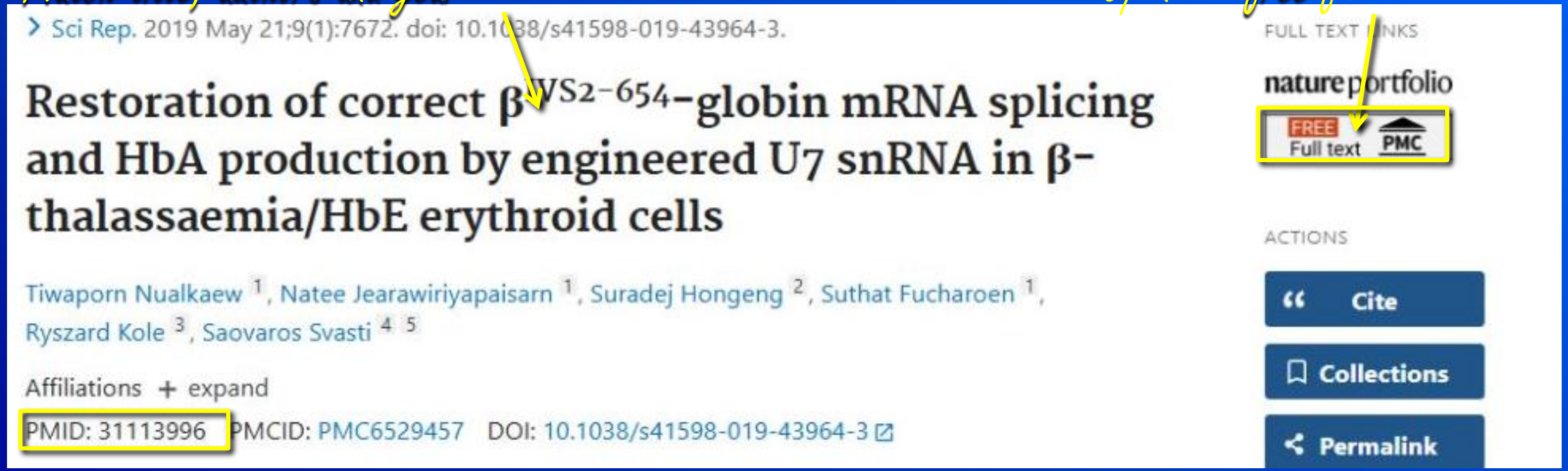
Check claims in the paper

- **Follow the citation (Nualkaew et al. (2019), PMID 31113996)**

AI summary: partially restored HBB splicing in cultured cells

Match title, authors and year

Open the free full text

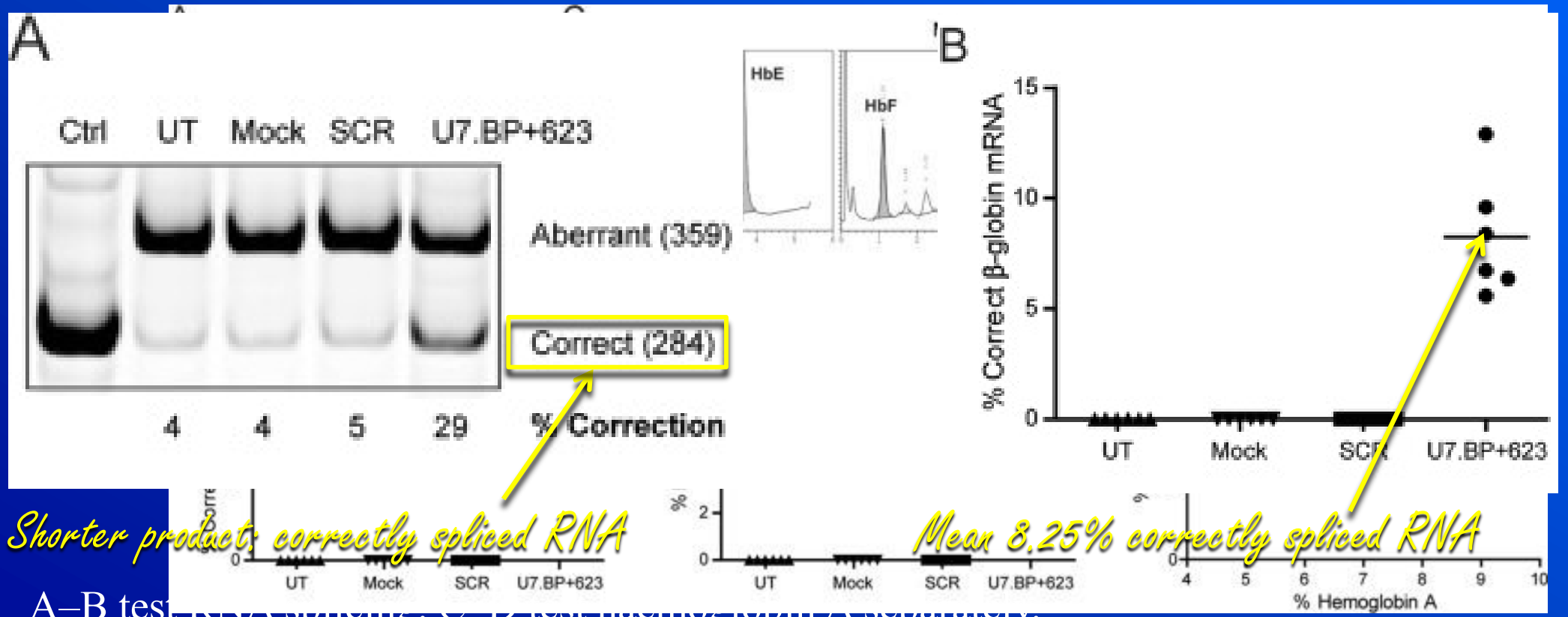


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Check claims in the paper

- **Find the measurement**



Shorter product: correctly spliced RNA

Mean 8.25% correctly spliced RNA

A–B test RNA splicing. C–D test haemoglobin A separately.

Nualkaew et al. (2019), Figure 3. CC BY 4.0.

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Check claims in the paper

- **State what the experiment supports**

- **Cells in culture**
  Patient-derived erythroid cells carrying IVS2-654

- **Engineered U7 RNA**
  U7.BP+623, delivered using a lentiviral vector

- **Partial restoration**
  Mean 8.25% correctly spliced β-globin RNA

In cultured patient-derived cells carrying IVS2-654, an engineered U7 RNA intervention partly restored correctly spliced HBB RNA.

Nualkaew et al. (2019), Results and Figure 3A–B

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Spot AI search pitfalls

- **Would you trust this reference?**



Model: Default (GPT-3.5)

Write a short paper intended for a general audience that summarizes prevention strategies within the field of dermatology. Include at least 3 references to papers that are indexed in Medline. Provide in-text citations and a complete reference list, including the PMID number for each.

Dermatology is the branch of medicine that deals with the diagnosis and treatment of skin, hair, and nail disorders. Skin disorders, in particular, are highly prevalent, and their incidence is increasing worldwide. Therefore, preventing skin disorders is crucial to maintaining skin health and preventing associated complications. This paper will summarize some of the prevention strategies within the field of dermatology.

Sun protection is one of the most important prevention strategies in dermatology. Prolonged sun exposure is the primary cause of skin cancer and skin aging. Therefore, using sunscreens, wearing protective clothing, and avoiding the sun during peak hours are essential in preventing skin damage. A study by Chen et al. (2021) found that sunscreen use significantly reduced the incidence of melanoma and squamous cell carcinoma, indicating the importance of sun protection.

Another important prevention strategy is early detection and treatment of skin disorders. Regular skin exams by a dermatologist can help detect and treat skin cancer and other skin disorders early. In a study by Telfer et al. (2009), regular skin exams led to the detection of more skin cancers at earlier stages, resulting in better treatment outcomes.

Maintaining a healthy lifestyle is also critical in preventing skin disorders. A balanced diet rich in vitamins and minerals, regular exercise, and stress management can help prevent skin aging and reduce the risk of skin cancer. A study by Kwon et al. (2015) found that a healthy lifestyle, including a balanced diet and regular exercise, was associated with a reduced risk of skin aging in middle-aged women.

In conclusion, prevention strategies in dermatology are crucial in maintaining skin health and preventing associated complications. Sun protection, early detection and treatment of skin disorders, and maintaining a healthy lifestyle are some of the key prevention strategies. By following these strategies, we can protect our skin and maintain its health.

References:

Chen, A. C., Martin, A. J., Choy, B., Fernandez-Penas, P., & Dalziell, R. A. (2021). A systematic review of the relationship between sunscreen use and the prevention of melanoma and non-melanoma skin cancer. European Journal of Dermatology, 31(1), 1-9. PMID: 33635047

Kwon, H. H., Yoon, J. Y., Park, S. Y., Min, S., Suh, D. H., & Kim, J. H. (2015). A comprehensive study on the association between diet and skin aging in Korean women. Journal of the American Academy of Dermatology, 73(2), 242-250. PMID: 26183923

Telfer, N. R., Colver, G. B., Morton, C. A., & British Association of Dermatologists Therapy Guidelines and Audit Subcommittee. (2009). Guidelines for the management of basal cell carcinoma. British Journal of Dermatology, 161(4), 643-659. PMID: 19663808

Authors and a plausible title

Chen, A. C., Martin, A. J., Choy, B., Fernandez-Penas, P., & Dalziell, R. A. (2021). A systematic review of the relationship between sunscreen use and the prevention of melanoma and non-melanoma skin cancer. European Journal of Dermatology, 31(1), 1-9. PMID: 33635047

Journal, year, volume and pages

PMID

Actual ChatGPT-3.5 output, 12 April 2023

Bhattacharyya et al. (2023), Figure 1. CC BY 4.0.

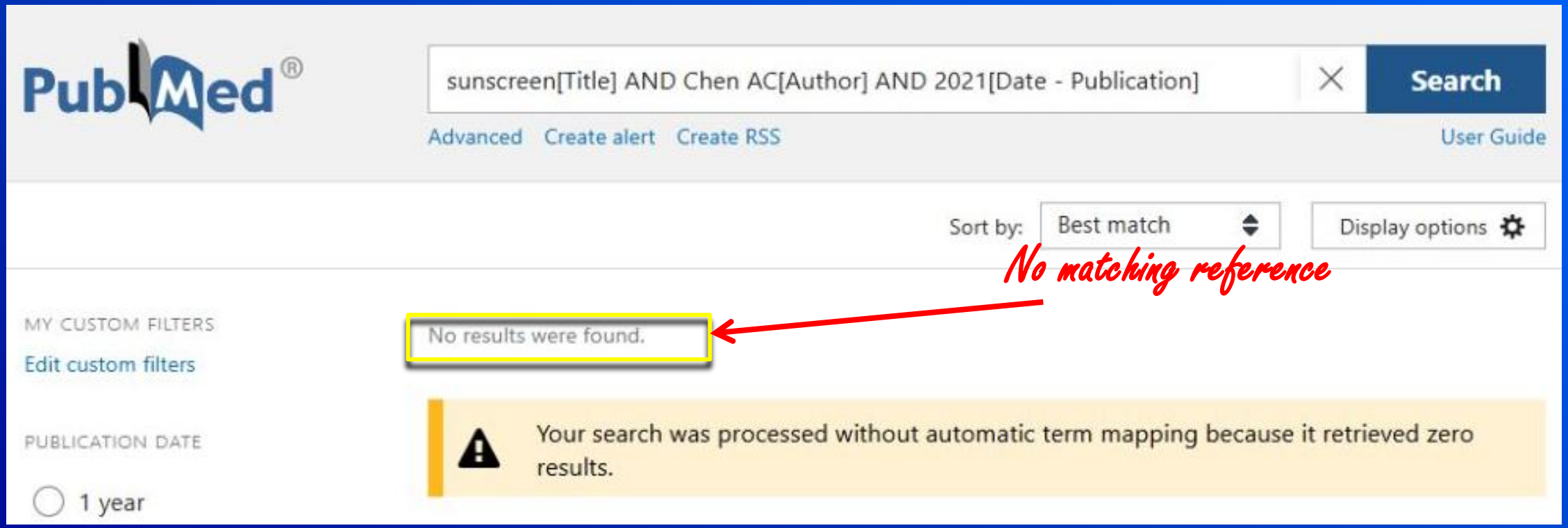
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Spot AI search pitfalls

- **Can we find the cited review?**

Title contains sunscreen; author Chen AC; publication year 2021



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Spot AI search pitfalls

- **Where does the PMID lead?**

The citation supplies PMID 33635047



This PMID identifies a different paper

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Spot AI search pitfalls

Why can references go wrong?

- ***Generating without checking records***
  A plausible citation can appear without a matching publication.
- ***Pressure to complete the list***
  A fixed quota and full metadata may exceed the available evidence.
- ***Assumptions in the question***
  The request may assume that supporting studies already exist.
- ***Incomplete or mixed source details***
  Authors, titles and identifiers can be incorrectly combined.

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Spot AI search pitfalls

- **Ask for references that can be checked**

Problematic prompt

From your existing knowledge, list **exactly ten** primary studies on intronic human HBB variants and RNA splicing. Include **complete citations, DOIs and PMIDs**. **Return only the bibliography**.

Improved prompt

Search for up to ten primary studies on intronic human HBB variants and RNA splicing. Include only references **verified against PubMed or publisher records**. Give **direct links** and **identify unverified details**. **Fewer papers are acceptable**.

A better prompt helps. The references still need checking.

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Spot AI search pitfalls

- **A real paper, the wrong conclusion**

Saline or albumin in patients with traumatic brain injury (NEJM, 2007. PMID 17761591)

ChatGPT's summary

"no significant difference in mortality between saline and albumin"

Recorded in Weinberg et al. (2024)

The paper's result

Mortality at 24 months

Albumin	**33.2%**
Saline	**20.4%**

P = 0.003

The summary lost the study's finding of higher mortality.

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From questions to evidence

Let the question guide the search



Choose the relevant biological level and type of information.

Connect the records



Use names for discovery, identifiers and cross-references for precision.

Choose a suitable access route



Record pages, programmatic access and genome browsers.

Check what supports the answer



Distinguish observations, predictions and interpretations.
Verify AI-generated claims.